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Electric Motors Drives

Graduation Project Title

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DECLARATION

We hereby declare that this graduation project titled "ELECTRIC MOTORS DRIVES" is our original work and has not been submitted previously, in whole or in part, for any degree or qualification at any other institution.
All the work presented in this report is the result of our team's joint efforts, except where otherwise acknowledged. Any contributions from other sources have been clearly referenced.

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ABSTRACT

This graduation project investigates electric motor drive systems, focusing on both theoretical research and practical implementation aspects. The primary objective is to explore efficient control strategies for modern electric motors using advanced gate driver circuits and inverter configurations.

The project begins with a comprehensive study of gate driver circuits, highlighting their importance in delivering appropriate switching signals to power semiconductor devices such as MOSFETs and IGBTs. Special attention was given to the isolation requirements, timing characteristics, and protection mechanisms necessary to ensure robust operation under high-speed switching conditions.

Inverter topologies were also thoroughly examined, particularly half-bridge and full-bridge configurations used in motor control applications. The principles of pulse-width modulation (PWM), dead-time insertion, and voltage vector synthesis were discussed to provide a solid foundation for subsequent control implementations.

Following the research phase, two advanced motor control strategies were designed, implemented, and tested:

- **Field Oriented Control (FOC)** for Permanent Magnet Synchronous Motors (PMSM): This method provides precise and dynamic torque and speed control by decoupling the stator currents into direct (d-axis) and quadrature (q-axis) components. Using Clarke and Park transformations, the motor is modeled in a rotating reference frame, allowing for independent control of flux and torque. A Proportional-Integral (PI) controller regulates the current loops, while a rotor position sensor enables accurate angle feedback.

- **Trapezoidal Control for Brushless DC (BLDC) Motors**: A simpler yet effective method was developed using Hall sensor feedback to determine rotor position and commutation timing. The approach utilizes six-step commutation, where voltage pulses are applied to the motor phases in a specific sequence to produce continuous rotation. This method is cost-effective and suitable for applications where high dynamic performance is not critical.

Custom gate driver circuits were designed and implemented to support both control strategies. For high-side and low-side switching, isolated flyback converters were developed to provide +20V/-5V gate drive voltages, improving switching noise immunity and reliability. The UCC28740 controller was used to manage flyback converter operation and maintain consistent output voltages across all six gate driver channels. Simulation and hardware validation were conducted to verify system performance. MATLAB/Simulink was used for control algorithm development and preliminary testing, while experimental setups confirmed the effectiveness of the designed circuits and control schemes. This project demonstrates a practical approach to integrating advanced motor control techniques with robust power electronics hardware, paving the way for further development in electric drive systems for automotive, industrial, and renewable energy applications.

Background and Motivation

1.1 Background

1.1.1 Market Trends and Industry Needs

In recent decades, there has been a dramatic shift towards electrification across multiple industries, including transportation, manufacturing, renewable energy, and consumer electronics. This shift is largely driven by the global demand for energy efficiency, sustainability, and reduced environmental impact. Electric motor systems account for nearly 50% of global electricity consumption, making efficiency improvements in motor control systems a key area of focus for both industry and research. As electric vehicles (EVs), hybrid electric vehicles (HEVs), industrial automation systems, and robotics become increasingly prevalent, the need for efficient, compact, and high-performance motor drives has grown exponentially. The modern market demands motors that not only deliver high torque and speed but also optimize energy consumption, reduce operational noise, and minimize maintenance requirements. To meet these growing demands, electric motors such as Permanent Magnet Synchronous Motors (PMSMs) and Brushless DC (BLDC) motors have emerged as industry favorites due to their:

- High power density.
- Superior efficiency.
- Wide speed range.
- Reduced maintenance (due to absence of brushes).

This has led to an urgent need for intelligent power electronic converters—specifically inverters—that can effectively control the power delivery to these motors. The inverter must generate precise voltage and current waveforms to ensure smooth motor operation, control the torque and speed, and protect both the power electronics and the motor from faults or overloads.

1.1.2 Inverter Topologies

The topology of the inverter plays a fundamental role in the performance of the motor drive. The most widely used inverter topologies include:

- **Half-Bridge Inverters:** Consist of one high-side and one low-side switch. They are cost-effective but limited to specific applications, typically where bidirectional power flow is not required.
- **Full-Bridge (Three-Phase) Inverters:** Consist of three half-bridge legs, providing better control over the motor's voltage and current waveforms, essential for three-phase PMSMs and BLDC motors.

These inverters employ advanced switching techniques such as Pulse Width Modulation (PWM) to regulate motor speed and torque efficiently. PWM allows for:

- Smooth control of voltage and current waveforms.
- Efficient operation across a wide speed range.
- Reduction of switching losses through optimal timing and dead-time insertion.

The increasing complexity of motor control strategies requires inverter hardware to be highly robust, capable of handling rapid switching while maintaining noise immunity and thermal stability.

1.1.3 Applications Driving Innovation

The practical applications that drive the need for sophisticated motor control systems include:

- Electric Vehicles (EVs) and Hybrid Vehicles (HEVs): Require high-efficiency, high-torque motors such as PMSMs for traction drives.
- Industrial Automation: Uses PMSMs and BLDCs in robotics, conveyor systems, and machine tools where precise positioning and speed control are essential.
- Renewable Energy Systems: Wind turbines and solar-powered systems often use electric drives for energy conversion and storage.
- Consumer Electronics: BLDC motors are increasingly used in household appliances, drones, and e-bikes due to their simplicity and low cost.

Each of these applications demands specific performance characteristics that can only be achieved through careful selection of control methods, inverter topology, and power electronics design.

1.2 Motivation

In light of these trends and technical challenges, this project was conceived with the clear motivation to:

1. Deepen theoretical understanding of modern electric motor control strategies..
2. Design and implement a working prototype that demonstrates full control over both PMSM and BLDC motors using advanced inverter hardware.
3. Bridge the gap between software simulation and real-world hardware validation, ensuring that the designed control strategies are not only theoretically sound but also practically feasible.

We decision to study this field arises from the realization that future engineers must be equally skilled in control theory, embedded programming, and hardware design. Merely understanding the control algorithms is not enough; one must also understand the practical limitations and challenges of real-world implementation, including switching noise, EMI, thermal management, and gate drive isolation. This led me to pursue a comprehensive development process, including:

- Control Software Design using MATLAB/Simulink for algorithm validation.
- Hardware Design of custom gate driver circuits with isolated power supplies.
- Experimental Testing to verify control strategies under actual load conditions.

The motivation extends beyond academic curiosity to a desire to create practical solutions applicable in real industrial environments, with an eye towards future employment or research in electric mobility, robotics, or power electronics.

1.3 Project Structure

To achieve these objectives, this project has been divided into four distinct but interrelated chapters:

1.3.1 Inverter Hardware Design

This chapter covers the design and implementation of the power electronics hardware necessary to drive both PMSM and BLDC motors:

- **Gate Drivers:** Responsible for translating microcontroller-level control signals to high-power switching signals. The design includes isolated drivers for both high-side and low-side MOSFETs/IGBTs.
- **Flyback Converter:**A dedicated isolated DC-DC converter provides stable +20V/-5V gate drive voltages, essential for safe and reliable switching.
- **Inverter Topology:**Design of a three-phase inverter capable of delivering the required voltage and current for the selected motors, with proper dead-time management and current sensing for feedback control.

The chapter also discusses key design challenges such as electromagnetic interference (EMI), thermal management, and switching losses.

1.3.2 Chapter: PMSM Control Using Field-Oriented Control (FOC)

In this chapter, the mathematical modeling and control of PMSMs are presented, including:

1.3.3 Chapter: BLDC Motor Control Using Trapezoidal Commutation

This chapter focuses on the control of BLDC motors using a six-step trapezoidal commutation scheme:

- Hall sensor feedback is used to determine rotor position and generate the correct commutation sequence.
- PWM modulation is applied to control the average voltage supplied to the motor phases.
- The simplicity of this method makes it suitable for cost-sensitive applications where extremely precise control is not required.

The hardware implementation of the BLDC control demonstrates the feasibility of using a custom inverter and microcontroller-based control in low-cost motor drive systems.

1.3.4 Chapter: Practical Implementation and Testing of the BLDC Motor Drive

The final chapter presents:

- The experimental setup used to validate the control algorithms on real hardware.
- Comparison of simulation and practical results for both PMSM and BLDC control strategies.
- Performance metrics: including speed response, current ripple, and efficiency.

- Discussion of challenges faced during testing, such as switching noise, current sensing errors, and thermal issues.

The practical phase of this project not only validated the theoretical control strategies but also enhanced the understanding of embedded systems design, power electronics hardware, and real-time control implementation.

1.4 Conclusion

The demand for efficient, flexible, and high-performance electric motor drives is rapidly increasing across multiple industries. By focusing on the development of both PMSM and BLDC control systems and implementing a custom inverter prototype, this project addresses both the theoretical understanding and the practical realization of modern electric drives. The knowledge gained from this work lays the foundation for future exploration into more complex topics such as sensorless control, model predictive control (MPC), and fault-tolerant motor drives, as well as the potential to contribute to ongoing technological advancements in electric vehicles, automation, and renewable energy systems.

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Chapter 1

Invertr

1. GATE DRIVES

1.1 Abstract

The main purpose of this report is to illustrate the behavior of the MOSFETs switching, including the effect of parasitic components, transients, and extreme operating conditions starting with an overview of MOSFET technology and switching operation.

1.2 Introduction

MOSFET (Metal Oxide Semiconductor Field Effect Transistor) is crucial in the electronics industry's high-frequency, high-efficiency switching applications.
FET technology was invented in 1930, about 20 years before the bipolar transistor.
The first power MOSFET has been available since the mid-1970s.
Today, millions of MOSFETs are integrated into modern electronic devices, ranging from microprocessors to discrete power transistors.

1.3 MOSFET Technology

Fundamentally, transistors are charge-controlled devices, which means that their output current is proportional to the charge established in the semiconductor by the control electrode. Theoretically, the switching speeds of MOSFET are close to identical, determined by the time required for the charge carriers to travel across the semiconductor region (approximately 20 to 200 picoseconds).
The MOSFET transistors are simpler to drive because a continuous ON current is not required. Once the MOSFET transistors are turned on, their drive current is practically zero. Furthermore, MOSFETs have a resistive nature. The voltage drop across the drain-source terminals of a MOSFET is a linear function of the current flowing in the semiconductor. This linear relationship is characterized by the $R_{DS(on)}$ of the MOSFET and is known as the on-resistance.

1.4 MOSFET Construction

The metallic gate terminal in the MOSFET is insulated from the semiconductor layer by a SiO_2 layer or dielectric layer. The MOSFET consists of three terminals, they are the source(S), Gate

(G), Drain (D), and the body which is called a substrate. The substrate is connected to the source internally

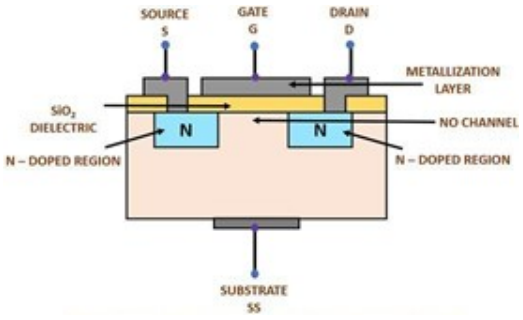


Figure 1.1: The constructional detail of enhancement type MOSFET

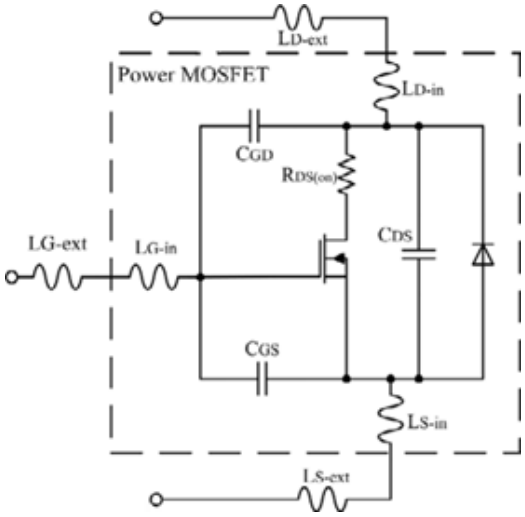


Figure 1.2: MOSFET switching model with key parasitic components

1.5 MOSFET Critical Parameters

When the MOSFET is considered to operate in switch mode the practical switching times of the MOSFETs (approximately 10 ns to 60 ns) are at least two to three orders of magnitude longer than the theoretical switching time (approximately 50 ps to 200 ps).
Referring to the MOSFET model, note that it includes three capacitors connected between the three terminals of the device. Ultimately, the switching performance of the MOSFET is determined by how quickly the voltages can be changed across these capacitors. Therefore, in high-speed switching applications, the most important parameters are the parasitic capacitances of the device Cgs, Cgd, and Cds. None of them is defined directly in the transistor data sheets. Their values are given indirectly by the Ciss, Crss, and Coss capacitor values.

From datasheet parameters:

$$C_{gd} = C_{rss} \tag{1.1}$$
$$C_{gs} = C_{iss} - C_{gd} \tag{1.2}$$
$$C_{ds} = C_{oss} - C_{gd} \tag{1.3}$$

Since the Cgd and Cds capacitors are voltage-dependent, the data sheet numbers are valid only at the test conditions listed. The relevant average capacitance for a certain application must be calculated based on the required charge to establish the actual voltage change across the capacitors.

$$C_{gd,avg} = 2 \cdot C_{rss,spec} \cdot \sqrt{\frac{V_{ds,spec}}{V_{ds,off}}} \tag{1.4}$$

$$C_{oss,avg} = 2 \cdot C_{oss,spec} \cdot \sqrt{\frac{V_{ds,spec}}{V_{ds,off}}} \tag{1.5}$$

The next important parameter is the gate internal resistance R_G, I.This parasitic resistance describes the resistance associated with the gate signal distribution within the device.Its importance is very significant in high-speed switching applications because it is in between the driver and the input capacitor of the device, directly impeding the switching times of the MOSFET. Other important parameters like the source inductance (L_S), gate inductance (L_g), and drain inductance (L_D) exhibit significant restrictions in switching performance. Typical L_S and L_D values are listed in the data sheets, and they are mainly dependent on the package type of the transistor.

1.5.1 Gate Inductance

If the current passing through an inductor changes very quickly, the inductor will generate a large voltage in an attempt to keep the current the same. Even though these parasitic inductance on the MOSFET are typically nH, the rate of change of the current is so high that can still be significant voltage generated.
Another similar problem is that when combined with the gate-source capacitance, this inductance can form an L_C tank which causes oscillations in the gate voltage. These oscillations can make the voltage exceed the threshold after the MOSFET is turned off, which will turn the MOSFET back on or partially on when it doesn't want to. Overshoots that happen during turn-on can exceed the maximum rating for the gate-source voltage.
So, because of these problems, an external gate resistor is added even though increased switching losses it acts to dampen these oscillations and overshoots by absorbing some of that energy and also reducing the rate of change of current through the inductor.

1.5.2 Source Inductance

The effect of it can be even worse than that of the gate. Because the source connection is used for both current through the channel from the drain and for driving the gate.
That means that when the MOSFET is turned ON/OFF the current through this inductance is changed from the full load current to zero. So, a higher voltage than that in the gate inductance is going to be generated.
And because the source inductance is part of the gate drive loop the voltage generated by it will appear on the gate which can again switch the MOSFET state to an undesired state.
The best way to mitigate against this is to use a MOSFET with a Kelvin source connection.
The second impact of the source inductance is on the drain-source voltage. But. This is also affected by the drain inductance.

1.5.3 Drain inductance

Source and drain inductance results in ringing and overshoots between the drain and source. These overshoots are typically much bigger than those on the gate because the currents involved are much higher.
In many applications, this inductance is more dependent on circuit layout and PCB design than the actual MOSFET itself. These overshoots can be controlled (or mitigated) by the gate resistor. Because if the gate resistor is increased, we slow down how fast the MOSFET is switching which reduces the rate of change of current. Can be controlled by snubber circuits as well.

1.5.4 Drain-Source capacitance

This capacitance is charged up every time the switch is turned off. When the switch is turned back on, the energy goes into the MOSFET. This is a loss and there is nothing that can be done about it.
The faster the switching, the worse it will be because the more often this capacitance will be charged up and discharged into the switch. These losses always exist even at low or zero drain current. Cds vary with the voltage across it and it drops significantly at high drain-source voltage

1.5.5 Gate-Drain capacitance

Known as reverse transfer capacitance. Despite being the smallest, this is the deadliest in fast-switching applications.When the MOSFET is turned off, the drain voltage will increase rise (to V_{DD} value). So, the voltage across the C_{gd} is being changed which causes a current to flow through this capacitor from the drain to the gate.
That means the gate will be pulled up to a very high voltage (V_{DD}) which will destroy the MOSFET instantly. This effect is known as the Miller turn-on. Because of that name, C_{gd} is often referred to as the (Miller capacitance). The effect of Miller capacitance has become a much bigger deal recently with the advent of Silicon-Carbide MOSFETs which are both higher voltage and fast switching. C_{gd} is very dependent on voltage as C_{ds}.

1.6 MOSFET switching

1.6.1 Turn-on procedure

The turn-on event of the MOSFET can be divided into four intervals.

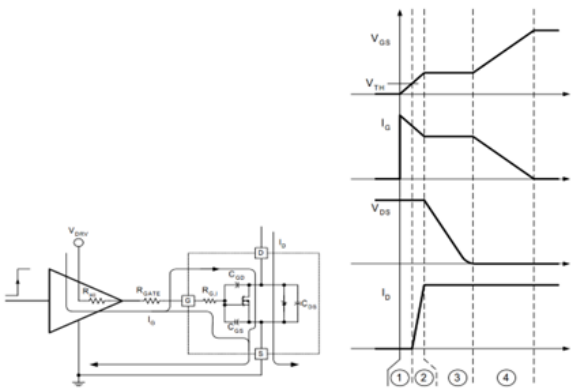


Figure 1.3: MOSFET Turn-on Time intervals

- **In the first interval:** The input capacitance of the device is charged from 0V to V_{TH}. During this interval, most of the gate current is charging the C_{gs} capacitor. A small current is flowing through the C_{gd} capacitor, too. This period is called the *turn-on delay* because both the drain current and the drain voltage of the device remain unchanged.
- **In the second interval:** The gate voltage rises from V_{TH} to the Miller plateau level, $V_{\mathrm{gs,Miller}}$. This is the linear operation region of the device when the current is proportional to the gate voltage. On the gate side, the current is flowing into the C_{gs} and C_{gd} capacitors just like in the first interval, and the V_{gs} voltage is increasing. On the output side of the device, the drain current is increasing, while the drain-to-source voltage stays at the previous level ($V_{\mathrm{ds,off}}$).
- **In the third interval:** The gate is already charged to a sufficient voltage ($V_{\mathrm{gs,Miller}}$) to carry the entire load current. That now allows the drain voltage to fall. While the drain voltage falls across the device, V_{gs} stays steady. This is the *Miller plateau region* in the gate voltage waveform. All the gate current available from the driver is diverted to discharge the C_{gd} capacitor to facilitate the rapid voltage change across the drain-to-source terminals.
- **In the fourth interval:** The final amplitude of V_{gs} determines the ultimate on-resistance of the device during its on-time. Therefore, in this fourth interval, V_{gs} is increased from $V_{\mathrm{gs,Miller}}$ to its final value, V_{DRV}. This is accomplished by charging the C_{gs} and C_{gd} capacitors, thus gate current is now split between the two components. While these capacitors are being charged, the drain current is still constant, and the drain-to-source voltage is slightly decreasing as the on-resistance of the device is reduced.

1.6.2 Turn-off procedure

The description of the turn-off procedure for the MOSFET transistor is backtracking the turn-on steps.

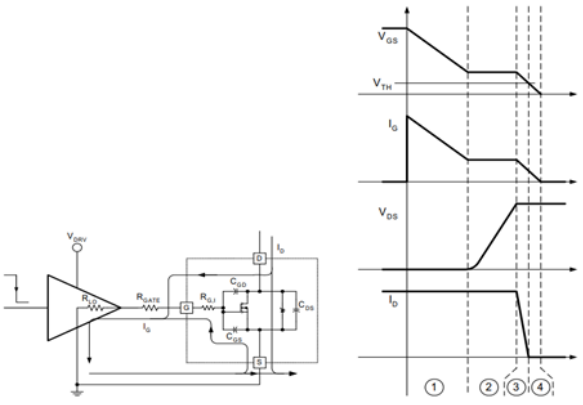


Figure 1.4: MOSFET Turn-off Time intervals

- **In the first interval:** This is the *turn-off delay* that is required to discharge the input capacitance (C_{ISS}) from its initial value to the Miller plateau level. During this time, the gate current is supplied by the C_{ISS} capacitor itself, and it flows through the C_{gs} and C_{gd} capacitors of the MOSFET. The drain voltage of the device is slightly increasing, while the current in the drain remains unchanged.
- **In the second interval:** The drain-to-source voltage of the MOSFET rises from $I_d \cdot R_{\mathrm{DS(on)}}$ to the final $V_{\mathrm{ds,off}}$ level. During this period, which corresponds to the *Miller*

plateau in the gate voltage waveform, the gate current is strictly the charging current of the C_{gd} capacitor because the gate-to-source voltage remains constant.

- **In the third interval:** The gate voltage resumes falling from $V_{\text{GS,Miller}}$ to V_{TH}. The majority of the gate current is coming out of the C_{gs} capacitor because the C_{gd} capacitor is virtually fully charged from the previous interval. The MOSFET is operating in its linear region, and the declining gate-to-source voltage causes the drain current to decrease, reaching near zero by the end of this interval. Meanwhile, the drain voltage remains steady at $V_{\text{ds,off}}$.
- **In the fourth interval:** V_{gs} is further reduced until it reaches 0V. A large portion of the gate current, similar to the third interval, is supplied by the C_{gs} capacitor. During this time, both the drain current and the drain voltage of the device remain unchanged.

The gate voltage at the Miller plateau can be expressed as:

$$V_{\text{gs,Miller}} = V_{\text{TH}} + \frac{I_d}{g_{fs}} \quad , \quad g_{fs} = \frac{dI_d}{dV_{gs}}$$

Summary: The MOSFET can be switched between its highest and lowest impedance states (turn-on or turn-off) in four time intervals. The lengths of these intervals depend on the parasitic capacitance values, the required voltage change across them, and the gate drive current.

1.7 Power loss

1.7.1 Gate drive losses

The switching action in a MOSFET results in some power losses, which can generally be divided into two categories.

The simpler of the two loss mechanisms is the **gate drive loss**. As described earlier, turning the MOSFET on or off involves charging and discharging the input capacitance, denoted as C_{ISS}.

When the voltage across a capacitor changes, a specific amount of charge must be transferred. The total gate charge Q_g required to switch the gate voltage from 0 V to V_{DRV} can be obtained from the *gate charge vs. gate-to-source voltage* curve.

This charge movement results in energy being dissipated in the gate driver circuitry during every switching event. The energy consumed per switching cycle is given by:

$$E_{\text{gate}} = Q_g \cdot V_{\text{DRV}}$$

Thus, the total gate drive power loss at a switching frequency f_{sw} is:

$$P_{\text{gate}} = Q_g \cdot V_{\text{DRV}} \cdot f_{\text{sw}}$$

This gate drive loss is usually small compared to switching and conduction losses but becomes significant at high switching frequencies.

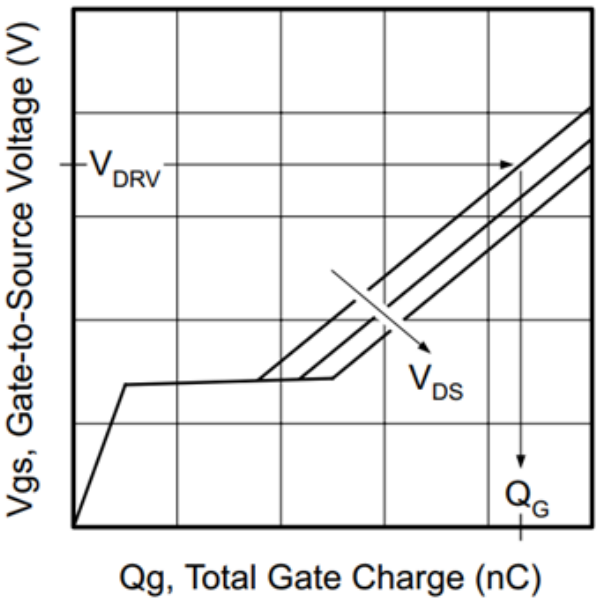


Figure 1.5: The variation of Vgs with Qg and highlighting the Miller plateau where Vds decreases

The power loss due to gate driving is:

$$P_{\text{gate}} = V_{\text{DRV}} \cdot Q_g \cdot f_{\text{sw}}$$

In every switching cycle, the required gate charge passes through the driver's output impedance, the external gate resistor R_G, and the internal gate resistance of the MOSFET.

Interestingly, the total gate drive power dissipation is independent of how quickly the charge is delivered through these resistive components. However, how this power is distributed among the components depends on their relative resistances.

The portion of gate drive loss dissipated in the driver itself is given by:

$$P_{\text{driver}} = P_{\text{gate}} \cdot \frac{R_{\text{Gdriver,out}}}{R_{\text{Gdriver,out}} + R_G}$$

In some gate drivers, the output resistance $R_{\text{Gdriver,out}}$ is well-defined, but in others, it is not constant. To simplify analysis, it is typically safe to assume the following:

- **Low Gate Resistance Case** ($R_G < 5ohm$):
  - Most gate drive power is dissipated in the driver IC
  - Occurs because the driver output impedance dominates the total resistance
  - Typical in high-speed switching applications where minimal R_G is desired
- **High Gate Resistance Case** ($R_G \geq 5ohm$):
  - Majority of losses occur in the external gate resistor R_G
  - Happens when R_G limits I_G below the driver's current capability
  - Common in designs prioritizing reduced EMI over switching speed

1.7.2 Switching losses

In addition to the gate drive power loss, the transistors accrue switching losses due to high current and high voltage being present in the device simultaneously for a short period. In order to ensure the least amount of switching losses, the duration of this time period must be minimized. This condition is limited to intervals 2 and 3 of the switching transitions in both turn-on and turn-off operations. These time intervals correspond to the linear operation of the device when the gate voltage is between V_{TH} and V_{gs} ,Miller, causing changes in the current of the device and to the Miller plateau region when the drain voltage goes through its switching transition

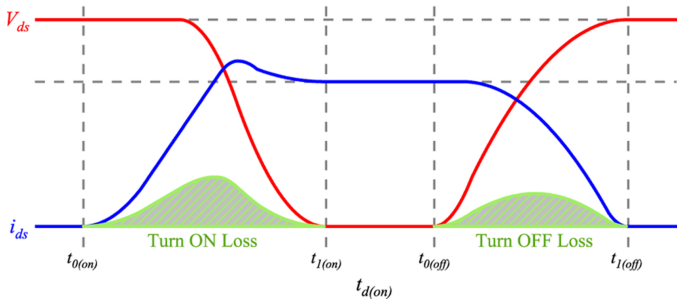


Figure 1.6: MOSFET switching losses during turn-on and turn-off events

A crude estimate of the MOSFET switching losses can be calculated using simplified linear approximations of the gate drive current, drain current, and drain voltage waveforms during intervals 2 and 3 of the switching transitions.

First, the gate drive currents must be determined for the second and third time intervals, respectively:

$$I_{G2} = \frac{V_{\text{DRV}} - 0.5(V_{\text{gs,Miller}} + V_{\text{TH}})}{R_{\text{Gdriver,out}} + R_g + R_{\text{G,int}}} \tag{1.6}$$

$$I_{G3} = \frac{V_{\text{DRV}} - V_{\text{gs,Miller}}}{R_{\text{Gdriver,out}} + R_g + R_{\text{G,int}}} \tag{1.7}$$

Assuming that I_{G2} charges the input capacitance of the device from V_{TH} to $V_{\text{gs,Miller}}$, and I_{G3} is the discharge current of the C_{rss} capacitor while the drain voltage changes from $V_{DS,\text{off}}$ to 0 V:

$$t_2 = C_{iss} \cdot \frac{V_{\text{gs,Miller}} - V_{TH}}{I_{G2}} \tag{1.8}$$

$$t_3 = C_{rss} \cdot \frac{V_{DS,\text{off}}}{I_{G3}} \tag{1.9}$$

During t_2, the drain voltage is $V_{DS,\text{off}}$ and the current ramps from 0 A to the load current I_L, while in the t_3 time interval, the drain voltage falls from $V_{DS,\text{off}}$ to near 0 V.

The power losses during these intervals are:

$$P_2 = \frac{1}{2} \cdot t_2 \cdot V_{DS,\text{off}} \cdot I_L \cdot f_{sw} \tag{1.10}$$

$$P_3 = \frac{1}{2} \cdot t_3 \cdot V_{DS,\text{off}} \cdot I_L \cdot f_{sw} \tag{1.11}$$

The total switching loss during turn-on is the sum of the two components:

$$P_{sw} = P_2 + P_3 = \frac{1}{2} \cdot V_{DS,\text{off}} \cdot I_L \cdot (t_2 + t_3) \cdot f_{sw} = \frac{1}{2} \cdot V_{DS,\text{off}} \cdot I_L \cdot t_{\text{on}} \cdot f_{sw} \tag{1.12}$$

Considering both turn-on and turn-off losses, the total switching power loss is:

$$P_{sw} = \frac{1}{2} \cdot V_{DS,\text{off}} \cdot I_L \cdot (t_{\text{on}} + t_{\text{off}}) \cdot f_{sw} \tag{1.13}$$

Calculating the exact switching losses is nearly impossible due to the significant influence of parasitic inductive components, which alter the current and voltage waveforms as well as switching times during the transition. Considering the impact of different source and drain inductances in a real circuit would require solving second-order differential equations to describe the actual waveforms.

Moreover, many of the variables involved—such as V_{TH}, MOSFET capacitance values, $R_{\text{Gdriver,out}}$, and so on—have wide tolerances. Therefore, these linear approximations provide a reasonable compromise for estimating the switching losses in MOSFETs.

1.7.3 Conduction losses

Conduction loss is a term given to the power loss or power dissipation that is due to the drain current and the on-state resistance. It is the power loss when the MOSFET is in on-state. When the VGS threshold requirement is achieved, the MOSFET will conduct, and the conduction loss will take effect.

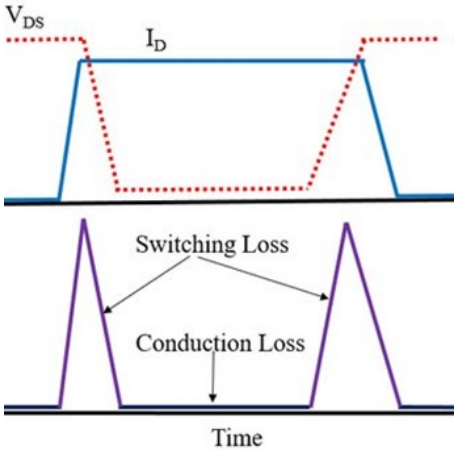


Figure 1.7: Breakdown of MOSFET power losses into switching and conduction losses over time.

$$P_{\text{cond}} = I_d^2 \times R_{\text{ds,on}}$$

1.8 Gate resistor

R_{gate} can be calculated for optimum performance

$$R_{\text{gate,opt}} = 2\sqrt{\frac{L_s}{C_{\text{iss}}}} - (R_{\text{Gdriver,out}} + R_{g,\text{int}})$$

An optimum value of the gate resistor can be obtained from the datasheet, the gate resistor used in the tests can be considered as a good starting point to try and start iterations till find

the most suitable gate resistor that can get the desired switching speed and acceptable EMI noise. Smaller resistor values will result in an overshoot in the gate drive voltage waveform but also result in a faster turn-on speed. Higher resistor values will underdamp the oscillation and extend the switching times.

1.9 Ground-Referenced Gate Drive

1.9.1 PWM Direct Drive

The simplest way of driving the gate of a MOSFET. The limited drive current capability of the PWM controllers is limiting the switching speeds. Very few integrated circuits offer more than 1-A peak gate drive capability

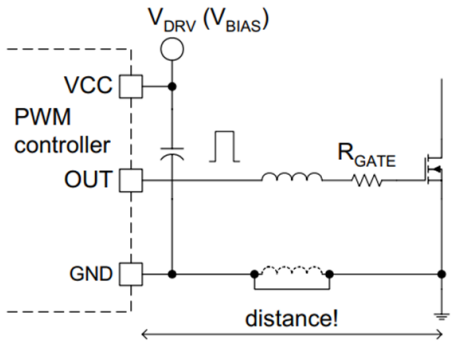


Figure 1.8: Direct Gate-Drive Circuit

1.9.2 Bipolar Totem-Pole

One of the most popular and cost-effective drive circuits for driving MOSFETs. This circuit handles the current spikes and power losses making the operating conditions for the PWM controller more favorable.

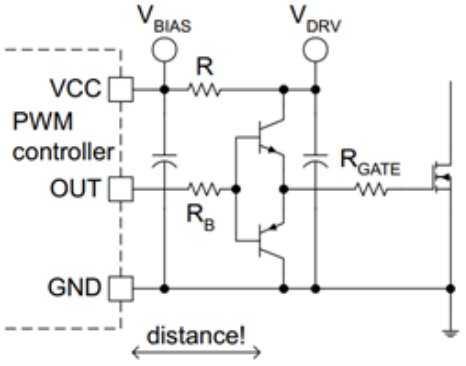


Figure 1.9: Bipolar Totem-Pole MOSFET Driver

1.10 Speed-Enhancement Circuits

When *speed enhancement circuits* are mentioned, the focus is usually on speeding up the **turn-off** process of the MOSFET. This is because the turn-on speed is often limited by the turn-off, or *reverse recovery speed*, of another switching element in the circuit.

The situation at turn-off is quite different. In theory, the turn-off speed of the MOSFET depends solely on the gate drive circuit. A higher current turn-off circuit can discharge the input capacitances faster, resulting in shorter switching times and consequently lower switching losses. This higher discharge current can be achieved by:

- Using a MOSFET driver with lower output impedance,
- Applying a negative turn-off voltage (commonly used for N-channel devices).

While faster switching can potentially reduce switching losses, turn-off speed-up circuits also increase the *ringing* observed in the voltage and current waveforms. This is due to the higher rates of change, di/dt and dv/dt, of the MOSFET during turn-off. Therefore, it is a trade-off between switching speed and electromagnetic noise or voltage stress.

The following examples illustrate turn-off circuits implemented on simple ground-referenced gate drive circuits.

1.10.1 Turn-Off diode

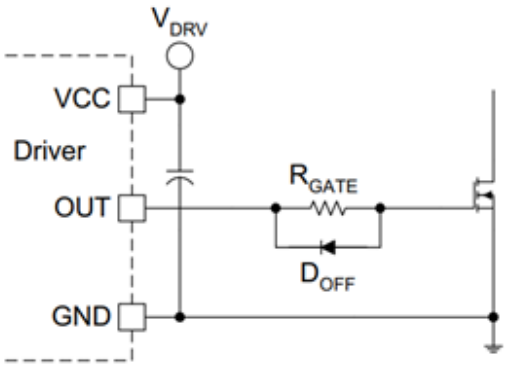


Figure 1.10: Simple Turn-Off Speed Enhancement Circuit

1.10.2 PNP Turn-off Circuit

With the help of the transistor Q_{OFF}, the gate and source terminals of the MOSFET are locally shorted during turn-off. This arrangement prevents the turn-off current from flowing back into the driver, effectively reducing the power dissipation of the driver by approximately a factor of two.

The turn-off transistor Q_{OFF} also shunts out the gate drive loop inductance, the potential current sense resistor, and the output impedance of the driver. Additionally, Q_{OFF} never saturates, which is crucial for enabling it to switch on and off rapidly.

The only known limitation of this circuit is that it cannot pull the gate all the way down to 0 V because of the voltage drop across the base-emitter junction of Q_{OFF}.

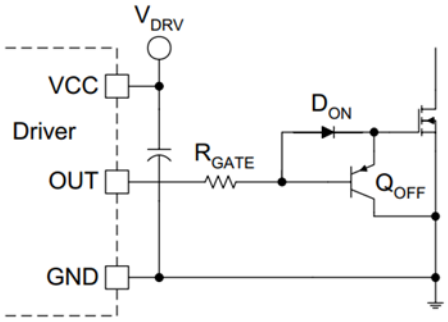


Figure 1.11: Local pnp Turn-Off Circuit

1.10.3 NPN Turn-off Circuit

The NPN transistor holds the gate closer to GND than its PNP counterpart.

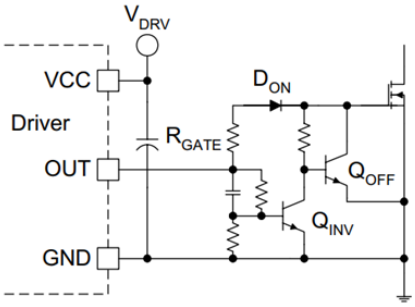


Figure 1.12: Local NPN Self-Biasing Turn-Off Circuit

1.10.4 NMOS turn-off circuit

This circuit offers very fast switching and complete discharge of the MOSFET gate to 0 V. The gate resistor R_{GATE} sets the turn-on speed as before, but it also serves an important role in preventing shoot-through currents between the two outputs of the driver in case of imperfect timing of the drive signals.
Another important consideration is that the output capacitance C_{OSS} of the transistor Q_{OFF} is connected in parallel with the input capacitance C_{ISS} of the main power MOSFET. This increases the effective *total gate charge* that the driver must supply.

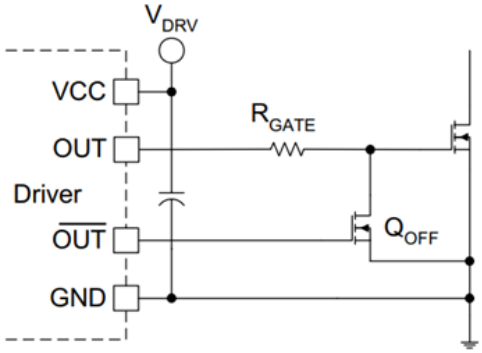


Figure 1.13: Improved N-Channel MOSFET-Based Turn-off Circuit

1.11 High side Gate Drives

In many switching power supply topologies, the power MOSFET source terminals are not connected to the circuit ground (GND). This precludes the use of conventional ground-referenced gate driver circuits.
Instead, a gate drive circuit is required where the input control signal is referenced to 0V (GND), but the output gate drive signal is referenced to the MOSFET source terminal, which can be at an arbitrary potential.
Several methods exist for implementing such isolated or floating gate drive circuits.

1.11.1 Bootstrap Gate-Drive Technique

A bootstrap circuit is used in half-bridge configurations to supply bias to the high-side FET

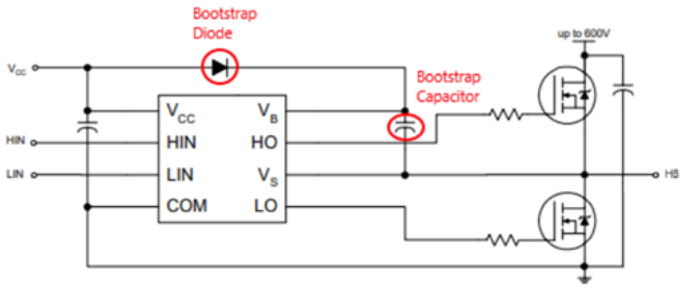


Figure 1.14: Diagram of a half-bridge driver circuit using bootstrap technique to drive the high-side MOSFET.

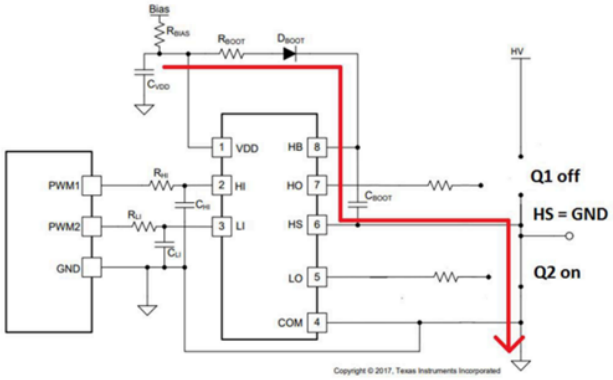


Figure 1.15: Bootstrap Capacitor Path

When the low-side FET is on (high-side FET is off), the VDD bias supply, through the bypass capacitor, charges the bootstrap capacitor through the bootstrap diode and resistor

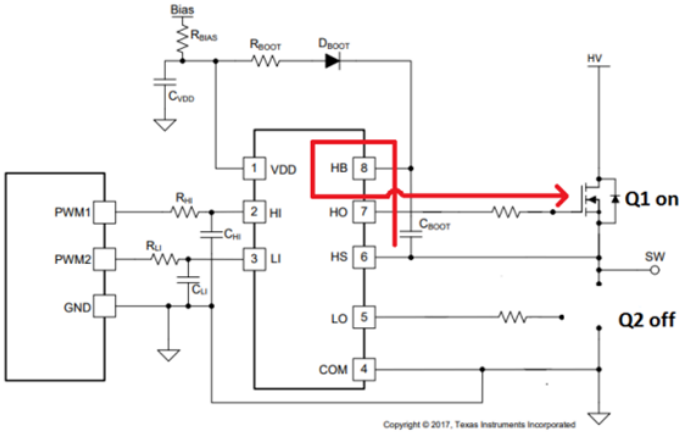


Figure 1.16: Bootstrap Capacitor Discharging Path

1.12 Transformer-Coupled Gate Drives

Transformers have at least two windings: a primary and a secondary. The use of separate windings facilitates galvanic isolation between the input and output. According to Faraday's law, the average voltage across any transformer winding must be zero over a complete period of operation. Even a small DC component can cause flux *walk* and eventual core saturation. Transformer-based gate drive circuits are commonly used to drive both high-side and low-side MOSFETs.

1.12.1 Single-Ended Transformer-Coupled Gate-Drive Circuits

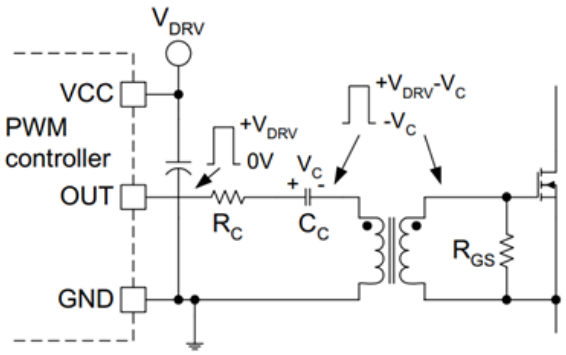


Figure 1.17: Single-Ended Transformer-Coupled Gate Drive

The coupling capacitor C_C must be placed in series with the primary winding of the gate drive transformer to provide the reset voltage for the magnetizing inductance. Without this capacitor, a duty ratio-dependent DC voltage would develop across the winding, causing transformer saturation.

The steady-state voltage across the coupling capacitor is given by:

$$V_C = D \times V_{\text{DRV}}$$

where D is the duty ratio and V_{DRV} is the driver voltage. The actual gate drive voltage V_C varies with the duty ratio. Moreover, sudden changes in duty ratio can excite the LC resonant

tank formed by the magnetizing inductance L_M of the transformer and the coupling capacitor C_C.

To damp this resonance, a low-value resistor R_C is usually inserted in series with C_C, satisfying:

$$R_C \geq 2\sqrt{\frac{L_M}{C_C}}$$

At wide duty cycles, such as in a buck converter, this circuit does not provide an adequate gate drive voltage. The coupling capacitor voltage V_C increases proportionally with the duty ratio, increasing the negative bias during off-time and reducing the effective turn-on voltage. Adding two small components on the secondary side of the gate drive transformer can prevent this issue.

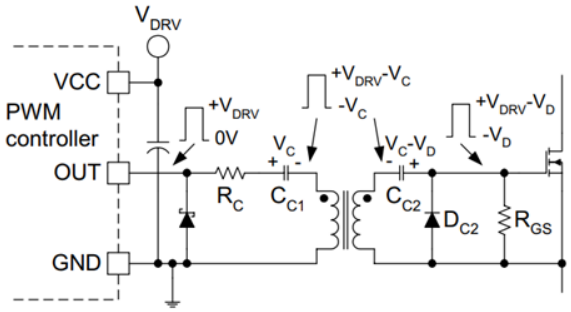


Figure 1.18: DC Restore Circuit in Transformer-Coupled Gate Drive

Here, a second coupling capacitor C_{C2} and a simple diode clamp D_{C2} are used to restore the original gate drive amplitude on the secondary side of the transformer. If a larger negative bias is desired during the off-time of the main switch, a Zener diode can be added in series with the diode clamp to set the clamp voltage accordingly.

1.12.2 Double-Ended Transformer-Coupled Gate Drives

In high-power half-bridge and full-bridge converters, it is often necessary to drive two or more MOSFETs, typically controlled by a push-pull (also called double-ended) PWM controller. In these applications, a dual-polarity symmetrical gate drive voltage is readily available. During the first clock cycle, OUTA is on, applying a positive voltage across the primary winding of the gate drive transformer. In the next clock cycle, OUTB is on for the same duration (steady-state operation), providing an opposite polarity voltage across the magnetizing inductance. Averaging the voltage across the primary winding over any two consecutive switching periods results in zero volts. Therefore, AC coupling is not required in push-pull type gate drive circuits.

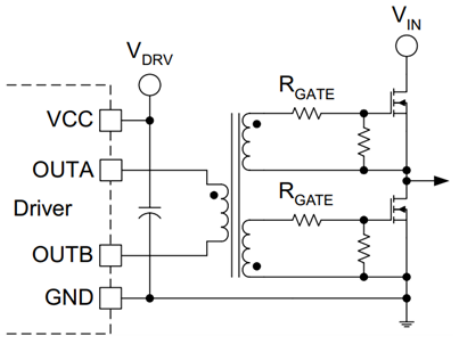


Figure 1.19: Push-Pull Type Half-Bridge Gate Drive

2. Inverter

1.13 Abstract

Inverters are indispensable in modern power electronics, converting direct current (DC) into alternating current (AC) for applications including solar energy systems, electric vehicles (EVs), and industrial motor drives. This document explores inverter architectures, working principles, and control strategies with an emphasis on enhancing efficiency, waveform quality, and grid compatibility. Detailed analysis includes Pulse Width Modulation (PWM), advanced multilevel topologies, loss mechanisms (conduction and switching), snubber circuit design, and thermal management. The future trajectory of inverter technology is highlighted through the integration of wide-bandgap semiconductors (SiC/GaN) and smart grid applications, pointing toward more efficient, compact, and intelligent systems. The practical implementation using CI60N120SM4 SiC MOSFETs and UCC215204A gate drivers is examined, demonstrating significant improvements in switching frequency, thermal performance, and overall system efficiency compared to traditional silicon-based solutions.

1.14 Introduction

The transition to sustainable energy systems and electrified transport has significantly increased the demand for efficient, reliable, and intelligent inverters. These devices form the backbone of systems converting DC energy from solar panels, batteries, or fuel cells to usable AC power. The progression from mechanical to solid-state inverters, leveraging IGBTs and MOSFETs, marks a significant leap in performance and miniaturization. This document investigates inverter functionality, control methods, loss mechanisms, thermal considerations, and advanced control of three-phase systems.

1.15 Working Principles

1.15.1 Basic Operation

Inverters operate by switching DC input through semiconductor devices to produce an AC output. The output waveform is shaped via techniques such as:

- **Square Wave** – Simplest and least efficient, with high harmonic distortion.
- **Modified Sine Wave** – Improved compatibility with general-purpose loads.
- **Pure Sine Wave** – Produced via PWM, offering low Total Harmonic Distortion (THD).

1.15.2 Pulse Width Modulation (PWM)

PWM compares a sinusoidal reference $v_{ref}(t) = V_m \sin(\omega t)$ with a high-frequency triangular carrier signal $v_{tri}(t)$ to generate switching signals.

1.15.3 Modulation Index

$$M = \frac{V_m}{V_{tri}} \tag{1.14}$$

Where:

- $M < 1$: Linear modulation (desired).
- $M > 1$: Overmodulation (distorted waveform).

1.16 Inverter Topologies

Three-Phase Bridge

A **Voltage Source Inverter (VSI)** converts an input DC voltage into an AC voltage with the desired amplitude and frequency. The capability to vary both the output frequency and amplitude while maintaining low harmonic distortion is one of the key characteristics of a high-performance inverter. The variable output voltage can be achieved either by adjusting the input DC voltage or by controlling the inverter gain, which is defined as the ratio of the output AC voltage amplitude to the input DC voltage. Typically, the inverter gain is modulated using various **Pulse Width Modulation (PWM)** techniques. Among these, **Space Vector Pulse Width Modulation (SVPWM)** has proven to be particularly effective in minimizing the harmonic distortion of the output voltage. Due to the inherent electrical limitations of power electronic devices, the use of **multilevel inverter topologies** has become essential for high-power applications. Multilevel converters serve as an interface between the power grid and the energy storage system (such as a capacitor bank), managing the bidirectional flow of electrical energy between them. In addition to enabling higher power handling capabilities, multilevel converters offer several advantages:

- Reduction of voltage stress on individual switching devices,
- Elimination or reduction of bulky passive filtering equipment,
- Lower switching frequency requirements,
- Reduced **Total Harmonic Distortion (THD)**,
- Decreased switching losses,
- Reduced output current ripple.

However, these benefits come with certain disadvantages. Multilevel topologies typically involve increased control system complexity, a higher number of power electronic devices, and challenges related to voltage balancing across capacitors during charge and discharge cycles. Despite these drawbacks, advancements in digital signal processing and the continuous reduction in the cost of power semiconductor devices have made multilevel converters increasingly attractive. The implementation of advanced modulation techniques such as SVPWM further enhances their viability and performance. In this work, we utilize the **Neutral Point Clamped (NPC)** multilevel inverter, which is one of the most widely adopted multilevel structures due to its favorable performance and reliability.

Space Vector Pulse Width Modulation

Figure 1a shows the schematic of a three-phase two-level inverter. The inverter feeds a three-phase R-L load through an LC filter. The filter is used to eliminate high-order harmonics (HOHs). It is important to note that some components of the harmonic spectrum are inherently mitigated by applying SVPWM.

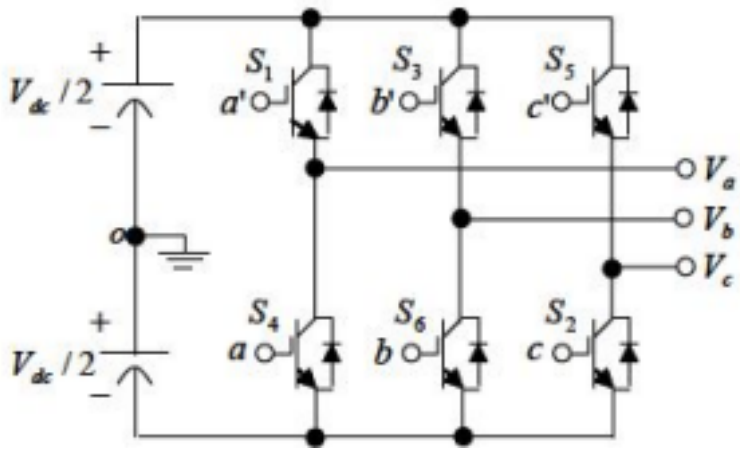


Figure 1.20: Figure 1a: Three-phase two-level inverter

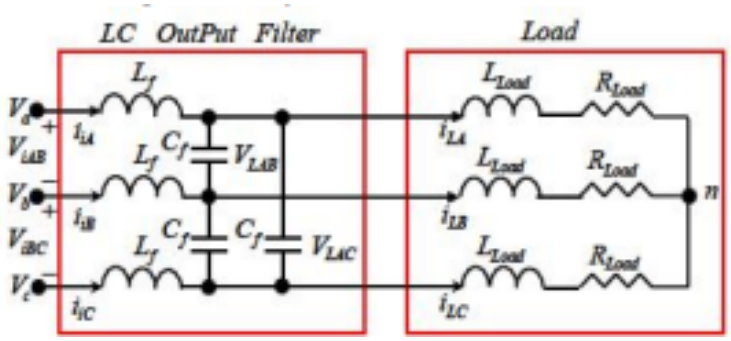


Figure 1.21: Figure 1b: LC filter and three-phase R-L load

$$V_{in} = (2 \times sw_i - 1) \times \frac{V_{dc}}{2}, \quad i = a, b, c \tag{1.15}$$

where sw_i represents the switching state of phase i (1 for upper switch ON, 0 for lower switch ON). Both switches in a single inverter leg must not be closed simultaneously to prevent a DC bus short circuit.

Assuming balanced and symmetrical output voltages:

$$V_{in} = V_{io} - V_{no} \tag{1.16}$$

$$V_{an} + V_{bn} + V_{cn} = 0 \tag{1.17}$$

where V_{in} is the inverter phase voltage and V_{io}, V_{no} are the inverter output and neutral point voltages respectively.

Substituting equations yields:

$$V_{no} = \frac{V_{an} + V_{bn} + V_{cn}}{3} \tag{1.18}$$

and:

$$\begin{bmatrix} V_{an} \\ V_{bn} \\ V_{cn} \end{bmatrix} = \frac{V_{dc}}{2} \times \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{bmatrix} \begin{bmatrix} 2sw_a - 1 \\ 2sw_b - 1 \\ 2sw_c - 1 \end{bmatrix} \tag{1.19}$$

Applying Park's transformation (with $\omega t = 0$):

$$V_{ref} = V_d + jV_q = \frac{2}{3}(V_{an} + aV_{bn} + a^2V_{cn}) \tag{1.20}$$

where:

$$a = e^{j2\pi/3} \tag{1.21}$$

The inverter's output voltage vector can then be written as:

$$V_{ref} = \frac{2}{3} V_{dc}(sw_a + a \times sw_b + a^2 \times sw_c) \tag{1.22}$$

A two-level inverter has eight possible switching combinations, shown in Figure .

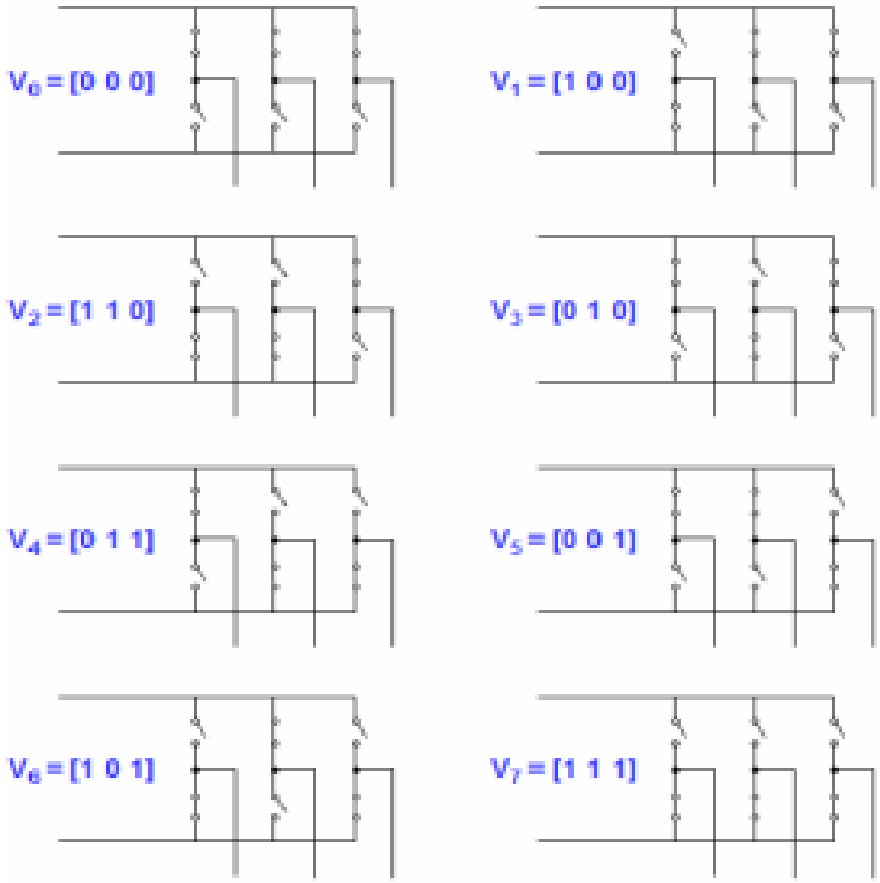


Figure 1.22: Different switching states of a three-phase two-level inverter

Table 1.1: Eight Switching States of a Two-Level Inverter

Voltage Vector	a	b	c	V_{an}	V_{bn}	V_{cn}	V_{ab}	V_{bc}
V_0	0	0	0	0	0	0	0	0
V_1	1	0	0	2/3	-1/3	-1/3	1	0
V_2	1	1	0	1/3	1/3	-2/3	0	1
V_3	0	1	0	-1/3	2/3	-1/3	-1	1
V_4	0	1	1	-2/3	1/3	1/3	-1	0
V_5	0	0	1	-1/3	-1/3	2/3	0	-1
V_6	1	0	1	1/3	-2/3	1/3	1	-1
V_7	1	1	1	0	0	0	0	0

The active voltage vectors divide the space vector diagram into six equal sectors, as shown in Figure 1.23.

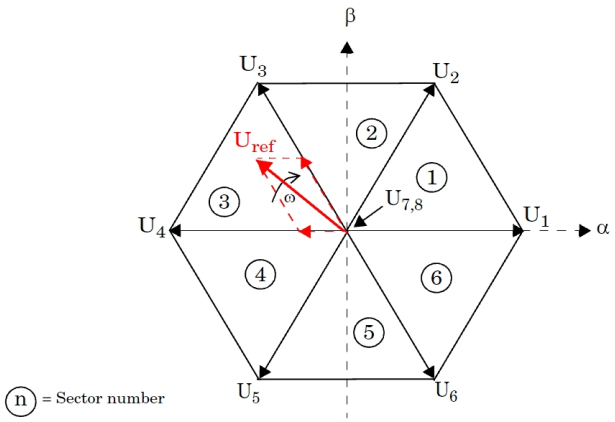


Figure 1.23: Space vector diagram of two-level inverter

To generate a desired output voltage vector within a sector, three vectors (two active and one zero vector) are used. For example, in the first sector:

$$T_s V_{ref} = T_1 V_1 + T_2 V_2 \tag{1.23}$$

The switching times are calculated as:

$$T_1 = \frac{\sqrt{3} T_s |V_{ref}| \sin\left(\frac{\pi}{3} - \alpha\right)}{V_{dc}} \tag{1.24}$$

$$T_2 = \frac{\sqrt{3} T_s |V_{ref}| \sin\left(\alpha\right)}{V_{dc}} \tag{1.25}$$

$$T_0 = T_s - T_1 - T_2 \tag{1.26}$$

where:

- MI is the modulation index,
- T_s is the switching period,
- f is the output frequency,
- n is the sector number,
- α is the angle of the reference vector.

Since all triangles in the space vector hexagon are equal, these equations can be generalized for all sectors by adjusting α accordingly.

Investigated Switching States

Depending on the distribution of T_0, T_1, and T_2, five different SVPWM switching strategies are considered in this research. These strategies have been summarized in Tables 2 to 6. To minimize switching losses, the distribution of T_0 is performed such that no more than one transition occurs in each switching period.

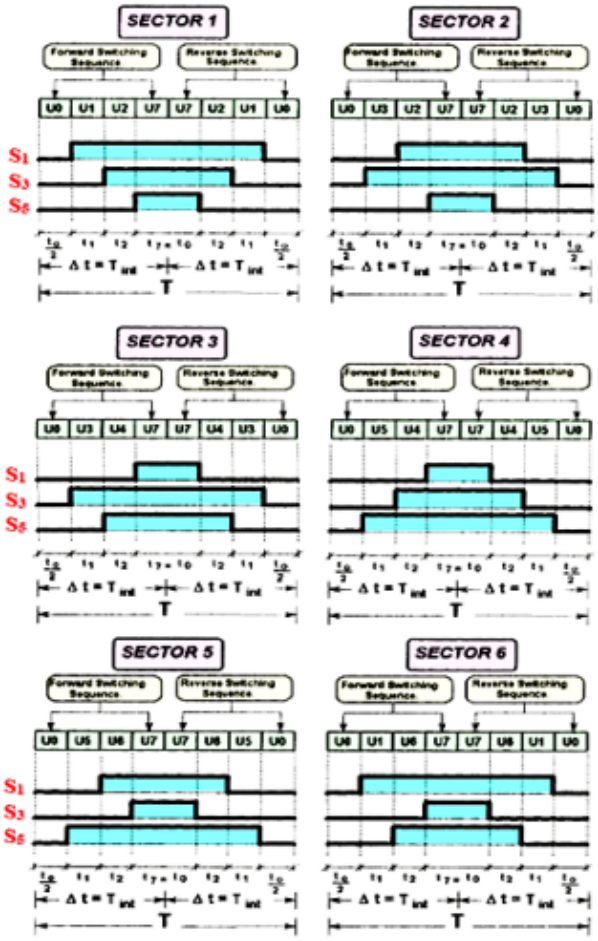


Figure 1.24: SVPWM switching states sector

1.17 Snubber Circuits

Snubber circuits protect semiconductor switches by suppressing voltage spikes and limiting the rate of change of voltage/current (dv/dt, di/dt).

1.17.1 Types

- **RC Snubber**: Dissipates energy across a resistor.
- **RCD Snubber**: Absorbs energy in a capacitor and discharges it through a diode and resistor.
- **Active Snubber**: Uses an auxiliary circuit for energy recovery (higher efficiency).

1.17.2 Equations

For an RC snubber:

$$E = \frac{1}{2} C V_{peak}^2 \tag{1.27}$$

$$R = \frac{1}{\omega C} \tag{1.28}$$

Where:

- C is the snubber capacitor
- V_{peak} is peak voltage across the switch
- ω is angular switching frequency

1.18 Losses in Inverters

1.18.1 Conduction Loss (P_{cond})

Occurs when the device is conducting current.

$$P_{cond} = I_{rms}^2 \cdot R_{on} \tag{1.29}$$

Where:

- I_{rms}: RMS load current
- R_{on}: On-state resistance (MOSFET) or Vce/R (IGBT)

1.18.2 Switching Loss (P_{sw})

Occurs during the transition between ON and OFF states.

$$P_{sw} = \frac{1}{2} V_{ds} \cdot I_d \cdot (t_{on} + t_{off}) \cdot f_{sw} \tag{1.30}$$

Where:

- V_{ds}: Drain-source voltage
- I_d: Drain current
- $t_{on/off}$: Switching times
- f_{sw}: Switching frequency

Total Power Loss:

$$P_{total} = P_{cond} + P_{sw} \tag{1.31}$$

1.19 Thermal Management

1.19.1 Heat Generation

Switches dissipate power, leading to junction temperature rise:

$$T_j = T_a + (P_{loss} \cdot R_{\theta ja}) \tag{1.32}$$

Where:

- T_j: Junction temperature
- T_a: Ambient temperature
- $R_{\theta ja}$: Thermal resistance from junction to ambient

1.19.2 Cooling Techniques

- **Passive Cooling**: Natural convection using heat sinks.
- **Forced Air Cooling**: Fans increase heat dissipation rate.
- **Liquid Cooling**: Circulated fluid removes heat—used in high-power systems.
- **Thermal Interface Materials (TIMs)**: Improve thermal conductivity between components.

1.20 Control of Three-Phase Inverters

Three-phase inverters use advanced techniques to manage switching for precise waveform generation, efficiency, and torque control.

1.20.1 Sinusoidal PWM (SPWM)

Generates three sinusoidal signals, each 120° apart, compared with a carrier signal.

1.20.2 Space Vector PWM (SVPWM)

Maximizes DC bus voltage utilization by using a space vector representation of three-phase signals.

- **Voltage Vectors**: V_1 to V_6, plus zero vectors V_0, V_7
- **Modulation Index**:

$$M = \frac{V_{ref}}{V_{dc}/\sqrt{3}} \tag{1.33}$$

Duty Cycles Calculation:

$$T_1 = T_s \cdot \frac{V_{ref} \cdot \sin(\pi/3 - \alpha)}{V_{dc}} \tag{1.34}$$

$$T_2 = T_s \cdot \frac{V_{ref} \cdot \sin(\alpha)}{V_{dc}} \tag{1.35}$$

Where:

- α: Angle between V_{ref} and active vector
- T_s: Sampling period

Power Devices and Gate Driver in Inverter Design

1.21 SiC Power MOSFET: CI60N120SM4

The **CI60N120SM4** is an N-channel Silicon Carbide (SiC) MOSFET rated for **1200 V** and **60 A**, designed for high-efficiency, high-speed switching applications. Leveraging the superior material properties of SiC technology, this device offers significantly lower switching losses, faster switching speeds, and enhanced thermal performance compared to traditional silicon-based MOSFETs.

Performance Characteristics

The key performance features of the **CI60N120SM4** include:

- **Lower conduction losses** due to the superior properties of SiC material.
- **Minimal temperature coefficient** of on-resistance ($R_{DS(on)}$), providing more stable performance over a wide temperature range.
- **Excellent reverse recovery behavior** when used as a synchronous rectifier, reducing switching losses in hard-switched applications.
- **Wide bandgap semiconductor** properties, enabling operation at higher junction temperatures (typically 150–200°C compared to 125°C for silicon devices).

Circuit Design Benefits

The fast switching capability and low gate charge of this SiC MOSFET provide significant benefits in circuit design:

- **Higher switching frequencies** (50–100 kHz and beyond), allowing for smaller passive components and more compact designs.
- **Improved power factor correction** (PFC) in AC-DC converters through efficient, high-speed switching.
- **Reduced EMI filtering requirements** due to smoother and more controlled switching transitions.
- **Parallel operation** capability for high-current applications, facilitated by good thermal and electrical characteristics.

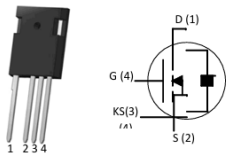


Figure 1.25: Package

1.21.1 Key Specifications

Table 1.2: CI60N120SM4 SiC MOSFET Specifications

Parameter	Value
$V_{DS\max}$ (Drain-Source Voltage)	1200 V
$V_{GS\text{op}}$ (Gate-Source Operating Voltage)	-5 V to 20 V
$V_{GS\max}$ (Gate-Source Maximum Voltage)	-10 V to 25 V
I_D (Continuous Drain Current)	60 A at $T_C = 25°C$ 40 A at $T_C = 100°C$
$I_{D(\text{pulse})}$ (Pulsed Drain Current)	160 A
P_D (Power Dissipation)	330 W at $T_C = 25°C$
T_J, T_{stg} (Operating Junction,Storage Temperature)	-55 to +175°C
$V_{GS(\text{th})}$ (Gate Threshold Voltage)	2.0 to 4.0 V at $V_{DS} = V_{GS}$, $I_D = 10$ mA
$R_{DS(\text{on})}$ (Drain-Source On-State Resistance)	45 to 55 mΩ
C_{iss} (Input Capacitance)	3550 pF
C_{oss} (Output Capacitance)	162 pF
C_{rss} (Reverse Capacitance)	29 pF
$R_G(int)$ (Internal Gate Resistance)	1.0 Ω
$R_{\theta JC}$ (Thermal Resistance, Junction to Case)	0.40 °C/W
$R_{\theta JA}$ (Thermal Resistance, Junction to Ambient)	38 °C/W

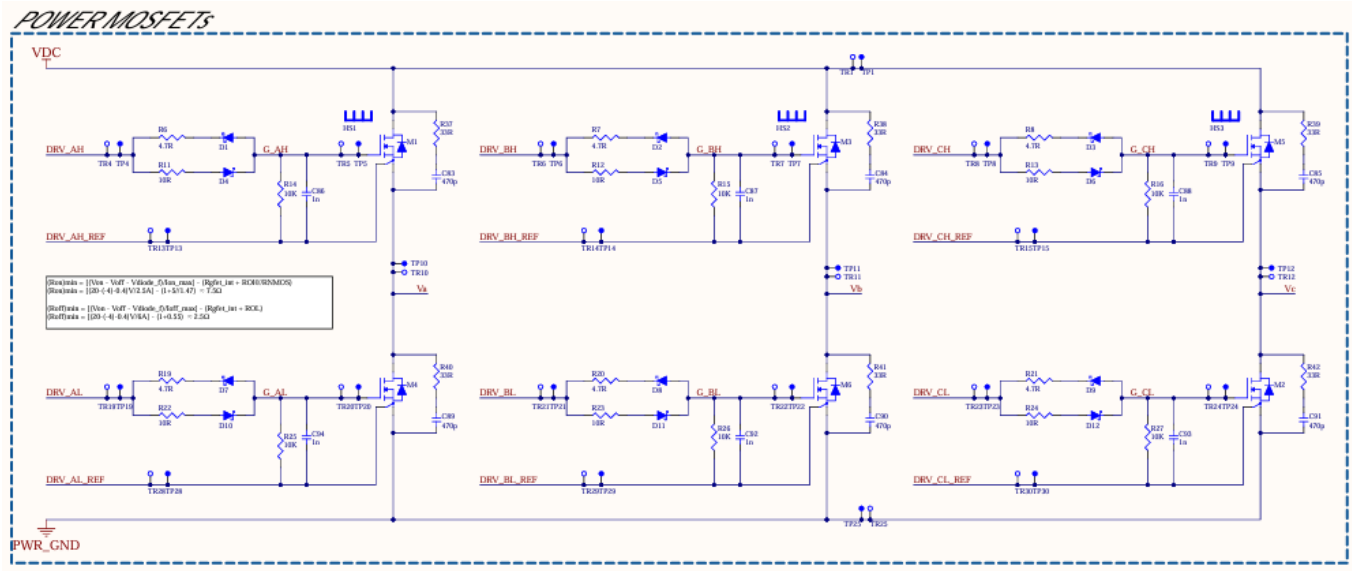


Figure 1.26: POWER MOSFET

1.22 Gate Driver: UCC215204A

The **UCC215204A** is a **6-A dual-channel isolated gate driver**, rated for **5.7 kV RMS isolation**, perfect for driving **high-speed SiC MOSFETs** in full-bridge and three-phase inverters.

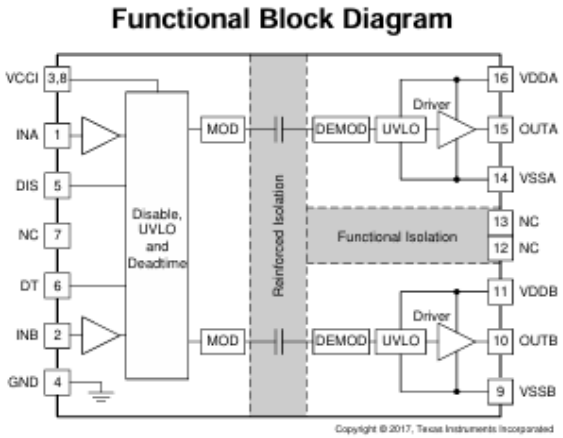


Figure 1.27: Functional Block Diagram

1.22.1 Key Specifications

Table 1.3: Recommended Operating Conditions

Parameter	Description	Unit	Min	Max
V_{CCI}	Input supply voltage	V	3	18
V_{DDA}, V_{DDB}	Driver output bias supply	V	6.5	25

Table 1.4: Isolation Characteristics

Parameter	Description	Test Conditions	Value
R_{IO}	Isolation resistance, input to output	$V_{\mathrm{IO}} = 500\,\mathrm{V}$ at $T_{\mathrm{A}} = 25\,^{\circ}\mathrm{C}$	$> 10^{12}\ \Omega$
		$V_{\mathrm{IO}} = 500\,\mathrm{V}$ at $100^{\circ}\mathrm{C} \le T_{\mathrm{A}} \le 125^{\circ}\mathrm{C}$	$> 10^{11}\ \Omega$
		$V_{\mathrm{IO}} = 500\,\mathrm{V}$ at $T_{\mathrm{S}} = 150^{\circ}\mathrm{C}$	$> 10^{9}\ \Omega$
UL 1577			
V_{ISO}	Withstand isolation voltage	$V_{\mathrm{TEST}} = V_{\mathrm{ISO}} = 5700\,\mathrm{V_{RMS}}$ $t = 60\,\mathrm{sec.}$ (qualification) $V_{\mathrm{TEST}} = 1.2 \times V_{\mathrm{ISO}} = 6840\,\mathrm{V_{RMS}}$ $t = 1\,\mathrm{sec.}$ (100% production)	5700 $\mathrm{V_{RMS}}$

Table 1.5: Output Electrical Characteristics

Parameter	**Test Conditions**	**Min**	**Typ**	**Unit**
I_{OAH+}, I_{OB+}	Peak output source current $C_{VDD} = 10\,\mu F$, $C_{LOAD} = 0.18\,\mu F$, $f = 1\,kHz$		4	A
I_{OAH-}, I_{OB-}	Peak output sink current $C_{VDD} = 10\,\mu F$, $C_{LOAD} = 0.18\,\mu F$, $f = 1\,kHz$		6	A
R_{OHA}, R_{OHB}	Output resistance at high state $I_{OUT} = -10\,mA$, $T_A = 25°C$		5	Ω
R_{OLA}, R_{OLB}	Output resistance at low state $I_{OUT} = 10\,mA$, $T_A = 25°C$		0.55	Ω

Table 1.6: Switching Characteristics of UCC21520

Parameter	**Test Conditions**	**Min**	**Typ**	**Max**
t_{RISE}	Output rise time, 20% to 80% measured points $C_{OUT} = 1.8\,nF$	-	6	16
t_{FALL}	Output fall time, 90% to 10% measured points $C_{OUT} = 1.8\,nF$	-	7	12
t_{PWmin}	Minimum pulse width, output off for less than minimum $C_{OUT} = 0\,pF$	-	-	20
t_{PDHL}	Propagation delay from INx to OUTx falling edges $C_{OUT} = 1.8\,nF$	14	19	30
t_{PDLH}	Propagation delay from INx to OUTx rising edges $C_{OUT} = 1.8\,nF$	14	19	30

Table 1.7: Input/Output Logic Truth Table

INA	**INB**	**DISABLE**	**OUTA**	**OUTB**	**Notes**
L	L	L/Open	L	L	If dead time function is used, output transitions occur after the dead time expires. See Programmable Dead Time (DT) pin section.
L	H	L/Open	L	H	Normal operation
H	L	L/Open	H	L	Normal operation
H	H	L/Open	L	L	DT pin left open or programmed with R_{DT}
H	H	L/Open	H	H	DT pin pulled to VCCI
Open	Open	L/Open	L	L	Default safe state
X	X	H	L	L	Disabled state (outputs forced low)

1.22.2 Input Stage

The input pins (INA, INB, and DIS) of the UCC21520 and the UCC21520A are based on a TTL and CMOS compatible input-threshold logic that is totally isolated from the VDD supply voltage. The input pins are easy to drive with logic-level control signals (such as those from 3.3-V micro-controllers), since the UCC21520 and the UCC21520A have a typical high threshold (VINAH) of 1.8 V and a typical low threshold of 1 V, which vary little with temperature (see Figures 22 and 23). A wide hysteresis (VINA_HYS) of 0.8 V makes for good noise immunity and stable operation.

If any of the inputs are ever left open, internal pull-down resistors force the pin low. These resistors are typically 200 kΩ (see Functional Block Diagram). However, it is still recommended to ground an input if it is not being used.

1.22.3 Device Functional Modes

Programmable Dead Time (DT) Pin

The UCC21520/UCC21520A provides flexible dead time (DT) configuration through the DT pin:

DT Pin to VCC Connection

- Outputs directly follow inputs with **no dead time**
- Allows output signal overlap when needed
- Maximum switching frequency operation

DT Pin with Programming Resistor (RDT)

- Connect resistor RDT between DT and GND
- Dead time calculated by:

$$t_{DT} \approx 10 \times R_{DT} \quad [\text{ns}] \tag{1.36}$$

  where R_{DT} is in kΩ
- DT pin characteristics:
  - Steady-state voltage: ≈0.8V
  - Current: <10μA (when R_{DT}=100kΩ)
- Recommended implementation:
  - For $R_{DT} > 5$kΩ, add parallel 2.2nF ceramic capacitor
  - Improves noise immunity and channel matching
  - Never leave DT pin floating

Dead Time Operation Logic

- Falling input edge activates programmed DT for complementary output
- Output DT uses the longer of:
  - Programmed driver DT
  - Input signal's inherent DT
- Shoot-through protection:
  - Both outputs forced low when inputs are simultaneously high
  - Doesn't affect normal operation DT setting

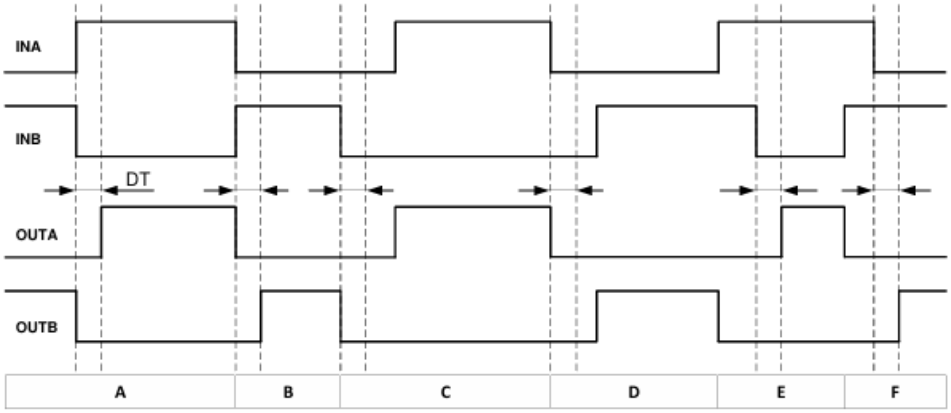


Figure 1.28: Input and output logic timing relationships showing dead time operation under various conditions (A-F)

1.22.4 Conditions for Input and Output Logic Relationship

- **Condition A:** INB goes low, INA goes high. INB sets OUTB low immediately and assigns the programmed dead time to OUTA. OUTA is allowed to go high after the programmed dead time.
- **Condition B:** INB goes high, INA goes low. Now INA sets OUTA low immediately and assigns the programmed dead time to OUTB. OUTB is allowed to go high after the programmed dead time.
- **Condition C:** INB goes low, INA is still low. INB sets OUTB low immediately and assigns the programmed dead time for OUTA. In this case, the input signal's own dead time is longer than the programmed dead time. Thus, when INA goes high, it immediately sets OUTA high.
- **Condition D:** INA goes low, INB is still low. INA sets OUTA low immediately and assigns the programmed dead time to OUTB. INB's own dead time is longer than the programmed dead time. Thus, when INB goes high, it immediately sets OUTB high.
- **Condition E:** INA goes high, while INB and OUTB are still high. To avoid overshoot, INA immediately pulls OUTB low and keeps OUTA low. After some time OUTB goes low and assigns the programmed dead time to OUTA. OUTB is already low. After the programmed dead time, OUTA is allowed to go high.

- **Condition F:** INB goes high, while INA and OUTA are still high. To avoid overshoot, INB immediately pulls OUTA low and keeps OUTB low. After some time OUTA goes low and assigns the programmed dead time to OUTB. OUTA is already low. After the programmed dead time, OUTB is allowed to go high.

1.22.5 Application Circuits with Output Stage Negative Bias

When parasitic inductances are introduced by non-ideal PCB layout and long package leads (e.g., TO-220 and TO-247 type packages), there could be ringing in the gate-source drive voltage of the power transistor during high di/dt and dv/dt switching. If the ringing exceeds the threshold voltage, there is a risk of unintended turn-on and even shoot-through. Applying a negative bias on the gate drive is a popular method to keep such ringing below the threshold voltage.

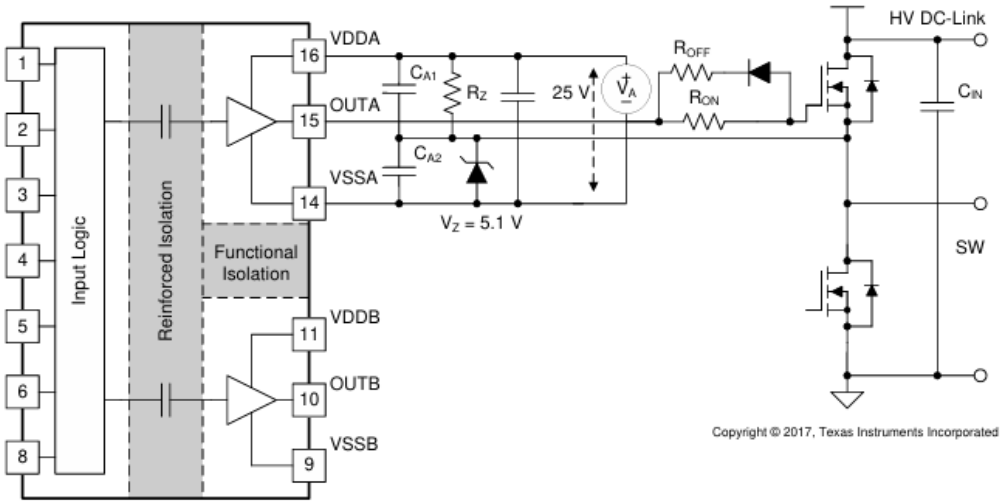


Figure 1.29: Negative Bias with Zener Diode on Iso-Bias Power Supply Output

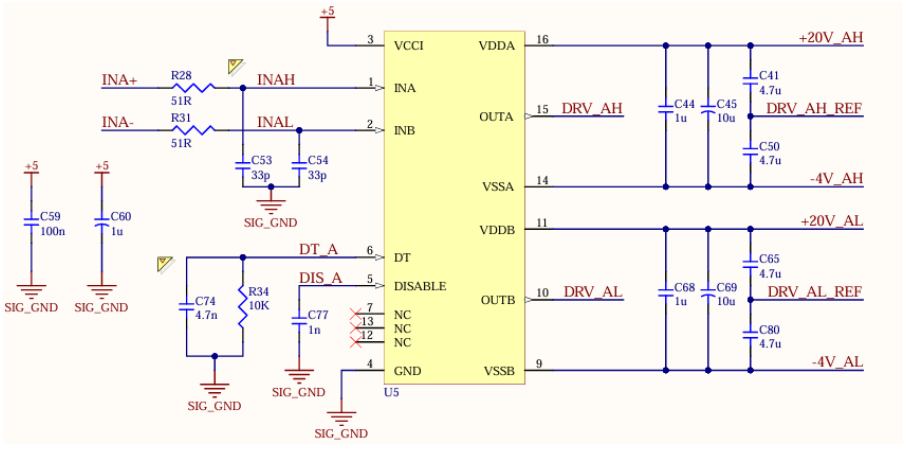


Figure 1.30: Schematic Gate Drive

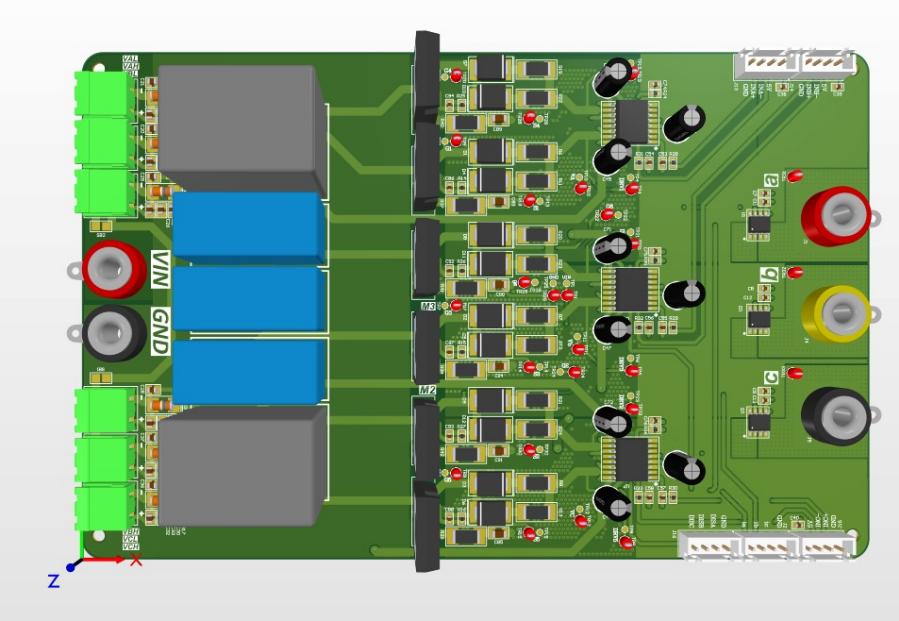


Figure 1.31: 3D inverter

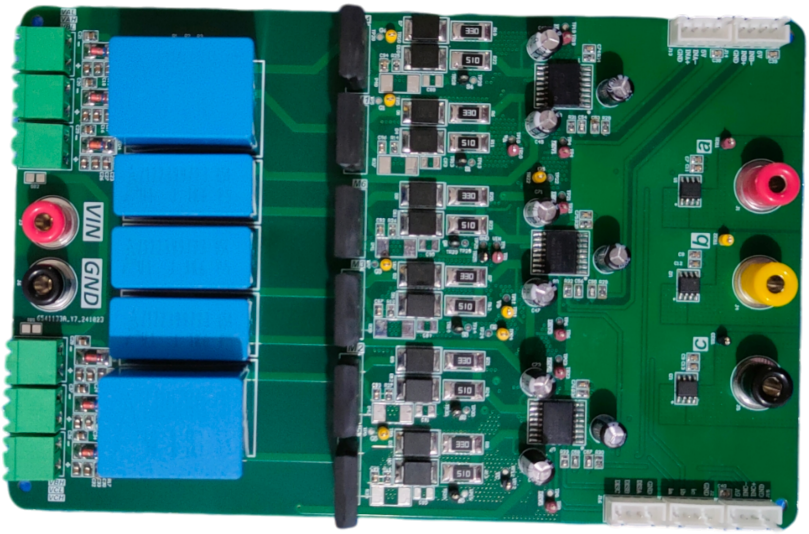


Figure 1.32: inverter

FLYBACK

1.23 Abstract

This chapter presents the design, development, and integration of a high-efficiency flyback converter tailored for gate driving applications in inverter-based motor control systems. The converter accepts a wide DC input voltage range of 100–480 V and provides six isolated 25 V outputs, each capable of sourcing up to 2.5 A peak current. Utilizing the UCC28740 controller

operating at a switching frequency of 69 kHz, the converter ensures stable gate voltages of +20 V for turn-on and -5 V for turn-off. This chapter outlines the converter's topology, control strategy, integration with the inverter, and compares it with conventional bootstrap capacitor techniques. Design decisions are guided by insights from the UCC28740DR reference design, with emphasis on the controller's contribution to overall performance and reliability.

1.24 Introduction

Gate driver circuits are fundamental in power electronics, particularly in inverter systems where high-speed switching and isolation are critical. In this project, a flyback converter was selected to supply isolated gate voltages for six MOSFETs used in a three-phase inverter. The chosen topology supports a wide input range (100–480 V DC), enabling it to directly draw power from the main DC bus.

The flyback topology was favored for several reasons:

- **Simplicity and Cost-Effectiveness:** It enables the generation of multiple isolated outputs using a single magnetic component, reducing both component count and system complexity.
- **Inherent Electrical Isolation:** Ensures the gate drive circuits are protected from high-voltage switching noise, enhancing system safety and electromagnetic compatibility.
- **Negative Turn-Off Voltage Capability:** By supplying -5 V for turn-off, the circuit improves noise immunity and reduces the risk of false turn-on due to Miller capacitance effects.
- **Design Integration:** The converter derives its input directly from the inverter's high-voltage DC source, eliminating the need for separate auxiliary supplies.

This chapter delves into the theoretical basis of the flyback converter, details the implementation using the UCC28740DR controller, and evaluates its performance in comparison to traditional gate supply methods such as bootstrap circuits.

1.25 Flyback Converter Description

The flyback converter employed in this project is a discontinuous conduction mode (DCM) topology controlled by the Texas Instruments UCC28740, a high-performance primary-side regulated controller. The key specifications of the converter are summarized in Table 1.8.

Table 1.8: Key Specifications of the Flyback Converter

Parameter	**Value**
Input Voltage (V_{in})	100–480 V DC
Output Voltage (V_{out})	25 V DC per output
Number of Outputs	6 isolated outputs
Switching Frequency ($f_{switching}$)	69 kHz
Peak Output Current (I_{peak})	2.5 A per output
Gate Drive Levels	+20 V (turn-on), -5 V (turn-off)
Controller	UCC28740DR

The converter is centered around a custom-designed high-frequency transformer with six secondary windings to provide isolation for each gate drive output. The UCC28740 controller

regulates the output voltage using primary-side feedback and frequency modulation, achieving precise voltage control with minimal external components.

1.25.1 PCB Layout and Implementation

The flyback converter was designed and implemented using Altium Designer. The layout emphasizes isolation between high-voltage and low-voltage sections, short loop areas for switching paths, and adequate thermal management for power components. The modular output stages simplify troubleshooting and allow for flexible testing of individual supplies.

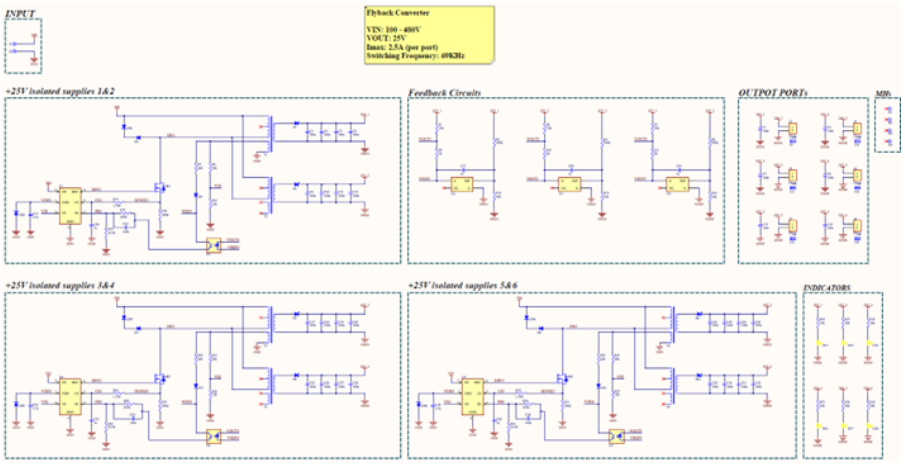


Figure 1.33: Schematic of the 25 V Flyback Converter for Inverter Gate Drive

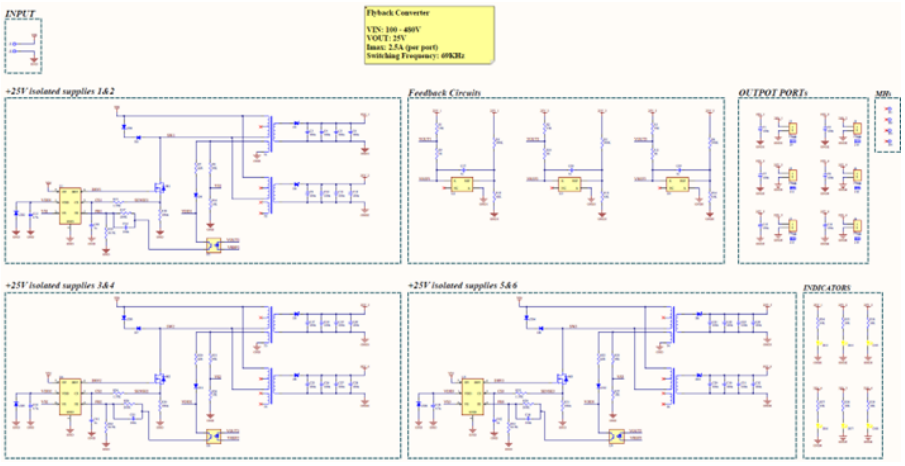


Figure 1.34: Schematic of the 25 V Flyback Converter for Inverter Gate Drive

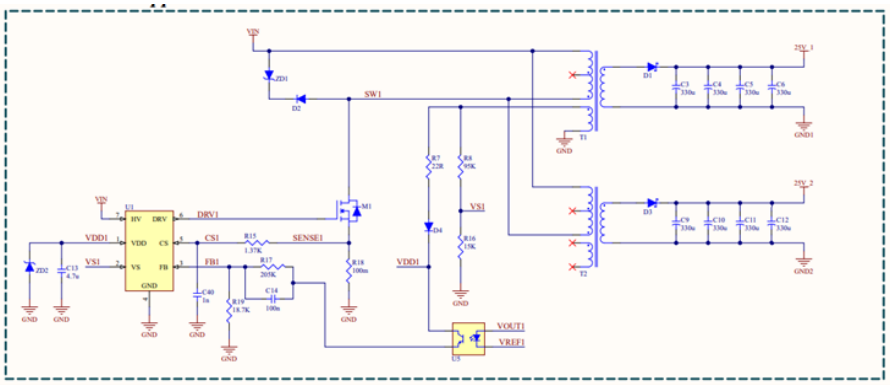


Figure 1.35: +25 V Isolated Supplies 1 & 2

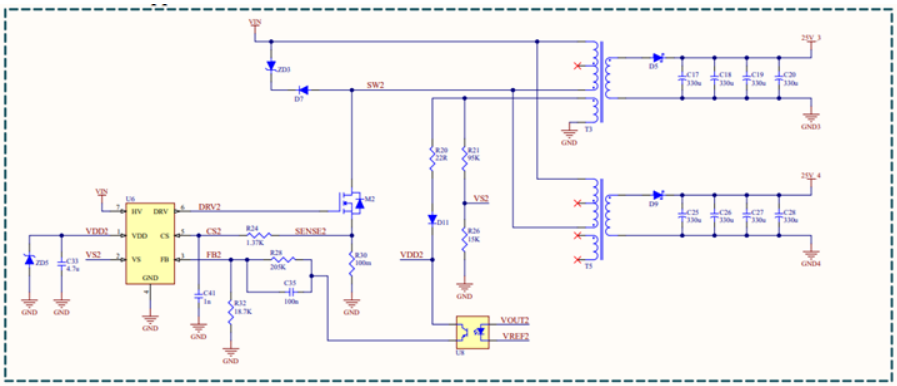


Figure 1.36: +25 V Isolated Supplies 3 & 4

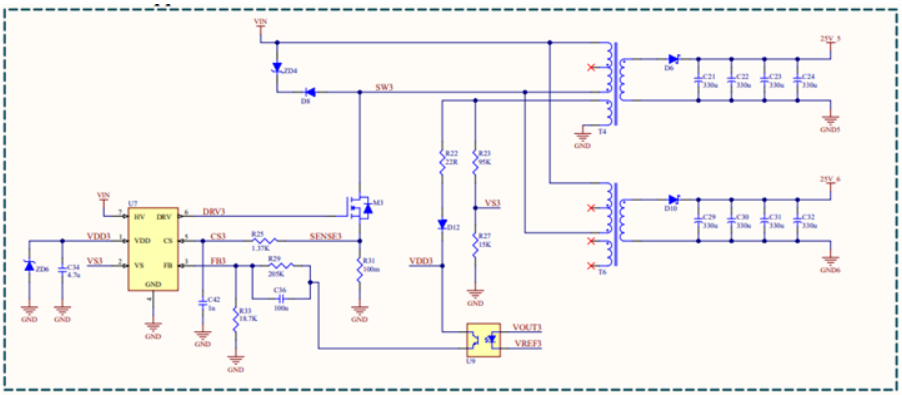


Figure 1.37: +25 V Isolated Supplies 5 & 6

1.25.2 Control and Feedback

The flyback converter utilizes an optocoupler and an auxiliary winding to provide primary-side regulation. Feedback and compensation components are optimized to ensure fast transient response and stable regulation across the full input range.

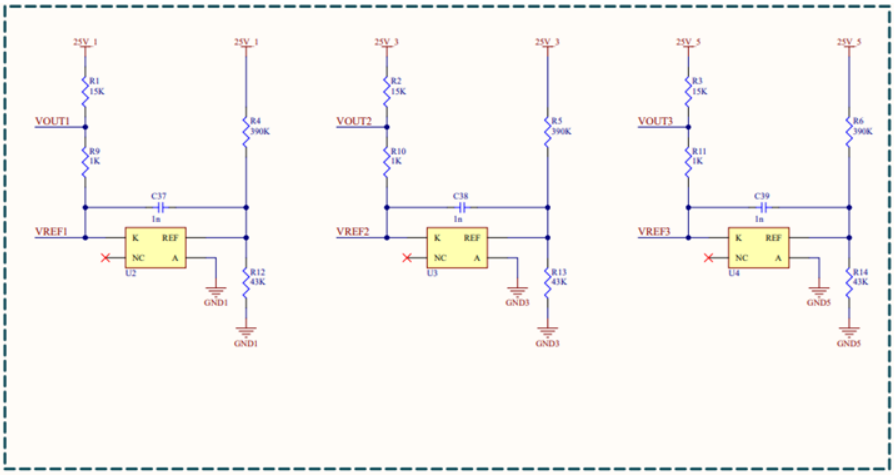


Figure 1.38: Feedback Circuits

1.25.3 Output Interface and Monitoring

Each output is equipped with connectors and test points for easy integration with gate driver circuits. Status indicators (LEDs) were added for visual confirmation of supply operation.

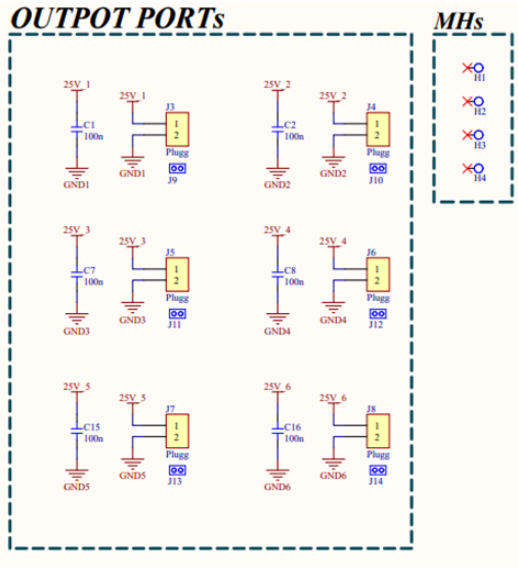


Figure 1.39: Output Ports and Mounting Holes

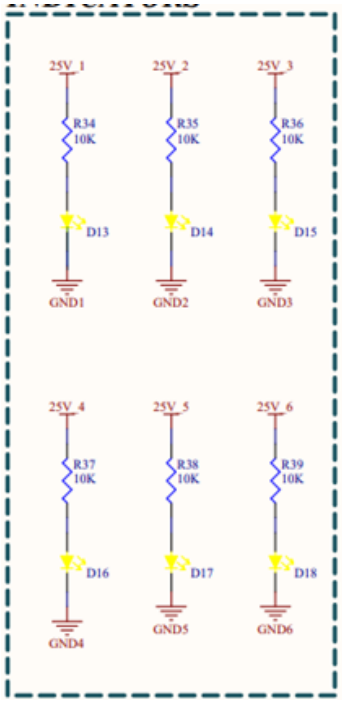


Figure 1.40: Supply Status Indicators

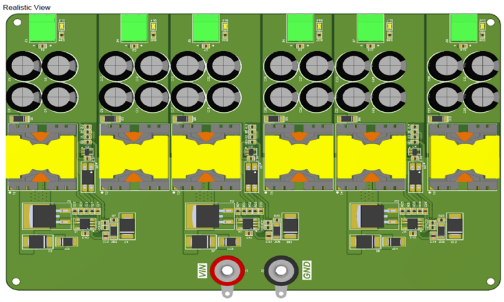


Figure 1.41: Primary PCB Layout of the Flyback Converter

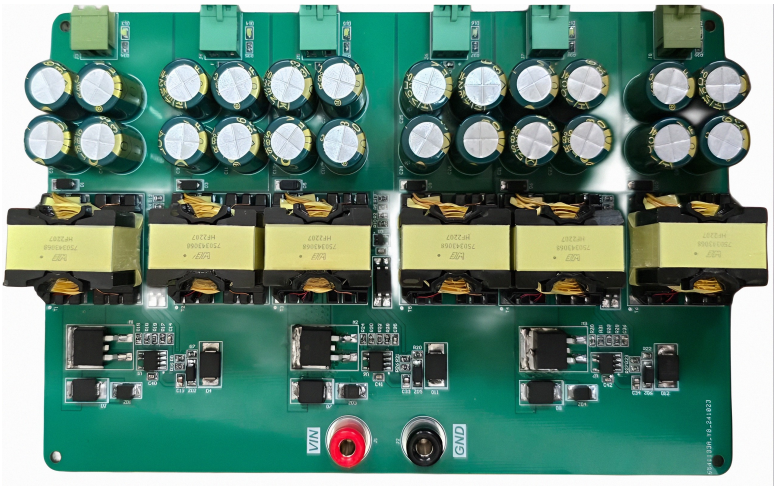


Figure 1.42: flyback

1.26 Comparison with Bootstrap Method

Traditional gate drive power is often derived from bootstrap capacitor techniques, which rely on switching action to charge capacitors connected to high-side drivers. While simpler, bootstrap circuits have limitations:

- **Limited Duty Cycle:** High duty cycles may prevent sufficient charging of the bootstrap capacitor.
- **No Negative Turn-Off Voltage:** Lacks the ability to actively pull the gate below source, reducing noise immunity.
- **No Start-Up at Low Frequency or DC:** Bootstrap supplies depend on switching activity, making them unsuitable for low-frequency or static applications.
- **No True Isolation:** Bootstrap circuits do not provide galvanic isolation between control and power stages.

In contrast, the flyback converter provides independent, isolated power supplies that operate regardless of switching frequency and offer robust noise immunity—making it a superior choice for advanced inverter applications.

1.27 Conclusion

The flyback converter developed in this project demonstrates a robust, efficient, and scalable solution for powering gate driver circuits in high-performance inverter systems. With six isolated

outputs derived from a single high-voltage input and managed by the UCC28740 controller, the design ensures safe and reliable operation under various load and switching conditions. The implementation showcases a practical integration of power electronics and control strategies suited for real-world motor drive applications.

Chapter 2

Permanent Magnet Synchronous Motors (PMSMs)

2.1 abstract

Permanent Magnet Synchronous Motors (PMSMs) represent a pinnacle of electric machine technology, combining the high efficiency of permanent magnets with the precise controllability of synchronous motors. Their advanced design enables exceptional power density, reliability, and dynamic performance, making them indispensable in critical applications including electric vehicles, industrial automation, robotics, and renewable energy systems. This comprehensive guide explores the theoretical foundations, design considerations, control strategies, and practical applications of PMSMs, with particular emphasis on modeling approaches and advanced control techniques.

2.2 Introduction

PMSMs have established themselves as premier electric machines in high-performance applications, outperforming traditional induction motors and brushed DC motors in numerous metrics. Their rise to prominence stems from several inherent advantages:

- High power density - More power output per unit volume/weight
- Superior efficiency - Often exceeding 95% at rated conditions
- Excellent dynamic response - Fast torque development and precise speed control
- Low maintenance requirements - No brushes, slip rings, or rotor windings
- Low acoustic noise - Smoother operation with reduced vibration

This document provides a comprehensive analysis of PMSMs, covering fundamental principles, mathematical modeling, control strategies, design considerations, practical applications, and emerging trends in PMSM technology.

2.3 Fundamental Working Principles

2.3.1 Operating Principle

PMSMs operate based on electromagnetic interaction between:

- A rotating magnetic field generated by the stator's three-phase windings
- A permanent magnetic field produced by rotor-mounted magnets

Unlike induction motors, PMSMs do not rely on slip for torque generation. The rotor rotates synchronously with the stator's rotating magnetic field, maintaining a constant phase relationship that enables precise control.

2.3.2 Stator and Rotor Interaction

- The stator is supplied with three-phase AC, generating a rotating magnetic field.
- The rotor, embedded with permanent magnets, aligns itself with the stator's rotating field, producing torque.
- The rotor speed is synchronized with the supply frequency:

$$N_s = \frac{120f}{P} \tag{2.1}$$

Where:

- N_s = synchronous speed (RPM)
- f = supply frequency (Hz)
- P = number of poles

2.3.3 Back-EMF Generation

As the rotor rotates, the permanent magnets induce a back-electromotive force (back-EMF) in the stator windings. This back-EMF is proportional to rotor speed:

$$E_b = k_e \omega_m \tag{2.2}$$

Where:

- k_e = back-EMF constant
- ω_m = mechanical rotor speed (rad/s)

The back-EMF waveform provides critical information about rotor position and is often utilized in sensorless control techniques.

2.3.4 Torque Production

The electromagnetic torque in a PMSM results from the interaction between the stator's magnetic field and the rotor's permanent magnets. The torque equation varies depending on the PMSM design:

For Surface-mounted PMSMs (SPMSMs):

$$T_e = \frac{3}{2} P \psi_m i_q \tag{2.3}$$

For Interior PMSMs (IPMSMs):

$$T_e = \frac{3}{2} P[\psi_m i_q + (L_d - L_q) i_d i_q] \tag{2.4}$$

Where:

- P = number of pole pairs
- ψ_m = permanent magnet flux linkage
- i_q = q-axis current component (torque-producing)
- i_d = d-axis current component (flux-controlling)
- L_d, L_q = d-axis and q-axis inductances

2.4 Key Components and Design Considerations

2.4.1 Stator Design

Core Material

- Laminated silicon steel - Minimizes eddy current losses
- Cobalt-iron alloys - For high-performance applications requiring higher flux density
- Amorphous metals - For ultra-low core loss applications

Winding Configurations

Distributed Windings

- Lower harmonic distortion
- Smoother torque production
- Better thermal dissipation
- More complex manufacturing process

Concentrated Windings

- Simpler construction
- Higher copper fill factor
- Lower end-winding losses
- Higher torque ripple and cogging torque
- Better fault tolerance

Slot-Pole Combinations

- Affects cogging torque, noise, vibration, and efficiency
- Common combinations include 12-slot/10-pole and 9-slot/8-pole designs
- Fractional slot combinations help reduce cogging torque

Stator Cooling Methods

- Natural convection
- Forced air cooling
- Liquid cooling (water, oil)
- Direct winding cooling techniques

2.4.2 Rotor Design

Surface-Mounted PMSM (SPMSM)

- Magnets glued or banded on the rotor surface
- Simple construction but limited mechanical strength at high speeds
- $L_d = L_q$ (no reluctance torque component)
- Higher back-EMF and lower inductance
- Better suited for low-speed, high-torque applications

Interior PMSM (IPMSM)

- Magnets embedded inside the rotor
- Higher mechanical robustness for high-speed operation
- Reluctance torque utilization due to $L_d < L_q$
- Better field-weakening capability for extended speed range
- Excellent for automotive and high-speed industrial applications

V-shaped/Multi-layered IPMSM)

- Enhanced reluctance torque component
- Improved field-weakening capabilities
- Optimized magnet utilization
- Better mechanical integrity at extreme speeds

Magnet Materials

- Neodymium-Iron-Boron (NdFeB) - Highest energy product, temperature-sensitive
- Samarium-Cobalt (SmCo) - Better temperature stability, expensive
- Ferrite magnets - Low cost, lower performance, better temperature stability
- AlNiCo - High temperature capability, demagnetization risk

2.4.3 Mechanical Considerations

- Rotor balancing for high-speed operation
- Bearing selection and lubrication
- Thermal expansion considerations
- Magnet retention methods
- Vibration damping
- Insulation system design

2.5 Mathematical Modeling of PMSMs

2.5.1 Three-Phase Model (ABC Frame)

The voltage equations in the natural three-phase reference frame:

$$v_a = R_s i_a + \frac{d\lambda_a}{dt} \tag{2.5}$$

$$v_b = R_s i_b + \frac{d\lambda_b}{dt} \tag{2.6}$$

$$v_c = R_s i_c + \frac{d\lambda_c}{dt} \tag{2.7}$$

Where flux linkages are position-dependent:

$$\lambda_a = L_{aa} i_a + L_{ab} i_b + L_{ac} i_c + \lambda_{am} \tag{2.8}$$

$$\lambda_b = L_{ba} i_a + L_{bb} i_b + L_{bc} i_c + \lambda_{bm} \tag{2.9}$$

$$\lambda_c = L_{ca} i_a + L_{cb} i_b + L_{cc} i_c + \lambda_{cm} \tag{2.10}$$

2.5.2 Stationary Reference Frame (α-β Model)

Using Clarke transformation to convert three-phase quantities to a two-phase stationary reference frame:

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

The voltage equations become:

$$v_\alpha = R_s i_\alpha + \frac{d\lambda_\alpha}{dt} \tag{2.11}$$

$$v_\beta = R_s i_\beta + \frac{d\lambda_\beta}{dt} \tag{2.12}$$

2.5.3 d-q Axis Transformation (Park's Model)

Park transformation converts the α-β quantities to a rotating reference frame synchronized with the rotor position:

$$i_d = i_\alpha \cos(\theta) + i_\beta \sin(\theta) \tag{2.13}$$

$$i_q = -i_\alpha \sin(\theta) + i_\beta \cos(\theta) \tag{2.14}$$

Voltage Equations

$$V_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \tag{2.15}$$

$$V_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \psi_m) \tag{2.16}$$

Torque Equation

$$T_e = \frac{3}{2} P \left[\psi_m i_q + (L_d - L_q) i_d i_q \right] \tag{2.17}$$

Mechanical Dynamics

$$J \frac{d\omega_r}{dt} = T_e - T_L - B\omega_r \tag{2.18}$$

Where:

- J = moment of inertia
- ω_r = rotor mechanical speed
- T_L = load torque
- B = viscous friction coefficient

2.5.4 Inverse Transformations

Inverse Park (to stationary frame)

$$v_\alpha = v_d \cos(\theta) - v_q \sin(\theta) \tag{2.19}$$

$$v_\beta = v_d \sin(\theta) + v_q \cos(\theta) \tag{2.20}$$

Inverse Clarke (to three-phase)

$$v_a = v_\alpha \tag{2.21}$$

$$v_b = -\frac{1}{2} v_\alpha + \frac{\sqrt{3}}{2} v_\beta \tag{2.22}$$

$$v_c = -\frac{1}{2} v_\alpha - \frac{\sqrt{3}}{2} v_\beta \tag{2.23}$$

2.5.5 Power Loss Models

Copper Losses

$$P_{cu} = \frac{3}{2} R_s (i_d^2 + i_q^2) \tag{2.24}$$

Iron Losses

$$P_{fe} = P_{hysteresis} + P_{eddy} \tag{2.25}$$

Where:

- $P_{hysteresis} = k_h f B_m^n$ (frequency and flux dependent)
- $P_{eddy} = k_e f^2 B_m^2$ (proportional to frequency squared)

Mechanical Losses

$$P_{mech} = P_{friction} + P_{windage} \tag{2.26}$$

Total Power and Efficiency

$$P_{in} = P_{out} + P_{cu} + P_{fe} + P_{mech} \tag{2.27}$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{T_e \omega_m}{T_e \omega_m + P_{losses}} \tag{2.28}$$

2.5.6 Thermal Modeling

Heat transfer equation:

$$\frac{dT}{dt} = \frac{P_{\text{losses}}}{C_{\text{th}}} - \frac{T - T_{\text{amb}}}{R_{\text{th}} C_{\text{th}}} \tag{2.29}$$

Where:

- T = motor temperature
- C_{th} = thermal capacitance
- R_{th} = thermal resistance
- T_{amb} = ambient temperature

2.6 Control Strategies for PMSMs

2.6.1 Field-Oriented Control (FOC)

Key Principles

- Decoupled control of torque and flux components
- Coordinate transformations for simplified control
- Current control in synchronously rotating reference frame

Control Loops Structure

- Inner current control loops (d-axis and q-axis)
- Outer speed/position control loop
- Optional flux weakening controller

MTPA Control Strategy

Maximum Torque Per Ampere (MTPA) strategy optimizes torque production efficiency:
For SPMSMs:

- Typically set $i_d = 0$ to maximize torque per ampere

For IPMSMs:

- Optimal current angle found from:
$$\frac{\partial T_e}{\partial i_d} = 0$$
- Results in negative i_d values to utilize reluctance torque

Field Weakening Operation

- Extends speed range beyond base speed by reducing magnetic flux
- Implemented by injecting negative i_d current
- Voltage and current constraints determine feasible operating region
- Constant power region operation

2.6.2 Direct Torque Control (DTC)

Basic Principles

- Direct control of stator flux and torque
- Hysteresis-based control for fast dynamic response
- No inner current control loops required

Advantages

- Fast torque response
- No coordinate transformations required
- Robust against parameter variations

Disadvantages

- Variable switching frequency
- Higher torque ripple
- Flux and torque estimation accuracy dependent

DTC Improvements

- Space vector modulation DTC
- Predictive direct torque control
- Sliding mode DTC

2.6.3 Sensorless Control Techniques

Back-EMF Based Methods

- Extended EMF estimators
- Sliding-mode observers
- Phase-locked loop techniques
- Adaptive observers

High-Frequency Signal Injection

- Carrier signal injection in d-axis
- Rotor position estimation from saliency effects
- Applicable at very low and zero speeds
- Secondary saliency effects and filtering techniques

Model Reference Adaptive Systems (MRAS)

- Reference model vs. adaptive model comparison
- Online parameter adaptation
- Speed and position estimation

Extended Kalman Filters

- Nonlinear state estimation
- Noise filtering capabilities
- Computational complexity considerations

2.6.4 Predictive Control Methods

Model Predictive Control (MPC)

- Finite control set MPC
- Continuous control set MPC
- Multi-objective optimization

Deadbeat Control

- Pole placement for fast response
- Digital implementation considerations

2.6.5 Why $i_d = 0$ in FOC for Surface-Mounted PMSMs

The $i_d = 0$ control strategy is commonly used for Surface-mounted PMSMs (SPMSMs) for several key reasons:

Maximizing Torque Production

For a SPMSM where $L_d = L_q$, the torque equation simplifies to:

$$T_e = \frac{3}{2} P \psi_m i_q$$

Setting $i_d = 0$ maximizes torque per ampere since:

- Only i_q directly contributes to torque production
- The reluctance torque component is eliminated
- Stator current is fully utilized for torque generation

Efficiency and Loss Minimization

- Zero i_d minimizes stator copper losses
- Avoids unnecessary magnetic field manipulation
- Reduces core losses
- Improves overall system efficiency

Simplified Control Implementation

- Control algorithm focuses only on i_q regulation
- Reduced computational complexity
- Simpler tuning and commissioning

Thermal Management

- Reduced current magnitude minimizes heating
- Better thermal management
- Extended motor lifespan

When $i_d \neq 0$ is Beneficial

For Interior PMSMs (IPMSMs), setting $i_d \neq 0$ is often advantageous:

- Utilizes reluctance torque component
- Enables field weakening for extended speed range
- Implements Maximum Torque Per Ampere (MTPA) trajectory
- Facilitates Maximum Torque Per Volt (MTPV) operation

2.7 Practical Implementation Aspects

2.7.1 Power Electronics Converters

Voltage Source Inverters (VSI)

- Two-level VSI topology
- Multi-level inverter configurations
- Common-mode voltage considerations

2.7.2 Modulation Techniques

Pulse Width Modulation (PWM) techniques for PMSM drives vary in implementation complexity, harmonic performance, and efficiency. The choice depends on application requirements including switching losses, harmonic distortion, and DC bus utilization.

2.7.3 Comparison of PWM Techniques

Table 2.1: Comparison of PWM Modulation Techniques for PMSM Drives

Parameter	**SVPWM**	**SPWM**	**DPWM**
Implementation Complexity	Moderate	Simple	Moderate
DC Bus Utilization (%)	100 (theoretical)	86.6 (max)	86.6
Harmonic Distortion (THD)	Lowest	High	Moderate
Switching Losses	Medium	High	Lowest (30–50% reduction)
Current Ripple	Low	Highest	Medium
Common Mode Voltage	Variable	Constant	Reduced
Acoustic Noise	Lowest	High	Medium

2.7.4 Space Vector PWM (SVPWM)

- **Principle**: Utilizes eight possible switching states (six active, two zero) of a three-phase inverter to synthesize the reference voltage vector.
- **Advantages**:

  - 15% better DC bus utilization compared to SPWM.
  - Lower harmonic distortion (THD typically 5-10% better than SPWM).
  - Better dynamic response due to optimal vector selection.
  - Balanced thermal distribution across switches.
- **Implementation Aspects**:
  - Sector identification based on voltage vector angle.
  - Dwell time calculation using $T_1 = T_s \frac{\sqrt{3}|V_{ref}|}{V_{dc}} \sin(60° - \theta)$.
  - $T_2 = T_s \frac{\sqrt{3}|V_{ref}|}{V_{dc}} \sin(\theta)$.
  - Dead-time compensation critical for low-speed operation.
- **Applications**: High-performance drives, servo systems, EV traction inverters.

2.7.5 Sinusoidal PWM (SPWM)

- **Principle**: Comparison of sinusoidal reference with triangular carrier wave.
- **Advantages**:
  - Simple implementation (analog or digital).
  - Predictable harmonic spectrum.
  - No sector identification required.
- **Limitations**:
  - Lower DC bus utilization (maximum 86.6% of available DC voltage).
  - Higher harmonic content, especially at low modulation indices.
  - Higher switching losses due to continuous modulation.
- **Variants**:
  - Symmetrical regular sampling.
  - Asymmetric regular sampling (improved harmonic performance).
  - Third-harmonic injection (for increased voltage utilization).
- **Applications**: Low-cost drives, HVAC systems, general-purpose inverters.

2.7.6 Discontinuous PWM (DPWM)

- **Principle**: Clamps one phase to positive or negative DC bus for 120° per cycle.
- **Types**:
  - DPWM0 (clamping at peak of reference).
  - DPWM1 (clamping at valley of reference).
  - DPWM2 (clamping at zero-crossing).
  - DPWM3 (minimum switching loss pattern).

- **Advantages**:
  - 30-50% reduction in switching losses compared to SVPWM.
  - Reduced common-mode voltage.
  - Better efficiency at high modulation indices.
- **Limitations**:
  - Increased current ripple (up to 20% higher than SVPWM).
  - Acoustic noise at certain speed ranges.
  - Complex implementation for field-weakening operation.
- **Applications**: High-power drives, applications where efficiency is critical.

2.7.7 Advanced PWM Techniques

- **Hybrid PWM**: Combines SVPWM and DPWM based on operating conditions.
- **3D-SVPWM**: For four-leg inverters in fault-tolerant applications.
- **Predictive PWM**: Model predictive control-based switching pattern.
- **Variable Frequency PWM**: Adaptive carrier frequency for acoustic noise reduction.
- **Overmodulation Techniques**: For extended voltage range (up to 15% beyond SVPWM).

2.7.8 Selection Criteria

The optimal PWM technique depends on:

- **Power Level**: DPWM preferred for high-power (>50kW), SVPWM for medium power.
- **Switching Frequency**: SVPWM better for high-frequency (>10kHz) operation.
- **Efficiency Requirements**: DPWM for maximum efficiency, SVPWM for balanced performance.
- **Acoustic Noise**: SVPWM or variable frequency PWM for low-noise applications.
- **Processor Capability**: SPWM for low-end controllers, SVPWM/DPWM for advanced processors.

2.7.9 Position/Speed Feedback Devices

Encoders

- Incremental vs. absolute encoders
- Resolution considerations
- Interface protocols (EnDat, BiSS, SSI)

Resolvers

- Robust and reliable
- High-temperature operation capability
- Signal processing requirements
- Resolver-to-Digital conversion

Hall Effect Sensors

- Low-cost positioning solution
- Typically provides 60° electrical resolution
- Integration considerations

2.7.10 Current and Voltage Measurements

Current Sensing Technologies

- Shunt resistors
- Hall-effect current sensors
- Fluxgate sensors
- Rogowski coils

Voltage Sensing

- Resistive dividers
- Isolation requirements
- Common-mode rejection considerations

Controller Tuning

- Current controller bandwidth selection
- Speed controller design considerations
- Anti-windup protection
- Observer tuning for sensorless operation

2.8 Modelling

2.8.1 Current Loop Of PMSM

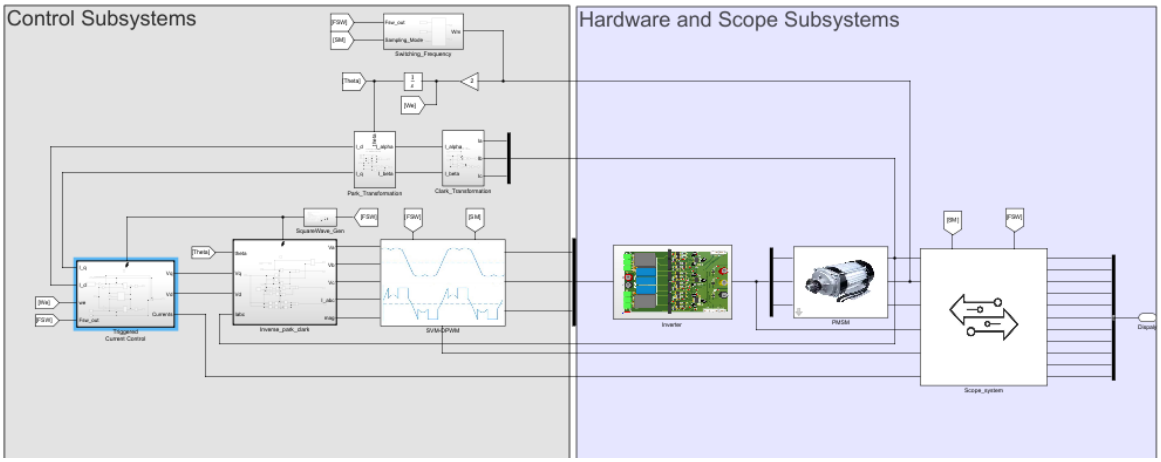


Figure 2.1: Current Loop Of PMSM

The developed system is a comprehensive **Hardware-in-the-Loop (HIL)** simulation platform intended for the real-time control, testing, and validation of advanced electric motor drive systems. Specifically, this architecture targets the control of a **Permanent Magnet Synchronous Motor (PMSM)**—a motor type widely used in high-performance industrial and automotive applications due to its high efficiency, power density, and precise controllability. The system is modeled using **MATLAB/Simulink**, which provides a flexible and powerful environment for implementing control algorithms, simulating real-time processes, and integrating with external hardware.

The platform is composed of two primary functional domains:

- **Control Subsystems:** A software-based environment responsible for implementing the entire control logic, signal processing, and modulation strategies.
- **Hardware and Scope Subsystems:** A virtual or physical representation of the power electronics, motor, and measurement systems, designed to mimic actual hardware behavior and facilitate real-time observation.

This modular architecture supports **rapid prototyping**, **algorithm tuning**, and **hardware integration**, enabling closed-loop control of the motor within a simulated or hybrid environment. It is particularly valuable in validating control performance before deploying on real-time embedded targets.

Current Control Loop

The current control loop, implemented in the `Triggered Current Control` block, serves as the **inner loop** of the FOC architecture. It ensures that the motor currents closely track their respective reference values.

Functional Details:

- **Reference Tracking:** Compares i_d^* and i_q^* with actual i_d and i_q feedback.

- **PI Controllers:** Two independent Proportional-Integral (PI) controllers generate V_d and V_q commands.
- **Feedforward Compensation:** Adds decoupling terms to counteract cross-axis effects and back-EMF.
- **Output:** Voltage commands are passed through inverse Park and Clarke transforms to produce three-phase voltage references (V_a, V_b, V_c).

2.8.2 Mathematical Model of PMSM

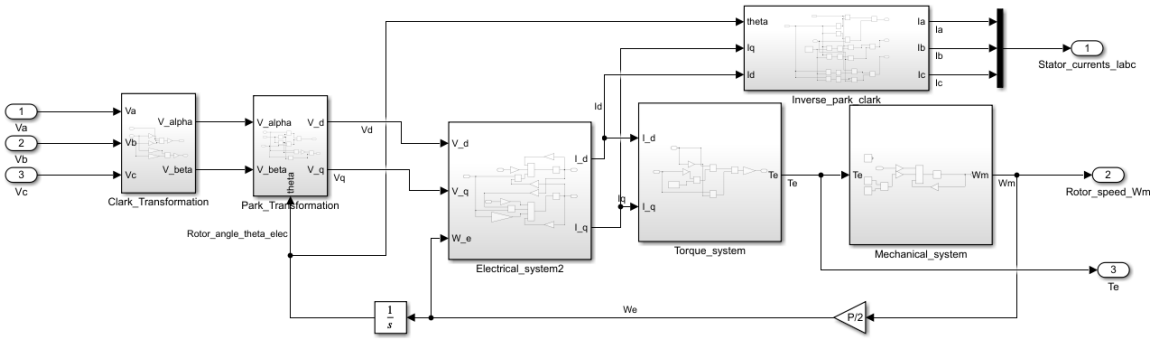


Figure 2.2: Mathematical Model of PMSM

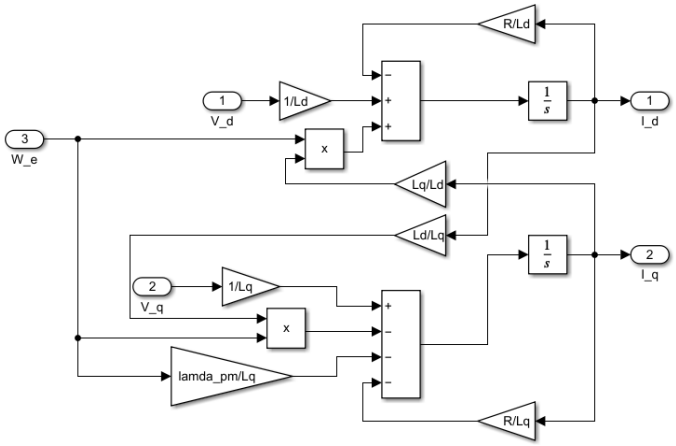


Figure 2.3: Electric System of PMSM

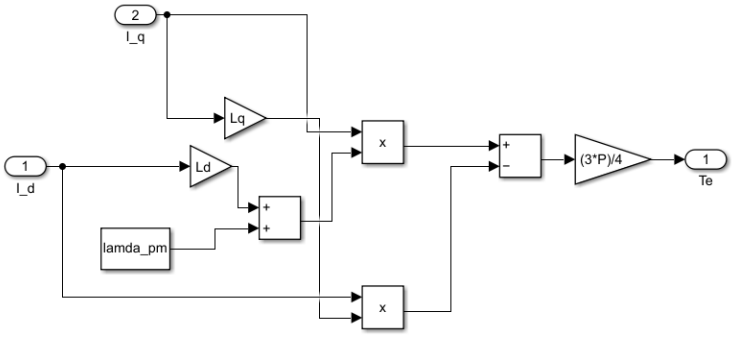


Figure 2.4: Torque System of PMSM

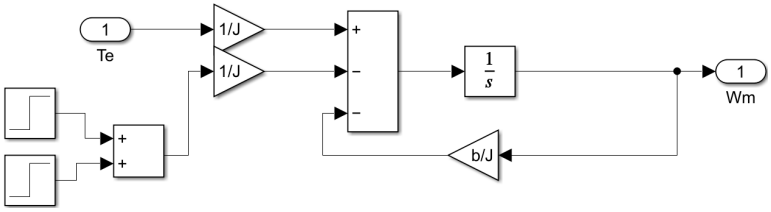


Figure 2.5: Mechanical System of PMSM

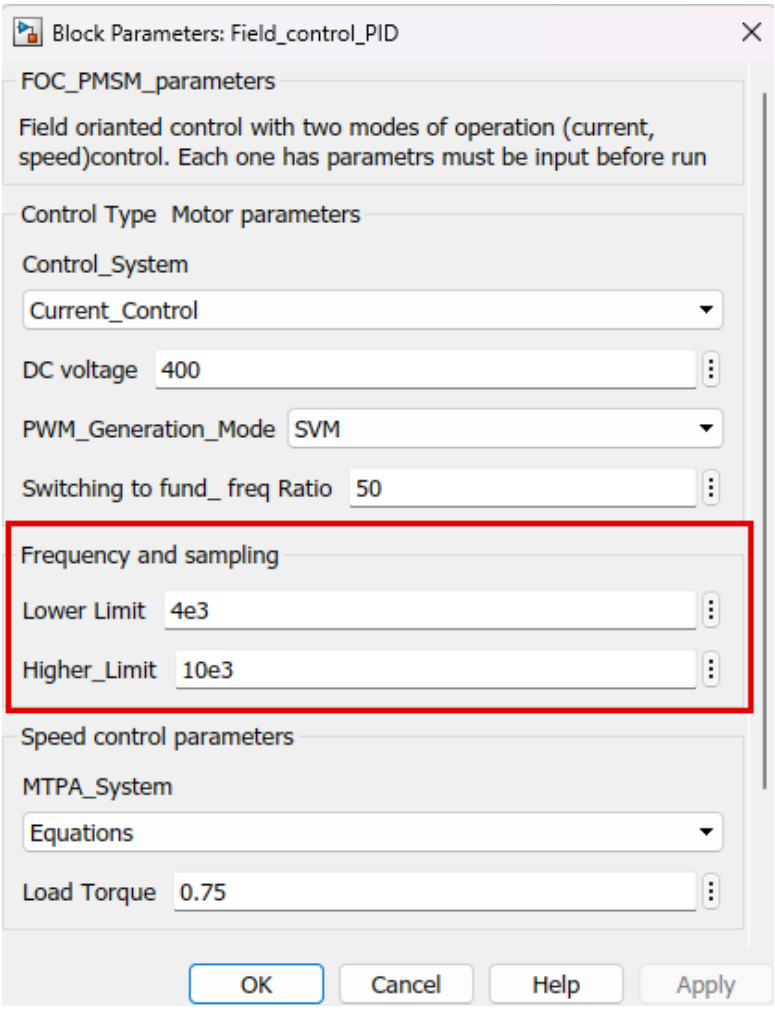


Figure 2.6: Parameters of PMSM

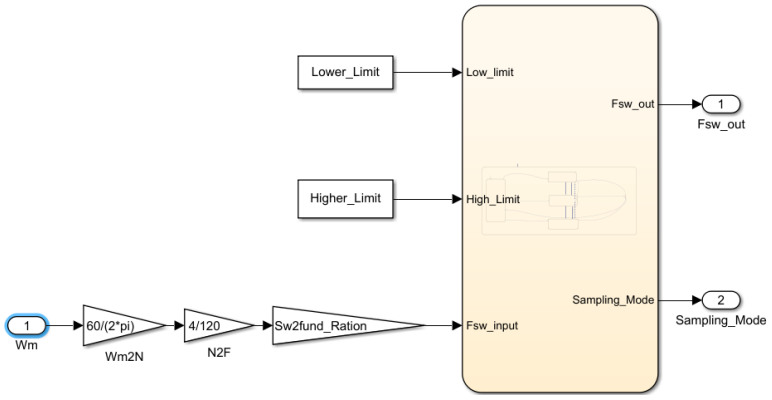


Figure 2.7: Switching Frequency

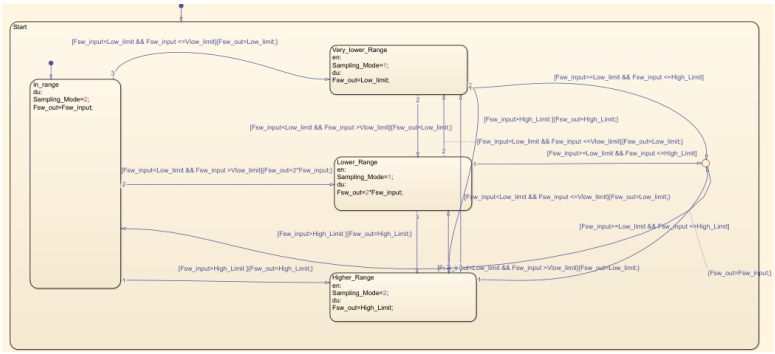


Figure 2.8: Chart of Switching Frequency

advanced adaptive control strategy for dynamically regulating the **switching frequency ($F_{\mathbf{sw}}$)** of a power electronic inverter in a motor drive system. The technique adjusts F_{sw} based on the **fundamental input frequency ($F_{\mathbf{input}}$)**, maintaining optimal efficiency, stability, and electromagnetic performance across wide operating conditions. The control scheme supports both **symmetric** and **asymmetric PWM** switching modes and intelligently transitions between them depending on load dynamics, motor speed, and system constraints.

2.9 Theoretical Foundations and Power Loss Analysis

2.9.1 Switching Losses in Power Devices

The total switching power loss in semiconductor devices (e.g., IGBTs, SiC-MOSFETs) is expressed as:

$$P_{\text{sw}} = \frac{1}{2} V_{\text{dc}} I_{\text{avg}} f_{\text{sw}} (t_{\text{on}} + t_{\text{off}}) \tag{2.30}$$

Where:

- V_{dc}: DC-link voltage
- I_{avg}: Average load current
- f_{sw}: Switching frequency
- $t_{\text{on}}, t_{\text{off}}$: Switching transition times

Insight: Increasing f_{sw} improves control accuracy and harmonic suppression but exponentially increases switching losses.

2.9.2 Motor Speed–Frequency Relationship

The synchronous speed of an AC motor is related to its fundamental frequency by:

$$N_s = \frac{120 f_{\text{input}}}{P} \tag{2.31}$$

Where:

- N_s: Synchronous speed (RPM)
- f_{input}: Fundamental electrical frequency (Hz)
- P: Number of motor poles

Implication: As motor speed varies, the inverter must adjust f_{sw} to maintain performance, minimize torque ripple, and reduce EMI.

2.10 Time Scheduling and Dynamic Model

2.10.1 Defined Limits

- **High Limit:** 10 kHz
- **Low Limit:** 4 kHz
- **Very Low Limit:** 2 kHz

2.10.2 Symmetrical Mode

This mode is employed when the switching frequency (f_{sw}) falls within or above standard operational boundaries. Both the controller and the PWM carrier operate at the same frequency, maintaining synchronization.

Case 1: Medium Frequency (Within Limits)

Condition:

$$\text{Low Limit} < f_{sw} < \text{High Limit}$$

Example: $f_{sw} = 5\,\text{kHz}$

Operation:

- Controller Frequency: f
- Carrier Frequency: f

Remarks:

- Allows flexible switching frequency selection based on application needs.
- Maintains precise symmetry between control loop execution and carrier signal updates.

Case 2: High Frequency (Above Upper Limit)

Condition:

$$f_{sw} \geq \text{High Limit}$$

Example: $f_{sw} = 12\,\text{kHz}$

Operation:

- Controller Frequency: Clamped at 10 kHz
- Carrier Frequency: Clamped at 10 kHz

Remarks:

- Prevents exceeding hardware capabilities or real-time processing constraints.
- Ensures safe and reliable operation within system-defined maximum limits.

Case 3: Low Frequency (Between Low Limit and Very Low Limit)

Condition:

$$\text{Very Low Limit} < f_{sw} < \text{Low Limit}$$

Example: $f_{sw} = 3.5\,\text{kHz}$

Operation:

- Controller Frequency: $2f$
- Carrier Frequency: f_{sw} (e.g., 3.5 kHz)

Remarks:

- Maintains control responsiveness by increasing controller update rate.
- Allows safe operation while preserving control quality.

2.10.3 Asymmetrical Mode

Used when the desired switching frequency (f_{sw}) falls below acceptable lower bounds. In this mode, the controller and carrier frequencies are decoupled to prevent performance degradation.

Case 4: Very Low Frequency (Below Very Low Limit)

Condition:

$$f_{sw} \leq \text{Very Low Limit}$$

Example: $f_{sw} = 2\,\text{kHz}$

Operation:

- Controller Frequency: Clamped at 4 kHz
- Carrier Frequency: Operates at f_{sw} (e.g., 2 kHz)

Remarks:

- Ensures controller operates at a sufficiently high frequency to maintain stability.
- Carrier frequency follows application needs while safeguarding control performance.

2.10.4 Summary Table

Case	Switching Frequency	Controller Frequency	Carrier Frequency	Mode
1	Low Limit $< f_{sw} <$ High Limit	f_{sw}	f_{sw}	Symmetrical
2	$f_{sw} \geq$ High Limit	10 kHz (clamped)	10 kHz (clamped)	Symmetrical
3	Very Low Limit $< f_{sw} <$ Low Limit	$2f_{sw}$	f_{sw}	Symmetrical
4	$f_{sw} \leq$ Very Low Limit	4 kHz (clamped)	f_{sw}	Asymmetrical

Key Points:

- **Symmetrical Mode:** Synchronizes controller and carrier frequencies.
- **Asymmetrical Mode:** Decouples control loop timing from carrier to ensure performance integrity at low frequencies.

2.11 Carrier-Based PWM Generation

A two-level carrier-based Pulse Width Modulation (PWM) generator performs the following steps:

- Samples a reference wave.
- Compares the sampled value to a triangle carrier wave.
- Generates:
  - A switch-on pulse if the reference sample is higher than the carrier signal.
  - A switch-off pulse if the reference sample is lower than the carrier wave.

To determine the switching behavior, the generator can employ different sampling techniques of the triangle wave:

- **Asymmetric Sampling:**
  - Sampling occurs at both the upper and lower boundaries of the carrier wave.
  - The comparison takes place at the intersection following the sampling point.
- **Symmetric Sampling:**
  - Sampling occurs only at the upper boundary of the carrier wave.
  - The comparison is made at the intersection that follows the sampling.

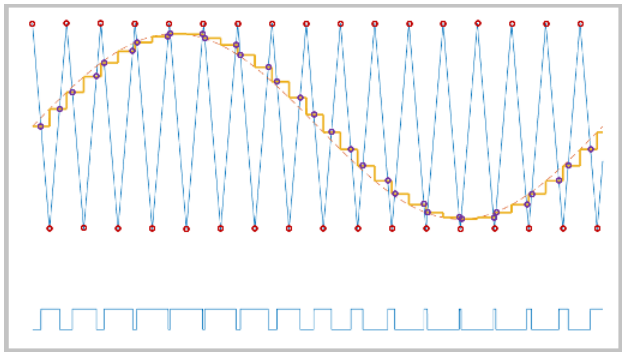


Figure 2.9: Asymmetric Sampling

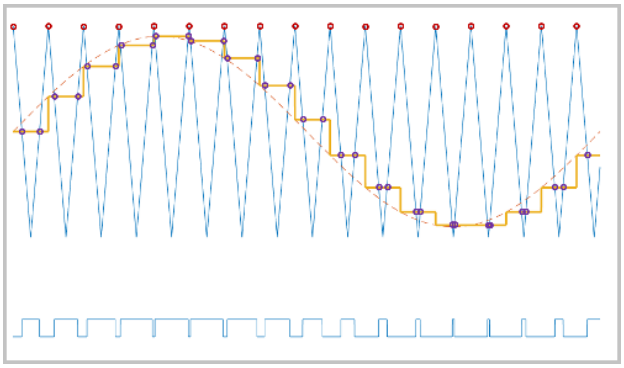


Figure 2.10: Symmetric Sampling

Parameter	Symmetric Mode	Asymmetric Mode
Switching Pattern	Triangular carrier, SVPWM	Randomized, modulated, phase-shifted
Frequency Stability	Constant	Variable, dynamic
Harmonic Content	Narrow-band, predictable	Spread-spectrum, low peaks
Efficiency	High (>95%)	Moderate (85–92%)
Use Case	Mid to high speeds	Low-speed or variable load
Noise Signature	Tonal (audible peaks)	“White noise” profile

Table 2.2: Comparison Between Symmetric and Asymmetric Switching Modes

2.11.1 System Protection and Stability

- **Pulse Skipping:** If current exceeds limit due to low frequency, pulse skipping is applied.
- **Thermal Derating:** Asymmetric mode triggers thermal monitoring due to higher losses.
- **Transition Hysteresis:** Prevents frequent switching between modes near thresholds.

2.12 Current Loop

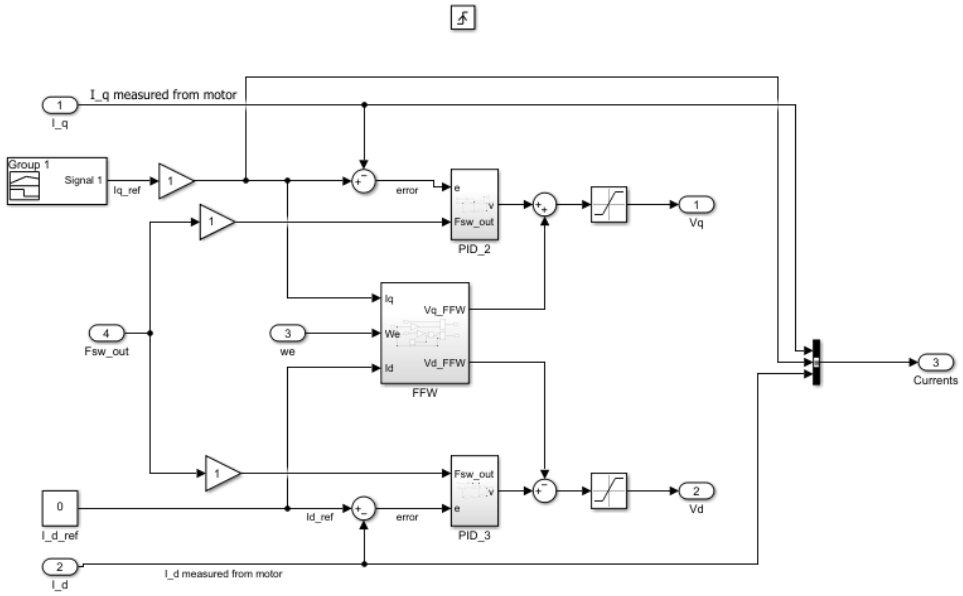
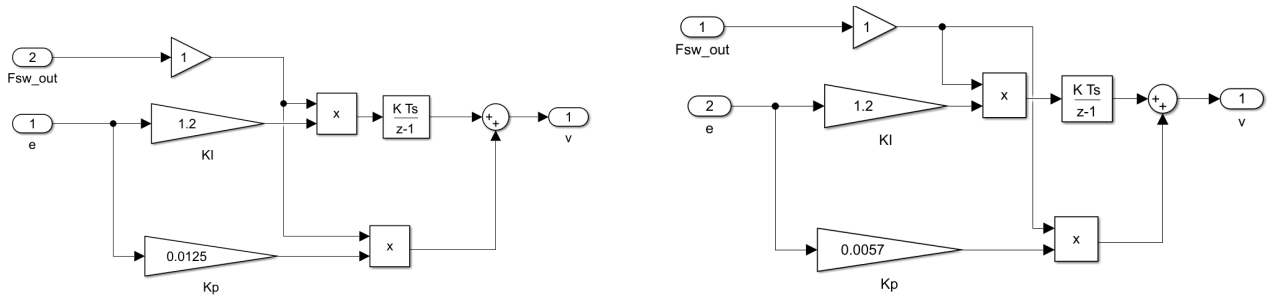


Figure 2.11: Current Loop



(a) PI_q Current Controller (b) PI_d Current Controller

Figure 2.12: Proportional-Integral current controllers for PMSM vector control showing (a) q-axis and (b) d-axis implementations

2.12.1 Magnitude Optimum Tuning Method

The **magnitude optimum** (MO) method is a classical controller tuning strategy used in automatic control systems, particularly in systems with dominant lag behavior such as electrical drives. The objective of this method is to shape the open-loop frequency response such that the magnitude of the system remains as flat as possible over a wide frequency range, thereby ensuring fast response and minimal overshoot.

Control Objective

To maximize the bandwidth and achieve optimal disturbance rejection while ensuring system stability, the controller is designed such that the magnitude of the open-loop transfer function approximates unity up to the desired cutoff frequency.

Typical Plant Model

For many drive systems, especially current control loops of PMSMs, the plant can be approximated as a first-order lag system:

$$G_p(s) = \frac{K}{1 + sT}$$

Where:

- K: DC gain of the plant
- T: Time constant of the system (e.g., $T = \frac{L}{R}$ in current loops)

Controller Structure

To compensate for the lag and flatten the gain, a proportional-integral (PI) controller is often used:

$$G_c(s) = K_p \left(1 + \frac{1}{sT_i}\right)$$

Where:

- K_p: Proportional gain
- T_i: Integral time constant

Magnitude Optimum Tuning Rules

To apply the magnitude optimum principle, the controller parameters are chosen as:

$$T_i = T \quad \text{(Match integral time constant to plant lag)}$$
$$K_p = \frac{1}{K} \cdot \frac{1}{T_f}$$

Where T_f is a desired filter time constant (typically much smaller than T) to shape the high-frequency gain roll-off.

Open-Loop Transfer Function

Substituting controller and plant into open-loop transfer function:

$$L(s) = G_c(s) \cdot G_p(s) = K_p \left(1 + \frac{1}{sT}\right) \cdot \frac{K}{1 + sT}$$

At low frequencies:

$$|L(j\omega)| \approx 1 \quad \text{for } \omega < \omega_c$$

This results in a flat magnitude response, improving tracking and disturbance rejection performance.

Table 2.3: PI Controller Parameters for d-axis and q-axis Current Control

Parameter	**Value**
d-axis Proportional Gain (K_p^d)	$\frac{L_d}{T_f}$
q-axis Proportional Gain (K_p^q)	$\frac{L_q}{T_f}$
d-axis Integral Gain (K_i^d)	$\frac{R}{T_f}$
q-axis Integral Gain (K_i^q)	$\frac{R}{T_f}$
Filter Time Constant (T_f)	$0.1T_e$ (where $T_e = \frac{L_d}{R}$)

Time-Domain Behavior

The MO-tuned controller ensures:

- Fast response time
- Moderate overshoot (typically ¡ 20%)
- Minimal steady-state error (due to integral action)

Application in Motor Drives

In field-oriented control (FOC) of PMSMs, the magnitude optimum method is particularly useful for:

- **Current control loops**: where the electrical dynamics are approximated as a first-order system

- **Speed loops**: after current loop decoupling

It ensures the loop bandwidth is maximized without compromising robustness, which is essential in high-performance servo and drive applications.

Advantages and Limitations

- **Advantages:**
  - Simple tuning based on physical parameters
  - Good disturbance rejection and fast response
  - Suitable for real-time embedded implementation
- **Limitations:**
  - Assumes linear, time-invariant systems
  - Not optimal for systems with strong nonlinearities or delays

2.13 Control System Design Methods Comparison

Magnitude Optimum (MO)

The **Magnitude Optimum (MO)** method is a **frequency-domain design technique** used to maximize the closed-loop bandwidth while maintaining stability. It is particularly effective for **inner control loops**, such as current or torque control in electric drives. The method shapes the open-loop transfer function to achieve a **flat magnitude response near the crossover frequency**, resulting in:

- **Fast dynamic response**
- **Minimal overshoot**
- **Excellent disturbance rejection**, especially when $\omega_{\text{controller}} \gg \frac{1}{\tau_{\text{system}}}$

The MO method offers direct analytical tuning as follows:

$$K_p = \frac{L}{2RT_\Sigma}, \quad K_i = \frac{1}{2T_\Sigma} \tag{2.32}$$

where:

- R: Stator resistance
- L: Stator inductance
- T_Σ: Total delay time constant (plant + controller)

Pole-Zero Cancellation (PZC)

Pole-Zero Cancellation (PZC) is a **time-domain method** that introduces a controller zero to cancel a dominant pole in the plant transfer function. The general controller form is:

$$C(s) = K \cdot \frac{s + p_{\text{plant}}}{s} \tag{2.33}$$

This method offers the following advantages:

- **Cancels dominant plant pole** p_{plant}
- **Simplifies dynamics to first-order**, ensuring fast response
- **Effective for thermal or mechanical systems** with dominant time constants

However, PZC has notable limitations:

- **Highly sensitive to parameter variation**
- **Reduced robustness** if pole estimation error exceeds 5%

Root Locus (RL)

The **Root Locus (RL)** method is a graphical technique used to analyze the trajectory of closed-loop poles in the complex s-plane as the gain K varies. It is based on the characteristic equation:

$$1 + KG(s)H(s) = 0 \tag{2.34}$$

Notable features include:

- **Visual insight** into system stability and pole migration
- **Fine-tuning of transient response**, including:
  - Rise time (t_r)
  - Settling time (t_s)
  - Overshoot (M_p)
- **Best suited for outer-loop control**, such as speed or position regulation

Comparison Summary

Table 2.4: Comparison of Control Design Methods

Criterion	**MO**	**PZC**	**Root Locus**
Design Domain	Frequency	Time	Graphical (s-domain)
Implementation Effort	Low	Medium	High
Robustness to Variation	High	Low	Medium
Best Application	Current Control	Speed/Thermal Control	Position Control
Model Dependency	Moderate	High	Moderate
Controller Type	PI	Lead or PI	PI, PID, or State-Space
Visualization Tool	Bode Plot	Analytical Solution	s-Plane Plot

2.14 Feed-Forward Equations

The feed-forward compensation block diagram implements the following equations:

2.14.1 d-axis Feed-Forward Voltage

$$V_{d_FFW} = I_q \cdot \omega_e \cdot L_q \tag{2.35}$$

2.14.2 q-axis Feed-Forward Voltage

$$V_{q_FFW} = \omega_e \cdot L_d \cdot I_d + L_d \cdot \frac{dI_d}{dt} + \Delta_{pm} \tag{2.36}$$

where:

$$\omega_e = \text{electrical frequency (rad/s)} \tag{2.37}$$
$$I_q, I_d = \text{q-axis and d-axis currents (A)} \tag{2.38}$$
$$L_q, L_d = \text{q-axis and d-axis inductances (H)} \tag{2.39}$$
$$\Delta_{pm} = \text{permanent magnet flux linkage (Wb)} \tag{2.40}$$
$$\frac{dI_d}{dt} = \text{rate of change of d-axis current} \tag{2.41}$$

From the block diagram:

$$L_q = 0.0125 \text{ H} \tag{2.42}$$
$$L_d = 0.0057 \text{ H} \tag{2.43}$$
$$\Delta_{pm} = 0.123 \text{ Wb} \tag{2.44}$$

2.15 Importance of Feed-Forward Control

2.15.1 System Decoupling

Feed-forward control eliminates cross-coupling between d and q axes in synchronous motors, where d-axis current affects q-axis voltage and vice versa through the coupling terms $\omega_e L_q I_q$ and $\omega_e L_d I_d$.

2.15.2 Dynamic Response Improvement

- **Fast dynamic response**
- **Minimal overshoot**
- **Strong disturbance rejection**, especially when $\omega_c \gg 1/\tau$ (controller bandwidth ω_c much greater than inverse system time constant)

2.15.3 Compensation Mechanisms

The feed-forward controller provides immediate compensation for:

1. **Back-EMF compensation**: Speed-dependent voltage drops
2. **Inductive voltage drops**: $L\frac{di}{dt}$ terms
3. **Cross-coupling effects**: Inter-axis magnetic coupling
4. **Permanent magnet flux**: Constant flux linkage effects

2.15.4 Performance Benefits

- **Reduced tracking error**: Minimizes steady-state error by compensating for predictable voltage requirements
- **Enhanced current regulation**: Enables more precise current control

- **Improved torque control**: Better current regulation translates to superior torque response
- **Reduced controller burden**: Allows feedback PI controllers to focus on unknown disturbances rather than predictable dynamics

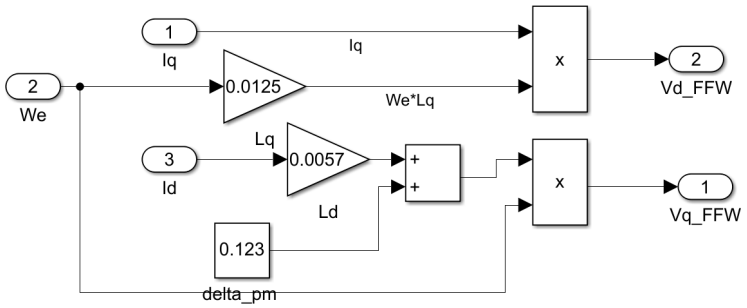


Figure 2.13: Feed Forward

2.16 Square Wave Triggering for Current Control Loops

Using a square wave generator as a **trigger input** for current control loops is a fundamental strategy in power electronics and motor drive applications. It enables deterministic, noise-resilient, and synchronized control, which is essential for maintaining system performance and stability.

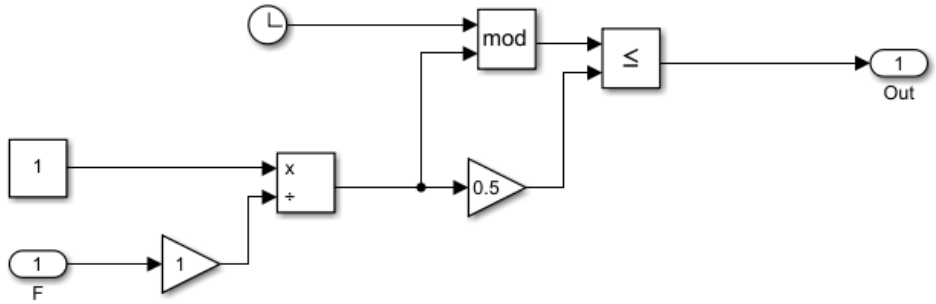


Figure 2.14: Square Wave

Primary Functions of Square Wave Triggering

1. Synchronization and Timing Control

- **PWM Synchronization:** Ensures that the current control loop operates in sync with the PWM switching frequency.
- **Sample-and-Hold Timing:** Triggers current sampling at optimal points (e.g., PWM peak or valley) to ensure accuracy.
- **Control Loop Scheduling:** Provides fixed, regular intervals for executing current control algorithms.

2. Switching Noise Avoidance

- **Optimal Sampling Points:** Sampling occurs when switching transients are minimal.
- **EMI Reduction:** Aligns control actions with switching states to minimize electromagnetic interference.
- **Clean Sensing:** Prevents current measurement during MOSFET/IGBT switching edges.

3. Multi-Phase Coordination

- **Phase Synchronization:** Aligns multiple current controllers in three-phase systems.
- **Interleaved Operation:** Supports coordinated switching across parallel converter legs.
- **Dead-Time Awareness:** Ensures control actions respect dead-time intervals to prevent shoot-through.

4. Real-Time Execution

- **Deterministic Timing:** Guarantees fixed-time execution of control loops for predictable system behavior.
- **Interrupt-Based Control:** Rising or falling edges of the square wave trigger real-time ISR (Interrupt Service Routine) execution.
- **Stable Loop Dynamics:** Regular triggering contributes to consistent control system response.

Importance of 50% Duty Cycle

- **Symmetry:** Provides equal high and low periods for balanced sampling in positive and negative current directions.

- **Consistent Timing:** Ensures uniform controller response in both directions of bidirectional current control.
- **Harmonic Suppression:** Reduces generation of even-order harmonics in control signals.

Technical Benefits

- **Improved Current Regulation:** Accurate sampling leads to enhanced current tracking and control precision.
- **Ripple Reduction:** Synchronized actions minimize current ripple and power losses.
- **Enhanced Stability:** Predictable timing improves phase margin and overall loop robustness.
- **Lower Total Harmonic Distortion (THD):** Cleaner current waveforms improve efficiency and reduce filter requirements.

2.17 Modulation Techniques

Discontinuous PWM (DPWM)

Power conversion systems commonly employ switched-mode approaches where a critical trade-off exists between switching frequency and switching losses. This trade-off is significantly influenced by the characteristics of the chosen PWM method. Unlike continuous PWM, where the pulse width is modulated throughout the entire switching cycle, Discontinuous PWM introduces breaks or gaps in the modulation process. These breaks are intentionally designed to:

- Reduce switching losses
- Improve efficiency
- Meet specific harmonic distortion requirements

The term "Discontinuous PWM" is coined due to the utilization of modulating signals characterized by their discontinuous nature over time. This unique characteristic is achieved through zero-sequence voltage injection.

2.17.1 Theoretical Foundation

2.17.2 DC-Bus Voltage Limitations

While producing sinusoidal phase voltage with a two-level inverter, the peak amplitude is limited by the DC-bus voltage. Each leg of the inverter can produce a leg voltage of $\pm V_{dc}/2$, corresponding to a relative amplitude of the output voltage (modulation index) of $m = \sqrt{3}/2 \approx 0.866$.

In Sinusoidal Pulse Width Modulation (SPWM), the output voltage is clipped for $m > 0.866$, and the converter enters the overmodulation region.

2.17.3 Zero-Sequence Voltage Injection Principle

Discontinuous PWM is a carrier-based method relying on the fundamental principle of saturating the reference voltage signals for 120° within a 360° cycle. This intentionally keeps one

of the three converter legs without commutation by injecting a zero-sequence common-mode voltage v_0 to the phase voltages.

The zero-sequence injection method adds a rectangular zero-sequence component at triple the fundamental frequency to the sinusoidal reference, which distorts the phase voltage references. This reduces the peak voltage of the reference compared to a pure sinusoidal reference, allowing the modulation index to be increased up to $m = 2/\sqrt{3} \approx 1.155$ before hitting the DC-bus limit.

2.17.4 Zero-Sequence Signal Generation

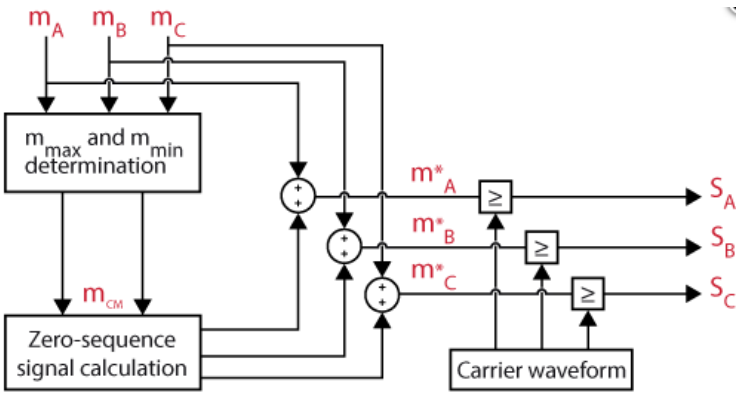


Figure 2.15: zero sequence

After normalizing the output reference voltage, three-phase sinusoidal modulation signals are generated:

$$v_a^* = m\sin(\omega t) \tag{2.45}$$
$$v_b^* = m\sin(\omega t - 2\pi/3) \tag{2.46}$$
$$v_c^* = m\sin(\omega t + 2\pi/3) \tag{2.47}$$

To clamp one of the phases to the positive or negative DC-bus, a common-mode signal is computed:

$$v_0 = -\frac{\max(v_a^*, v_b^*, v_c^*) + \min(v_a^*, v_b^*, v_c^*)}{2} \tag{2.48}$$

The injection of the common-mode signal into the initial modulation signals results in the altered modulating signals:

$$v_a^{**} = v_a^* + v_0 \tag{2.49}$$
$$v_b^{**} = v_b^* + v_0 \tag{2.50}$$
$$v_c^{**} = v_c^* + v_0 \tag{2.51}$$

The resulting modulating signals are then compared with carrier waveforms to generate the gate signals S_a, S_b, and S_c.

2.17.5 Classification of Discontinuous PWM Methods

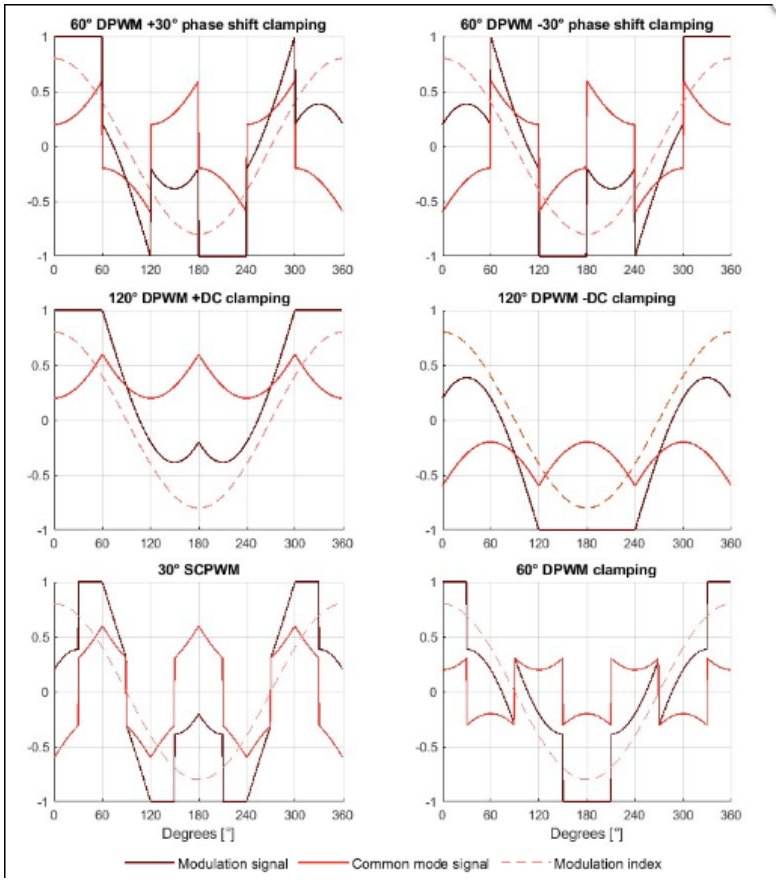


Figure 2.16: DPWM

2.17.6 Discontinuous PWM (DPWM)

Discontinuous PWM (DPWM) is a class of advanced PWM techniques that modify the standard modulation signals to intentionally create regions where certain switches remain continuously ON or OFF for a portion of the cycle. This eliminates unnecessary switching, thus reducing switching losses and improving overall inverter efficiency.

Basic Concept

Unlike Sinusoidal PWM (SPWM), where all switches toggle every cycle, DPWM "clamps" one of the inverter legs to the DC rail (either V_{dc}^{+} or V_{dc}^{-}) during specific intervals.

Types of DPWM Strategies

Several DPWM variants exist, based on which phase is clamped and during which interval:

- **DPWM0**: Clamping occurs at the bottom (minimum) of the sine wave.
- **DPWM1**: Clamping occurs at the top (maximum) of the sine wave.
- **DPWMMIN**: Clamps the phase with the minimum reference voltage.
- **DPWMMAX**: Clamps the phase with the maximum reference voltage.
- **DPWM3 or Third Harmonic Injection PWM (THIPWM)**: Adds a third harmonic component to flatten and extend modulation.

Advantages of DPWM

- **Reduced Switching Losses**: Because one leg remains unswitched during clamped intervals.
- **Lower EMI**: Due to fewer transitions.
- **Improved Efficiency**: Especially under light-load or low-power-factor conditions.
- **Flexible Control**: Can be tailored depending on power factor and modulation index.

Relation to Power Factor and Modulation Index

- At **low power factor** (leading or lagging), careful DPWM design helps avoid distortion near zero-crossings.
- At high modulation index ($M_a > 0.8$), DPWM helps reduce overmodulation effects by clamping the waveform near peaks.
- The **clamping strategy** may be selected dynamically based on load conditions and desired waveform shape.

Space Vector Modulation (SVM)

Introduction

Space Vector Modulation (SVM) is a modern pulse-width modulation (PWM) technique used for controlling Voltage Source Inverters (VSIs). It offers efficient synthesis of sinusoidal output voltages using the available DC bus voltage. Unlike carrier-based PWM, which operates in the three-phase (abc) domain, SVM functions in the stationary $\alpha\beta$ (Clarke) reference frame.

Active and Zero Space Vectors

For a two-level three-phase inverter, there are eight possible switching states that correspond to eight space vectors in the $\alpha\beta$ plane:

- **Six active vectors** (V_1 to V_6): These generate non-zero phase voltages.
- **Two zero vectors** (V_0, V_7): These result in zero phase voltage.

Each inverter leg can be either ON (1) or OFF (0). To avoid shoot-through, the upper and lower switches of a leg must operate in a complementary manner, leading to only eight valid switching states.

Table 2.5: Switching States and Corresponding Space Vectors

Vector	A	B	C	v_a	v_b	v_c
V_0	0	0	0	0	0	0
V_1	1	0	0	$\frac{2V_{DC}}{3}$	$-\frac{V_{DC}}{3}$	$-\frac{V_{DC}}{3}$
V_2	1	1	0	$\frac{V_{DC}}{3}$	$\frac{V_{DC}}{3}$	$-\frac{2V_{DC}}{3}$
V_3	0	1	0	$\frac{V_{DC}}{3}$	$\frac{2V_{DC}}{3}$	$-\frac{V_{DC}}{3}$
V_4	0	1	1	$-\frac{2V_{DC}}{3}$	$\frac{V_{DC}}{3}$	$\frac{V_{DC}}{3}$
V_5	0	0	1	$-\frac{V_{DC}}{3}$	$-\frac{V_{DC}}{3}$	$\frac{2V_{DC}}{3}$
V_6	1	0	1	$\frac{V_{DC}}{3}$	$-\frac{2V_{DC}}{3}$	$\frac{V_{DC}}{3}$
V_7	1	1	1	0	0	0

Voltage Synthesis Using SVM

Although only eight vectors are available, SVM can synthesize any desired reference vector $\vec{V}_{\text{ref}}$ within the modulation hexagon by a time-weighted average of:

- Two adjacent active vectors
- One or both zero vectors

The modulation period T_s is divided into three time slots:

$$\vec{V}_{\text{ref}} \cdot T_s = \vec{V}_1 \cdot T_1 + \vec{V}_2 \cdot T_2 + \vec{V}_0 \cdot T_0$$

where T_1, T_2, and T_0 are the dwell times for each vector.

The $\alpha\beta$ plane is divided into six sectors of 60° each. In each sector, a pair of adjacent active vectors and zero vectors is used to construct the reference vector.

Optimal Switching Patterns

To minimize switching losses, a symmetric switching sequence is applied within each modulation period T_s:

$$[\text{Zero}] \rightarrow \text{Active 1} \rightarrow \text{Active 2} \rightarrow \text{Zero}$$

Two common strategies:

- **Optimal**: Reduces the number of switch transitions.
- **Sub-optimal**: Easier to implement but with higher switching losses.

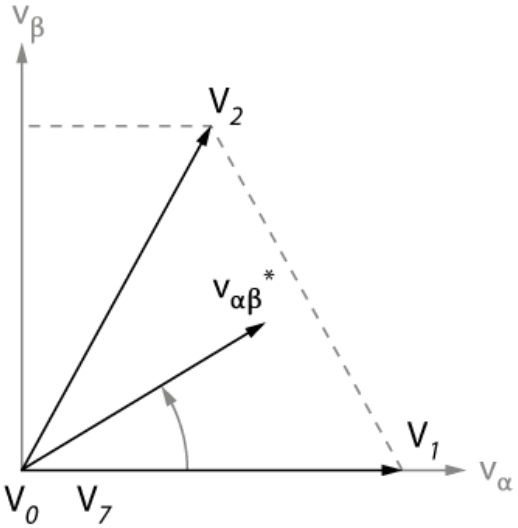


Figure 2.17: Reference vector synthesis using active and zero vectors (V1, V2, V0/V7).

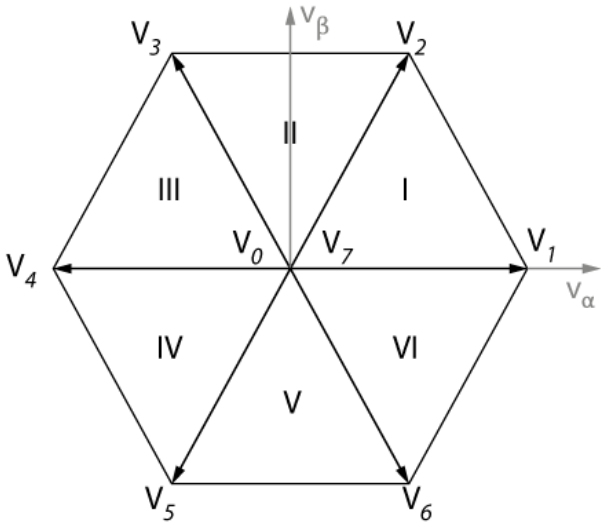


Figure 2.18: Space vector hexagon division into six sectors in $\alpha\beta$ frame.

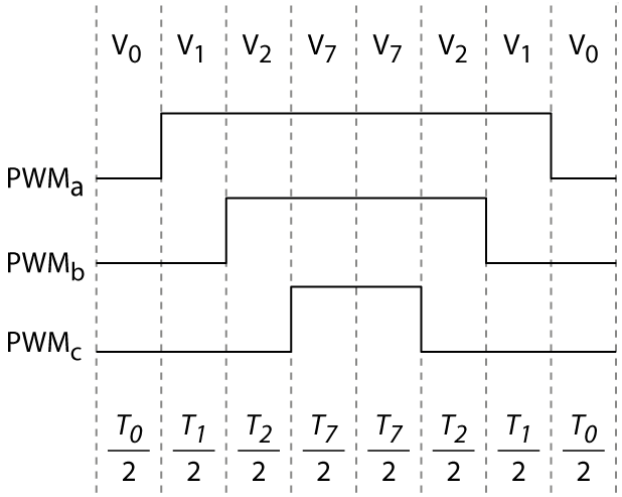


Figure 2.19: Optimal switching pattern

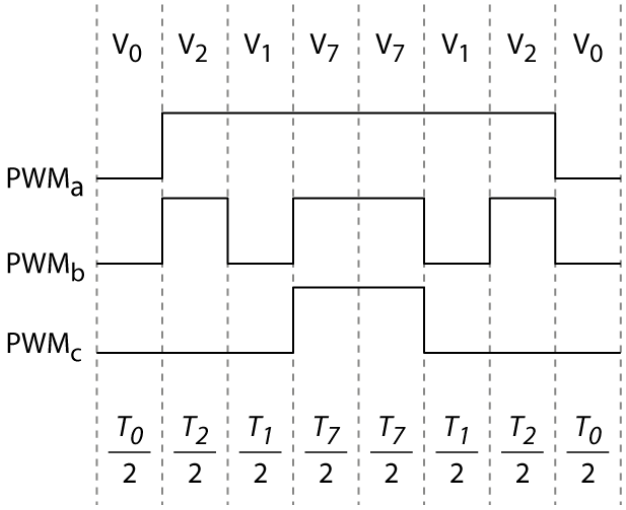


Figure 2.20: Sub-optimal switching patterns

SVM for Three-Level Inverters

SVM can be extended to multilevel topologies such as the Neutral Point Clamped (NPC) inverter. Each inverter leg can now take three states: Positive (P), Zero (0), and Negative (N), resulting in $3^3 = 27$ switching states.

- **Higher voltage resolution** and smoother waveforms
- **Lower Total Harmonic Distortion (THD)**
- **Reduced filter requirements**

A hexagonal coordinate system is used to select the optimal combination of vectors. The basic control principle remains the same: alternate between vectors to synthesize the desired voltage with minimal switching.

Comparative Analysis

Table 2.6: Comparison of SPWM and DPWM Techniques

Parameter	**SPWM**	**DPWM**
Max. Modulation Index	0.866	1.155
DC-Bus Utilization	86.6%	90.7%
Switching Losses	Higher	Lower
Harmonic Performance	Good	Excellent (at high m)
Implementation	Simple	Moderate

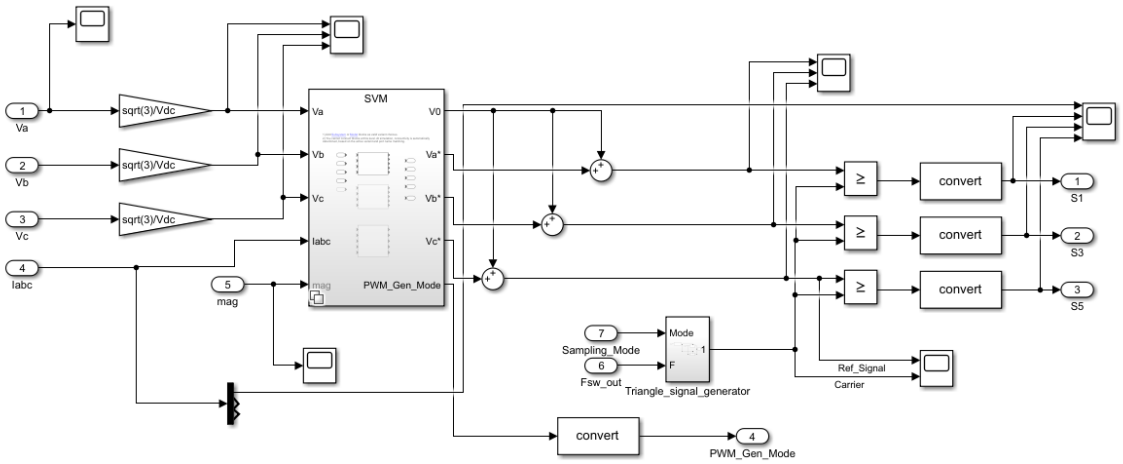


Figure 2.21: SVM-DPWM

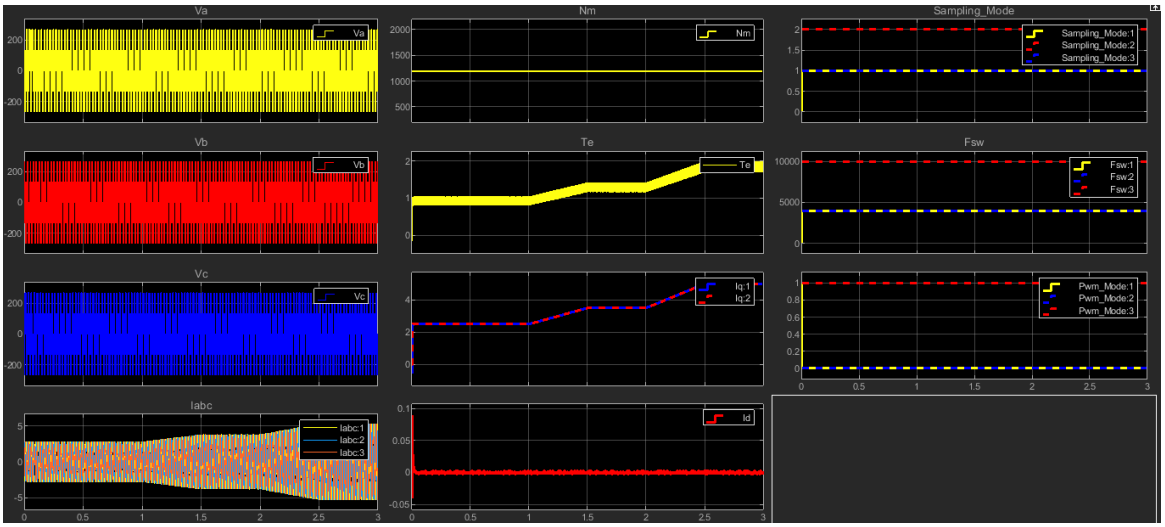


Figure 2.22: Output From Current Loop

2.18 Speed Loop

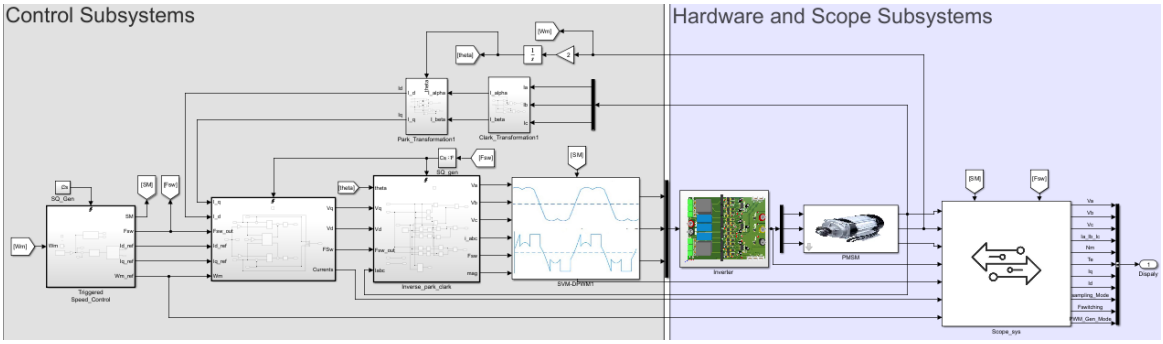


Figure 2.23: Speed and Current Loop

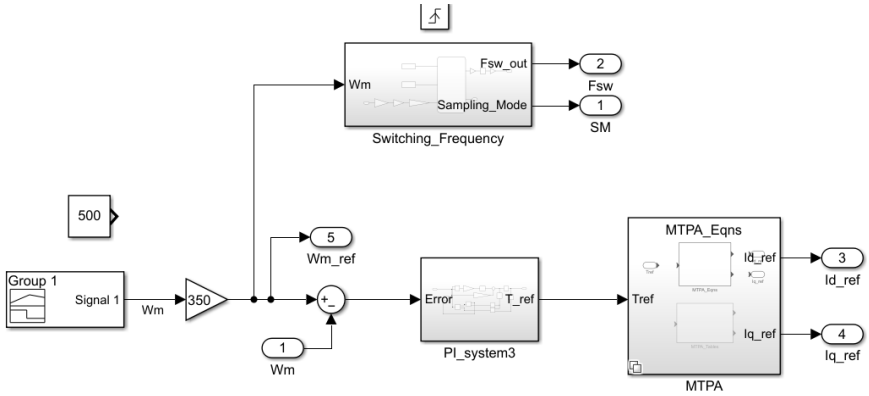


Figure 2.24: Speed Loop

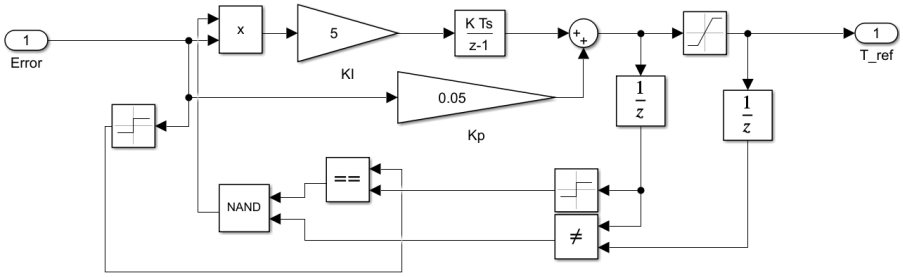


Figure 2.25: PI Speed Loop

2.19 Anti-Windup Mechanism for PI Controller

2.19.1 Introduction

In control systems, a PI (Proportional-Integral) controller is widely used to regulate system behavior. However, when the actuator reaches its physical limits (e.g., torque saturation in a motor), the integral term may continue to accumulate error, leading to what is known as **integral windup**. This causes overshoot or sluggish response in the closed-loop system.

To prevent this, an **anti-windup clamping mechanism** is implemented. It detects when the actuator is saturated and temporarily disables the integration process until the error is corrected and the output is back within allowable limits.

2.19.2 PI Controller Overview

The discrete-time PI control law is given by:

$$u[k] = K_p \cdot e[k] + K_i \sum_{i=0}^{k} e[i] \cdot T_s$$

Where:

- $u[k]$: Controller output at sample k
- $e[k]$: Error signal at sample k
- K_p: Proportional gain
- K_i: Integral gain
- T_s: Sampling time

The integral term in discrete form is implemented as:

$$\text{Integral}[k] = \text{Integral}[k-1] + K_i \cdot e[k] \cdot T_s$$

2.19.3 Anti-Windup Clamping Logic

Saturation Detection

The PI output is passed through a saturation block that limits $u[k]$ within the actuator's allowable range:

$$-1 \leq u[k] \leq 1$$

Clamping Conditions

To determine whether the integrator should be active, two conditions are evaluated:

Condition A (Saturation Detection):

$$u_{\text{saturated}}[k] \neq u_{\text{unsaturated}}[k]$$

Condition B (Error Change Detection):

$$e[k] \neq e[k-1]$$

Clamping Decision: These two conditions are combined using a NAND gate:

Saturation Occurred	Error Changing	Integrator Enabled
Yes	Yes	No (Clamp)
Yes	No	Yes
No	Yes	Yes
No	No	Yes

If both saturation has occurred and the error is changing, the integrator is disabled to avoid windup. Otherwise, integration proceeds normally.

2.19.4 Implementation Details

Signal Flow Summary

- **Error Signal** $e[k]$ is split into two paths:
  - Proportional Path: $K_p = 0.05$
  - Integral Path: $K_i = 5$
- **Integrator:**
$$\text{Integral}[k] = \text{Integral}[k-1] + K_i \cdot e[k] \cdot T_s$$
- **Saturation Block:**
$$u[k] = \text{sat}(u_{\text{PI}}[k])$$
- **Saturation Detection:** Compare $u[k]$ vs. $u_{\text{PI}}[k]$
- **Error Memory:** Use a delay block to obtain $e[k-1]$
- **NAND Logic:** Generate enable/disable signal for the integrator

2.19.5 Importance of Anti-Windup

Without Anti-Windup

- Integrator continues accumulating error even when output is saturated
- Stored integrator value causes large overshoot when actuator recovers
- System becomes sluggish or unstable

With Anti-Windup

- Integration is paused during saturation
- Controller resumes normal operation once back in range
- System exhibits improved stability, faster recovery, and reduced overshoot

2.19.6 Conclusion

The anti-windup clamping strategy effectively prevents integrator windup by:

- Monitoring actuator saturation
- Detecting dynamic changes in the error signal
- Disabling integration when both saturation and changing error are detected

This approach improves system stability and transient performance, making it essential in high-performance applications such as motor drives and process control systems.

2.20 Maximum Torque Per Ampere (MTPA)

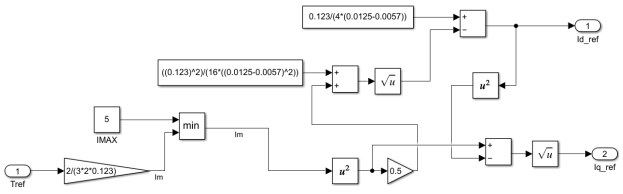


Figure 2.26: MTPA control implementation in FOC loop in Equations

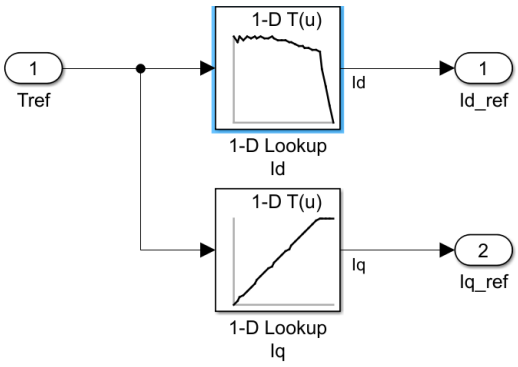


Figure 2.27: MTPA control implementation in FOC loop in Table

2.20.1 Fundamental Concept

The **Maximum Torque Per Ampere (MTPA)** strategy is an optimization method used in Field-Oriented Control (FOC) of Permanent Magnet Synchronous Motors (PMSMs). The goal is to maximize the torque output for a given stator current magnitude, thereby improving the overall system efficiency and minimizing copper losses.

2.20.2 Torque Equation and Optimization Condition

The electromagnetic torque T_e of an interior PMSM is expressed as:

$$T_e = \frac{3}{2} P \left(\lambda_{pm} i_q + (L_d - L_q) i_d i_q \right) \tag{2.52}$$

Where:

- P — Number of pole pairs
- λ_{pm} — Permanent magnet flux linkage (Wb)
- i_d, i_q — d- and q-axis currents (A)
- L_d, L_q — d- and q-axis inductances (H)

The current magnitude constraint is:

$$i_s^2 = i_d^2 + i_q^2 \tag{2.53}$$

To find the MTPA condition, we maximize torque T_e with respect to i_d under a constant i_s:

$$\frac{\partial T_e}{\partial i_d} = 0 \tag{2.54}$$

This leads to the MTPA current trajectory:

$$i_d = \frac{\lambda_{pm}}{L_q - L_d} - \sqrt{\left(\frac{\lambda_{pm}}{L_q - L_d} \right)^2 + i_q^2} \tag{2.55}$$

Note: Valid only for $L_q > L_d$, which holds for interior PMSMs (IPMSM). For surface-mounted PMSMs (SPMSM) where $L_q = L_d$, the optimal $i_d = 0$.

2.20.3 Benefits of MTPA Control

- **Efficiency Optimization:** Minimizes I^2R losses in the stator windings.
- **Torque Maximization:** Achieves higher torque output for the same current.
- **Reduced Heat Generation:** Lower copper losses translate to better thermal management.
- **Extended Operating Range:** Maximizes torque capability under current constraints.

2.20.4 Numerical Example from System Parameters

Assuming:

- $\lambda_{pm} = 0.123$ Wb
- $L_d = 5.7$ mH, $L_q = 12.5$ mH
- $i_s = 6.4$ A (total current magnitude)

We compute:

$$i_d^{\text{opt}} \approx \frac{0.123}{0.0125 - 0.0057} - \sqrt{\left(\frac{0.123}{0.0125 - 0.0057}\right)^2 + i_q^2} \tag{2.56}$$

Solving iteratively with the constraint $i_d^2 + i_q^2 = i_s^2$, we get:

$$i_d^{\text{opt}} \approx -4.52 \text{ A}, \quad i_q^{\text{opt}} \approx 4.52 \text{ A} \tag{2.57}$$

2.20.5 Implementation in Control System

1. Torque command is input to the MTPA block.
2. MTPA logic outputs optimal i_d^{ref}, i_q^{ref}.
3. These references are tracked by the inner current controllers.
4. Space Vector PWM (SVPWM) generates the required gate signals.

2.20.6 Saliency Ratio and Control Implications

The saliency ratio affects MTPA performance:

$$\rho = \frac{L_q - L_d}{L_d} \tag{2.58}$$

Higher ρ implies:

- Greater deviation from circular current locus
- Higher potential torque boost via MTPA

2.20.7 MTPA vs. Conventional Control

Table 2.7: Comparison of MTPA and Standard FOC

Feature	**MTPA**	**Standard (e.g. $i_d = 0$)**
Torque per Ampere	High (Optimal)	Lower
Stator Losses	Reduced	Higher
Flux Utilization	Better	Limited
Implementation	Complex (Lookup/Real-time Calc)	Simpler
Sensitivity to Parameters	High	Moderate

2.21 MTPA control implementation in FOC loop in Table

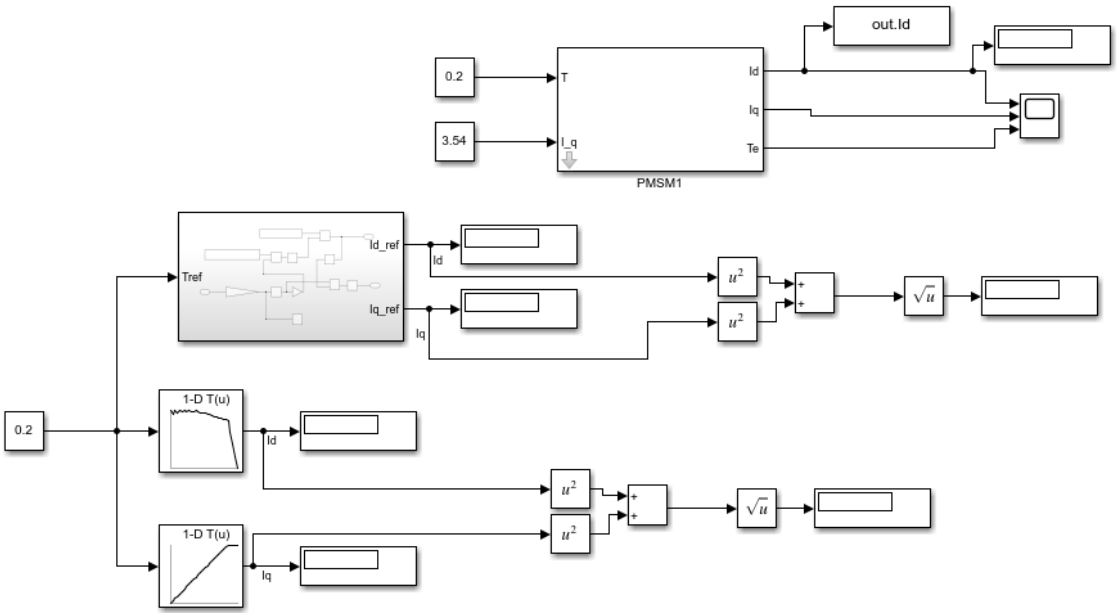


Figure 2.28: MTPA Calculations

To evaluate the motor's electromagnetic torque generation under varying load conditions, a systematic current-based analysis was conducted. For each sampling instant, corresponding values of the direct-axis current i_d and quadrature-axis current i_q were recorded. Using these, the stator current magnitude i_m was calculated according to the relation:

$$i_m = \sqrt{i_d^2 + i_q^2} \tag{2.59}$$

This metric serves as a key indicator of the total stator current effort required to produce a given torque. By analyzing the dataset over the complete operating range, the torque output was mapped against i_q variations, which predominantly influences torque in Permanent Magnet Synchronous Motors (PMSMs), especially under the $i_d = 0$ condition (as in standard FOC) or when negative i_d is used for MTPA operation. From the computed torque values, the maximum torque point was identified, and the corresponding optimal set of i_d, i_q, and i_m was extracted. This point represents the best operating condition in terms of torque-per-ampere ratio, and is

critical for efficient control design. This analysis aids in validating control strategies such as Maximum Torque Per Ampere (MTPA) and highlights how current vector orientation directly impacts torque production efficiency in the field-oriented control framework.

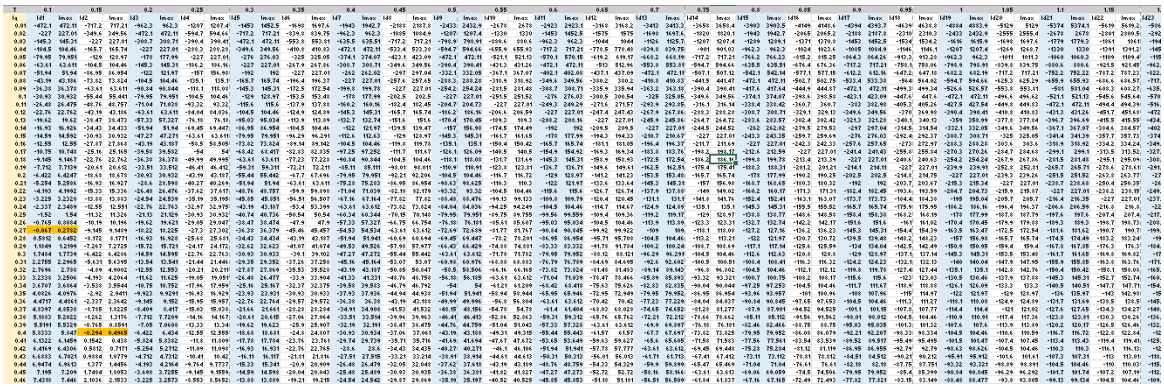


Figure 2.29: Excel Sheet

Table 2.8: Torque vs. i_q, i_d, and Resultant Stator Current Magnitude i_m

Torque T **(Nm)**	i_q **(A)**	i_d **(A)**	$i_m = \sqrt{i_q^2 + i_d^2}$ **(A)**
0.10	0.270	-0.06718	0.27823
0.15	0.4	-0.29412	0.49649
0.20	0.540	-0.06718	0.54416
0.25	0.670	-0.20266	0.69998
0.30	0.810	-0.06718	0.81278
0.35	0.940	-0.16375	0.95416
0.40	1.080	-0.06718	1.08209
0.45	1.210	-0.14220	1.21832
0.50	1.350	-0.06718	1.35167
0.55	1.480	-0.1285	1.48556
0.60	1.485	-0.12851	1.49055
0.70	1.610	-0.17994	1.62002
0.75	1.750	0.11905	1.75404
0.80	1.880	-0.16375	1.88712
0.85	2.010	-0.20266	2.02019
0.90	2.140	-0.23680	2.15306
0.95	2.270	-0.26712	2.28566
1.00	2.650	-0.40900	2.68138
1.05	2.780	-0.42600	2.81245
1.10	2.910	-0.44151	2.94330
1.15	3.030	-0.51657	3.07372
1.20	3.160	-0.52600	3.20347
1.25	3.280	-0.59302	3.33331
1.30	3.410	-0.59900	3.46221
1.35	3.530	-0.65800	3.59080
1.40	3.540	-1.29000	3.76772
1.45	3.540	-1.99000	4.06100
1.50	3.540	-2.68000	4.44005
1.55	3.540	-3.37000	4.88759

2.22 Output

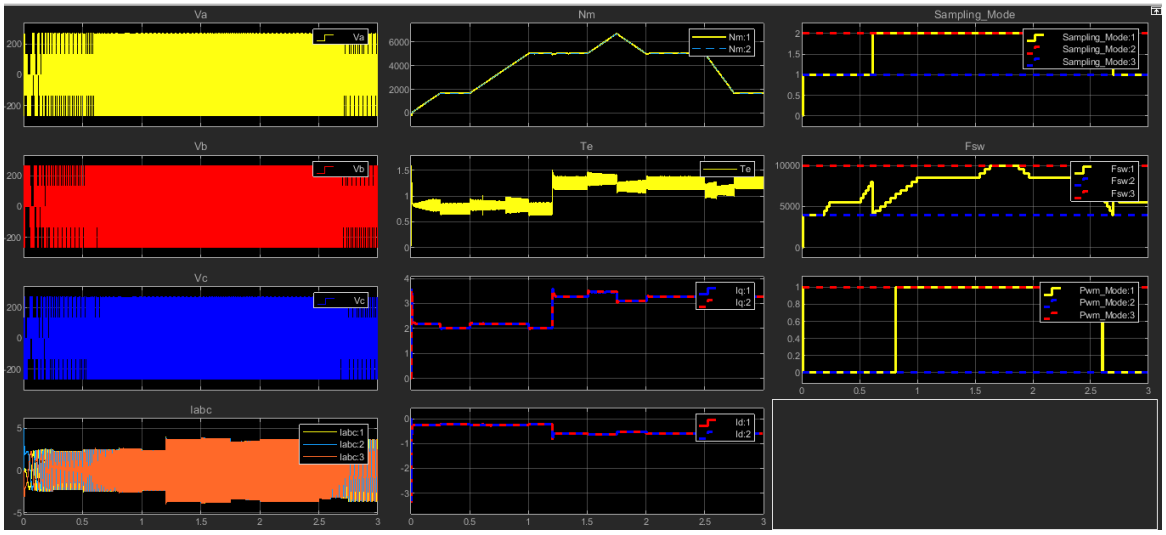


Figure 2.30: Output From Speed Loop

2.23 Model Predictive Control (MPC)

2.23.1 Introduction

Model Predictive Control (MPC) is an advanced control technique that has attracted increasing interest in the control of electrical machines, particularly Permanent Magnet Synchronous Motors (PMSMs). Owing to its ability to incorporate system constraints, handle nonlinearities, and optimize performance over a prediction horizon, MPC offers significant advantages over traditional control strategies such as Field-Oriented Control (FOC). This chapter provides a comprehensive discussion on the implementation, benefits, and challenges of applying MPC to PMSM drives, suitable for high-performance industrial and transportation applications.

Fundamentals of Model Predictive Control

2.23.2 Concept of MPC

Model Predictive Control is a discrete-time optimal control strategy that predicts future system behavior using a dynamic model of the system. At each control interval, MPC solves a constrained optimization problem to compute the optimal control inputs that minimize a defined cost function over a finite prediction horizon.

The key elements of MPC are:

- **Prediction Model**: Describes how the system evolves over time.
- **Optimization Problem**: Minimizes a cost function subject to system dynamics and constraints.
- **Receding Horizon Principle**: Only the first input of the optimal control sequence is applied. The optimization is repeated at the next time step with updated system states.

2.23.3 Advantages of MPC in Motor Drives

- Handles multi-objective optimization (e.g., torque tracking, loss minimization).
- Incorporates input and state constraints naturally.
- Offers fast transient performance and excellent steady-state accuracy.
- Capable of managing nonlinearities and model uncertainties.
- Provides a unified framework for multi-loop control (current, speed, and position).

PMSM Modeling in d-q Reference Frame

2.23.4 Electrical Model

The electrical dynamics of a surface-mounted PMSM in the d-q frame are:

$$v_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \tag{2.60}$$

$$v_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \lambda_{pm} \tag{2.61}$$

Where:

- v_d, v_q: d-q axis voltages [V]
- i_d, i_q: d-q axis currents [A]
- R_s: stator resistance [Ω]
- L_d, L_q: d-q axis inductances [H]
- ω_e: electrical angular speed [rad/s]
- λ_{pm}: permanent magnet flux linkage [Wb]

2.23.5 Mechanical Model

$$J\frac{d\omega_m}{dt} = T_e - T_L - B\omega_m \tag{2.62}$$

Where:

- ω_m: mechanical angular speed [rad/s]
- T_e: electromagnetic torque [Nm]
- T_L: load torque [Nm]
- J: rotor moment of inertia [kg·m^2]
- B: viscous friction coefficient [Nm·s]

2.23.6 Electromagnetic Torque Expression

$$T_e = \frac{3}{2}P\left[\lambda_{pm}i_q + (L_d - L_q)i_d i_q\right] \tag{2.63}$$

Where P is the number of pole pairs.

2.24 MPC Implementation for PMSM Drives

2.24.1 Discrete-Time State-Space Model

Using Euler's forward method, the continuous model is discretized:

$$i_d(k+1) = i_d(k) + \frac{T_s}{L_d}\left[v_d(k) - R_s i_d(k) + \omega_e(k)L_q i_q(k)\right] \tag{2.64}$$

$$i_q(k+1) = i_q(k) + \frac{T_s}{L_q}\left[v_q(k) - R_s i_q(k) - \omega_e(k)L_d i_d(k) - \omega_e(k)\lambda_{pm}\right] \tag{2.65}$$

2.24.2 Prediction Model

The predicted state over horizon N_p:

$$x(k+j|k) = f\left(x(k+j-1|k), u(k+j-1|k)\right) \tag{2.66}$$

Where:

- $x = [i_d, i_q]^T$
- $u = [v_d, v_q]^T$

2.24.3 Cost Function

A quadratic cost function is used to minimize tracking error and input variation:

$$J = \sum_{j=1}^{N_p} \left[\lambda_d (i_d^{ref} - i_d(k+j|k))^2 + \lambda_q (i_q^{ref} - i_q(k+j|k))^2\right] + \lambda_u \sum_{j=0}^{N_c-1} \Delta u(k+j)^2 \tag{2.67}$$

Where:

- λ_d, λ_q: weighting factors for current tracking
- λ_u: weighting factor for control effort
- $\Delta u(k) = u(k) - u(k-1)$

2.24.4 Constraint Handling

- **Voltage constraints**:

$$|v_{abc}| \leq \frac{V_{dc}}{\sqrt{3}} \tag{2.68}$$

- **Current constraints**:

$$\sqrt{i_d^2 + i_q^2} \leq I_{max} \tag{2.69}$$

- **Input rate constraints**:

$$|\Delta u| \leq \Delta u_{max} \tag{2.70}$$

Types of MPC Strategies for PMSM Control

2.24.5 Continuous Control Set MPC (CCS-MPC)

- Control variables are continuous (e.g., v_d, v_q).
- Requires PWM modulation.
- Fixed switching frequency, low harmonics.

2.24.6 Finite Control Set MPC (FCS-MPC)

- Uses inverter switching states directly.
- No PWM stage.
- Simpler implementation but variable switching frequency.

2.24.7 Deadbeat Predictive Control

- Special case with prediction horizon = 1.
- Fast dynamics, simple controller.
- Limited robustness to constraints and modeling errors.

CasADi Library for MPC Implementation

2.24.8 Overview

CasADi (**C**omputer **A**lgebra **S**ystem with **A**utomatic **Di**fferentiation) is an open-source software framework designed for efficient algorithmic differentiation and numeric optimization. It is widely used in optimal control, nonlinear model predictive control (NMPC), and parameter estimation applications. CasADi provides symbolic expressions and automatic differentiation capabilities that allow users to define dynamic models and optimization problems in a high-level and compact form.

2.24.9 Key Features of CasADi

- **Automatic Differentiation (AD)**: Enables exact derivatives of complex functions with machine precision.
- **Symbolic Expressions**: Allows users to define mathematical expressions symbolically, which can be easily evaluated and differentiated.
- **Integration with NLP Solvers**: Interfaces with high-performance nonlinear programming (NLP) solvers such as `IPOPT`, `KNITRO`, and `SNOPT`.
- **Support for Optimal Control**: Provides tools for constructing and solving optimal control problems using direct multiple shooting or collocation methods.
- **Code Generation**: Enables generation of efficient C code for deployment on embedded platforms.

2.24.10 Why Use CasADi for MPC?

Model Predictive Control often involves solving constrained optimization problems in real-time. CasADi simplifies this process through its:

- **Concise Problem Definition**: Models and cost functions are defined symbolically and concisely.
- **Efficient Derivative Computation**: Required Jacobians and Hessians are computed automatically, improving solver performance.

- **Seamless Integration with Solvers**: CasADi's solver interfaces make it easy to formulate and solve MPC problems using state-of-the-art algorithms.

2.24.11 Basic Workflow with CasADi

A typical CasADi-based MPC workflow consists of the following steps:

1. **Define the system dynamics**: Specify the state-space equations (e.g., PMSM model) symbolically using CasADi variables.
2. **Discretize the dynamics**: Use numerical methods like Euler or Runge-Kutta to obtain a discrete-time model.
3. **Formulate the cost function**: Define the objective function using symbolic expressions for state and control trajectories.
4. **Add constraints**: Symbolically specify state and input constraints (e.g., current/voltage limits).
5. **Solve the optimization problem**: Use a solver like `IPOPT` to solve the nonlinear programming (NLP) problem at each sampling instant.
6. **Apply control input**: Use the first optimal control action, then repeat the process at the next time step (receding horizon).

2.25 MPC with CasADi (`mpc_cas_ad3`)

```
function output = mpc_cas_ad3(input)
    % Add CasADi to path
    addpath('C:\Users\musta\Downloads\casadi-3.7.0-windows64-matlab2018b')
    import casadi.*

    % Extract input parameters
    id = input(1);
    iq = input(2);
    id_ref = input(3);
    iq_ref = input(4);
    W = input(5);
    T_s = 0.5 / input(6); % Sampling time

    % Motor parameters
    L_d = 0.0057;
    L_q = 0.0125;
    R = 1.2;
    psi_f = 0.123;

    % MPC parameters
    N = 15;

    % Decision variables
    vd = MX.sym('vd', N, 1);
    vq = MX.sym('vq', N, 1);

    % Cost initialization
    J = 0;
    id_k = id; iq_k = iq;
    e_id = 0; e_iq = 0;
    lambda_int = 10;
    g = [];

    % Prediction loop
    for k = 1:N
        id_k1 = id_k + (T_s/L_d)*(vd(k) - R*id_k + W*L_q*iq_k);
        if abs(W) < 1e-3
```

```
            iq_k1 = iq_k + (T_s/L_q)*(vq(k) - R*iq_k - W*L_d*id_k);
        else
            iq_k1 = iq_k + (T_s/L_q)*(vq(k) - R*iq_k - W*L_d*id_k - W*psi_f);
        end
        e_id = e_id + (id_k - id_ref)*T_s;
        e_iq = e_iq + (iq_k - iq_ref)*T_s;
        J = J + (id_k1 - id_ref)^2 + (iq_k1 - iq_ref)^2 + lambda_int*(e_id^2 + e_iq^2);
        g = [g; id_k1^2 + iq_k1^2 - 25];
        id_k = id_k1;
        iq_k = iq_k1;
    end

    % Voltage constraints
    Vmax = 500 / sqrt(3);
    Vmin = -Vmax;
    g = [g; vd - Vmin; Vmax - vd; vq - Vmin; Vmax - vq];

    % Solver setup
    opts = struct('ipopt', struct('tol', 1e-3, 'max_iter', 100, 'print_level', 0), 'print_time', false);
    nlp = struct('x', [vd; vq], 'f', J, 'g', g);
    solver = nlpsol('solver', 'ipopt', nlp, opts);

    % Bounds
    lbg = [-inf * ones(N,1); zeros(4*N,1)];
    ubg = [zeros(N,1); inf * ones(4*N,1)];

    % Solve
    sol = solver('x0', zeros(2*N,1), 'lbg', lbg, 'ubg', ubg);

    % Extract first control action
    vd_opt = full(sol.x(1));
    vq_opt = full(sol.x(N+1));

    output = [vd_opt; vq_opt];
end
```

Listing 2.1: mpc_cas_ad3.m - MPC using CasADi symbolic optimization

2. MPC with FMINCON (mpc_cas)

```
function output = mpc_cas(input)
    id = input(1); iq = input(2);
    id_ref = input(3); iq_ref = input(4);
    W = input(5); T_s = 0.5 / input(6);

    L_d = 0.0057; L_q = 0.0125; R = 1.2; psi_f = 0.123;
    N = 15;

    Vmax = 500 / sqrt(3); Vmin = -Vmax;
    x0 = zeros(2*N, 1);
    lb = [Vmin*ones(N,1); Vmin*ones(N,1)];
    ub = [Vmax*ones(N,1); Vmax*ones(N,1)];

    options = optimoptions('fmincon', 'Display', 'off', 'Algorithm', 'sqp', ...
                        'MaxIterations', 100, 'OptimalityTolerance', 1e-3);

    try
        [x_opt, ~] = fmincon(@(x) cost_function_local(x, id, iq, id_ref, iq_ref, ...
                            W, T_s, L_d, L_q, R, psi_f, N), ...
                            x0, [], [], [], [], lb, ub, ...
                            @(x) constraint_function_local(x, id, iq, W, T_s, ...
                            L_d, L_q, R, psi_f, N), options);
        vd_opt = x_opt(1);
        vq_opt = x_opt(N+1);
    catch
        vd_opt = 0;
        vq_opt = 0;
    end

    output = [vd_opt; vq_opt];

    function J = cost_function_local(x, id, iq, id_ref, iq_ref, W, T_s, ...
                                L_d, L_q, R, psi_f, N)
        vd = x(1:N); vq = x(N+1:2*N);
        J = 0; id_k = id; iq_k = iq; e_id = 0; e_iq = 0; lambda_int = 10;
        for k = 1:N
            id_k1 = id_k + (T_s/L_d)*(vd(k) - R*id_k + W*L_q*iq_k);
            if abs(W) < 1e-3
                iq_k1 = iq_k + (T_s/L_q)*(vq(k) - R*iq_k - W*L_d*id_k);
            else
                iq_k1 = iq_k + (T_s/L_q)*(vq(k) - R*iq_k - W*L_d*id_k - W*psi_f);
            end
            e_id = e_id + (id_k - id_ref)*T_s;
            e_iq = e_iq + (iq_k - iq_ref)*T_s;
            J = J + (id_k1 - id_ref)^2 + (iq_k1 - iq_ref)^2 + lambda_int*(e_id^2 + e_iq^2);
            id_k = id_k1; iq_k = iq_k1;
        end
    end

    function [c, ceq] = constraint_function_local(x, id, iq, W, T_s, ...
                                                L_d, L_q, R, psi_f, N)
        vd = x(1:N); vq = x(N+1:2*N);
        c = zeros(N, 1); id_k = id; iq_k = iq;
        for k = 1:N
            id_k1 = id_k + (T_s/L_d)*(vd(k) - R*id_k + W*L_q*iq_k);
            if abs(W) < 1e-3
                iq_k1 = iq_k + (T_s/L_q)*(vq(k) - R*iq_k - W*L_d*id_k);
            else
                iq_k1 = iq_k + (T_s/L_q)*(vq(k) - R*iq_k - W*L_d*id_k - W*psi_f);
            end
            c(k) = id_k1^2 + iq_k1^2 - 25;
            id_k = id_k1; iq_k = iq_k1;
        end
        ceq = [];
    end
end
```

Listing 2.2: mpc_cas.m - MPC using FMINCON

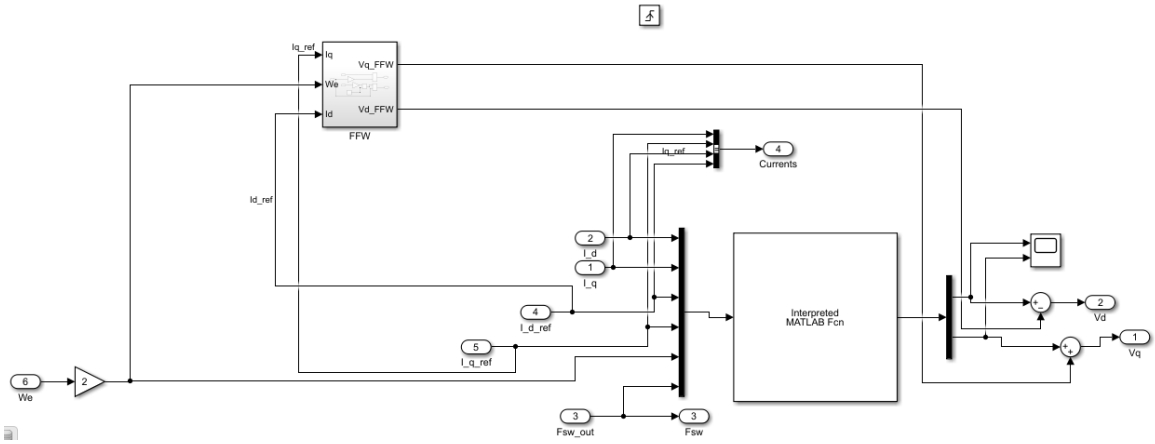


Figure 2.31: MPC For Current Loop

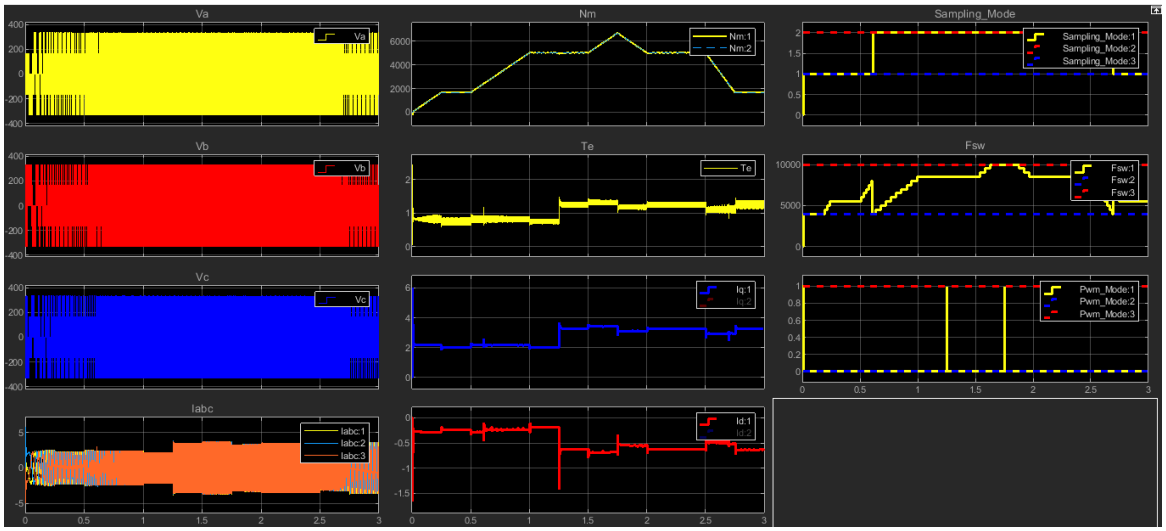


Figure 2.32: Output MPC

2.25.1 Integration with Real-Time Systems

CasADi can generate C code for real-time execution. This is essential for deploying MPC algorithms on embedded platforms, such as digital signal processors (DSPs) used in motor drives.

2.26 Advantages of MPC for PMSM Drives

- **Multi-objective Optimization**: Current tracking, torque control, efficiency.
- **Constraint Handling**: Voltage, current, thermal and mechanical limits.
- **Fast Dynamic Response**: Predictive capabilities ensure quick reaction.
- **Nonlinearity Handling**: Effective for magnetic saturation, parameter drift.
- **Tuning Flexibility**: Cost function weights are intuitive and modular.

2.27 Implementation Challenges

- **Computational Burden**: Real-time optimization can be intensive.
- **Parameter Sensitivity**: Requires accurate models or estimation.
- **Sampling Rate**: Trade-off between accuracy and CPU load.
- **Tuning**: Requires careful selection of prediction/control horizons and weights.

2.28 Real-Time Considerations

- Use of DSPs or FPGAs is recommended.
- Fast QP solvers or explicit MPC can improve speed.
- State observers or sensorless techniques can enhance robustness.

2.29 Conclusion

Model Predictive Control provides a powerful and flexible framework for the control of Permanent Magnet Synchronous Motors. Its ability to handle system constraints, nonlinearities, and multiple objectives makes

Chapter 3

brushless DC motor (BLDC)

3.1 abstract

Brushless DC (BLDC) motors have emerged as a cornerstone technology in modern electromechanical systems, offering superior performance characteristics including high efficiency (>90%), exceptional power-to-weight ratio, and maintenance-free operation. This paper presents a comprehensive analysis of BLDC motor technology, covering fundamental operating principles, detailed mathematical modeling, and advanced control strategies. We examine trapezoidal commutation, sinusoidal control, field-oriented control (FOC), and emerging sensorless techniques, providing comparative performance analysis and practical implementation guidelines. The results demonstrate that modern control approaches like FOC can achieve torque ripple below 5% while maintaining efficiency above 93%, making BLDC motors ideal for high-performance applications in electric vehicles, aerospace systems, and industrial automation.

3.2 Introduction

Brushless DC (BLDC) motors are a class of synchronous electric motors powered by DC electricity and electronically commutated instead of using brushes. Their design ensures low maintenance, high efficiency, and excellent torque-to-weight ratio. With the rise of embedded systems and power electronics, BLDC motors have become prevalent in many applications where high performance and reliability are critical.

Traditional DC motors suffer from wear and tear of brushes and commutators, which limits their lifespan and increases maintenance costs. BLDC motors overcome these issues by replacing the mechanical commutator with a solid-state inverter, enabling precise control of the motor's speed and torque.

3.3 Fundamental Working Principles

A BLDC motor consists of:

- **Stator**: wound with three-phase windings (like an AC motor).
- **Rotor**: equipped with permanent magnets.
- **Electronic commutator**: replaces mechanical brushes using semiconductor switches.

3.3.1 Commutation Principle

- The rotor's position is detected using **Hall effect** sensors or **sensorless techniques**.
- Based on rotor position, the controller energizes the appropriate stator windings to produce a rotating magnetic field.
- This magnetic field interacts with the rotor magnets, generating torque.

BLDC motors can be:

- **Trapezoidal commutated**: uses six-step switching based on Hall signals.
- **Sinusoidally commutated**:uses smooth sine-wave currents for better efficiency and low torque ripple.

3.4 Mathematical Model of BLDC Motor

3.4.1 Back-EMF Equations

The back-EMFs of the three phases are defined as functions of the rotor electrical angle θ_e and electrical angular speed ω_e:

$$E_a = K_e \cdot \omega_e \cdot f(\theta_e) \tag{3.1}$$

$$E_b = K_e \cdot \omega_e \cdot f\left(\theta_e - \frac{2\pi}{3}\right) \tag{3.2}$$

$$E_c = K_e \cdot \omega_e \cdot f\left(\theta_e - \frac{4\pi}{3}\right) \tag{3.3}$$

Here, $f(\theta)$ represents a unit trapezoidal waveform defining the back-EMF shape over one electrical cycle. Alternatively, using flux linkage ψ notation, the back-EMFs can also be expressed as:

$$E_a = \omega_e \cdot \psi_a \tag{3.4}$$

$$E_b = \omega_e \cdot \psi_b \tag{3.5}$$

$$E_c = \omega_e \cdot \psi_c \tag{3.6}$$

3.4.2 Phase Current Differential Equations

Using Kirchhoff's voltage law and assuming a Y-connected winding with symmetrical phases, the differential equations for the phase currents are:

$$\frac{di_a}{dt} = \frac{1}{3L_s}\left(2V_{ab} + V_{bc} - 3R_s i_a + p \cdot \phi_m \cdot \omega_e(\psi_c + \psi_b - 2\psi_a)\right) \tag{3.7}$$

$$\frac{di_b}{dt} = \frac{1}{3L_s}\left(2V_{bc} + V_{ab} - 3R_s i_b + p \cdot \phi_m \cdot \omega_e(\psi_a + \psi_c - 2\psi_b)\right) \tag{3.8}$$

Since $E = \omega_e \cdot \psi$, we can rewrite the above equations in terms of back-EMFs:

$$\frac{di_a}{dt} = \frac{1}{3L_s}\left(2V_{ab} + V_{bc} - 3R_s i_a + p \cdot \phi_m(E_c + E_b - 2E_a)\right) \tag{3.9}$$

$$\frac{di_b}{dt} = \frac{1}{3L_s}\left(2V_{bc} + V_{ab} - 3R_s i_b + p \cdot \phi_m(E_a + E_c - 2E_b)\right) \tag{3.10}$$

Due to the constraint:

$$i_a + i_b + i_c = 0 \quad \Rightarrow \quad i_c = -i_a - i_b \tag{3.11}$$

3.4.3 Torque and Mechanical Dynamics

Electromagnetic Torque

The electromagnetic torque produced by the motor is given by:

$$T_e = p \cdot \phi_m \cdot (i_a\psi_a + i_b\psi_b + i_c\psi_c) \tag{3.12}$$

Where:

- p: Number of pole pairs
- ϕ_m: Permanent magnet flux linkage

Mechanical Dynamics

The mechanical equation describing rotor dynamics is:

$$\frac{d\omega_m}{dt} = \frac{1}{J}\left(T_e - T_l - B \cdot \omega_m\right) \tag{3.13}$$

Where:

- ω_m: Mechanical angular velocity
- T_l: Load torque
- B: Viscous friction coefficient
- J: Moment of inertia

The relation between mechanical and electrical angular velocity is:

$$\omega_e = p \cdot \omega_m \tag{3.14}$$

3.5 Types of Control for BLDC Motors

The performance and efficiency of a BLDC motor depend significantly on the employed control strategy. Each technique has trade-offs between complexity, cost, precision, and dynamic behavior. Below are the primary types of control strategies used in practice:

3.5.1 Six-Step Commutation (Trapezoidal Control)

Six-step commutation, also known as trapezoidal or block commutation, is a widely used technique to control three-phase BLDC motors due to its simplicity and cost-effectiveness.

Operating Principle

The BLDC motor is driven by electronically switching the inverter in six discrete steps per electrical revolution. At each step, two of the three motor windings are energized while the third is left unpowered (floating). The switching sequence is determined by the rotor position, which is typically sensed using three Hall effect sensors spaced 120° electrical apart.

Commutation Table

Table 3.1: Six-Step Commutation Sequence

Step	Hall Sensors (A,B,C)	Active Phases	Current Direction
1	101	A+, B−	A → B
2	100	A+, C−	A → C
3	110	B+, C−	B → C
4	010	B+, A−	B → A
5	011	C+, A−	C → A
6	001	C+, B−	C → B

Current and Torque Characteristics

- Each phase conducts current for 120°.
- The back-EMF waveform is trapezoidal; current waveform is quasi-square.
- The torque is ideally constant when back-EMF and current are aligned, but in practice exhibits ripple.
- Torque ripple arises during commutation due to imperfect transitions.

Control Process

1. Read rotor position from Hall sensors.
2. Determine sector (1 to 6) based on the Hall code.
3. Apply the appropriate voltage to the corresponding phase pairs.
4. Optionally use PWM on the high-side switch for voltage control.

Advantages

- Easy to implement and low cost.
- Robust in mid-to-high speed applications.
- No need for complex transformations or sensorless observers.

Limitations

- Significant torque ripple (up to 15%).
- Noisy operation due to abrupt switching.
- Poor low-speed performance.
- Not suitable for high-performance or quiet applications.

Typical Applications

- Cooling fans and pumps
- Home appliances and power tools
- Drones and RC motors
- Automotive cooling or HVAC subsystems

3.5.2 Sinusoidal Commutation

This control strategy applies sinusoidal currents that are synchronized with the rotor position.

- **Working Principle:** The motor phases are driven with sinusoidal current waveforms, usually using PWM to approximate the desired current shape.
- **Position Feedback:** High-resolution encoder or sensorless observer
- **Back-EMF Type:** Sinusoidal (ideal case)
- **Advantages:**
  - Smoother torque with reduced ripple
  - Lower vibration and acoustic noise
  - Better low-speed performance compared to six-step
- **Limitations:**
  - Requires accurate position estimation
  - Less efficient than FOC in some operating regions
  - Increased control complexity
- **Applications:** Medical equipment, robotics, quiet appliance motors

3.5.3 Field-Oriented Control (FOC)

Also known as vector control, FOC is a sophisticated and high-performance technique that allows independent control of torque and flux.

- **Working Principle:**
  1. 3-phase currents are transformed into 2-phase (α-β) quantities using the Clarke transformation.
  2. Then transformed into a rotating d-q frame using the Park transformation aligned with the rotor flux.
  3. Independent PI controllers regulate i_d (flux-producing) and i_q (torque-producing) components.
  4. The output voltages are converted back to three-phase using inverse transforms and fed to a space vector PWM inverter.
- **Advantages:**

  - Near-ideal torque control with minimal ripple
  - High efficiency across wide speed range
  - Superior dynamic response
  - Smooth operation at low speeds and during transients

- **Limitations:**
  - Requires precise rotor angle estimation (encoder or observer)
  - High computational burden
  - Complex implementation and tuning

- **Applications:** Electric vehicles, robotics, servo drives, high-end appliances

3.5.4 Direct Torque Control (DTC)

DTC is a high-performance control strategy that regulates torque and stator flux directly without using PWM or coordinate transformations.

- **Working Principle:** Torque and flux errors are calculated using estimated back-EMF, and hysteresis comparators decide the inverter switching state in real time.

- **Advantages:**
  - Extremely fast torque response
  - No need for complex transformations or modulation
  - Robust to parameter variations

- **Limitations:**
  - More difficult to implement with BLDC due to non-sinusoidal EMF
  - Higher torque and current ripple compared to FOC
  - Sensitive to noise in low-speed operation

- **Applications:** High-performance motor drives, industrial control, precision actuation

3.5.5 Sensorless Control

Sensorless techniques eliminate the need for physical position sensors, improving reliability and reducing cost.

- **Approaches:**
  - Back-EMF Zero-Crossing Detection (for six-step control)
  - Observer-based methods (Luenberger, Sliding Mode, Kalman Filter)
  - High-frequency signal injection (for low speed detection)

- **Advantages:**
  - Lower system cost and size
  - Improved reliability in harsh environments

  - No mechanical alignment or wear issues

- **Challenges:**
  - Back-EMF is weak at low speeds ⇒ poor low-speed resolution
  - Observer tuning is sensitive to motor parameters
  - Requires fast and precise ADC and processing units

- **Applications:** Electric vehicles, HVAC systems, drones, pumps

3.6 Modling Of BLDC Motor

Mathematical Modeling

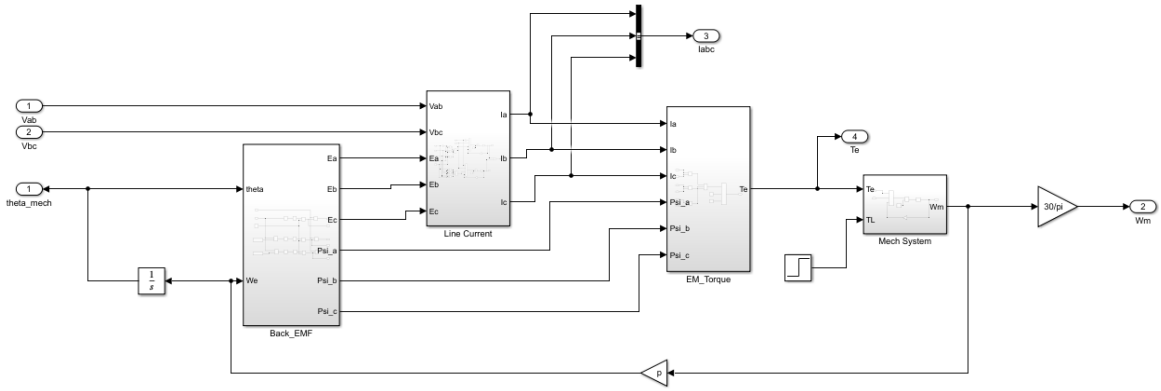


Figure 3.1: BLDC mathematical modeling

3.6.1 Line Current Block

The line current block represents the three-phase stator windings of the BLDC motor. The electrical equations governing the stator currents are:

Voltage Equations:

$$V_a = R_a \cdot i_a + L_a \cdot \frac{di_a}{dt} + e_a \tag{3.15}$$

$$V_b = R_b \cdot i_b + L_b \cdot \frac{di_b}{dt} + e_b \tag{3.16}$$

$$V_c = R_c \cdot i_c + L_c \cdot \frac{di_c}{dt} + e_c \tag{3.17}$$

Where:

- V_a, V_b, V_c = Phase voltages
- i_a, i_b, i_c = Phase currents
- R_a, R_b, R_c = Phase resistances
- L_a, L_b, L_c = Phase inductances
- e_a, e_b, e_c = Back EMF voltages

Input Signals:

- V_{ab}, V_{bc}: Line-to-line voltages
- E_a, E_b, E_c: Enable signals for electronic commutation
- θ: Rotor position (feedback)

Output Signals:

- i_a, i_b, i_c: Three-phase currents

3.6.2 Back EMF Block

The Back EMF (Electromotive Force) block calculates the voltage induced in the stator windings due to rotor motion. For a BLDC motor, the back EMF is trapezoidal in shape.

Mathematical Expression:

$$e_a = K_e \cdot \omega_m \cdot f_a(\theta) \tag{3.18}$$
$$e_b = K_e \cdot \omega_m \cdot f_b(\theta) \tag{3.19}$$
$$e_c = K_e \cdot \omega_m \cdot f_c(\theta) \tag{3.20}$$

Where:

- K_e = Back EMF constant
- ω_m = Mechanical angular velocity
- $f_a(\theta), f_b(\theta), f_c(\theta)$ = Position-dependent functions (trapezoidal waves)
- θ = Rotor electrical position

Input Signals:

- W_e: Electrical angular velocity
- Pal_a, Pal_b, Pal_c: Position signals

3.6.3 Electromagnetic Torque Block

The electromagnetic torque is generated by the interaction between stator currents and rotor magnetic field.

Torque Equation:

$$T_e = \frac{e_a \cdot i_a + e_b \cdot i_b + e_c \cdot i_c}{\omega_m} \tag{3.21}$$

Or alternatively:

$$T_e = K_t \cdot (i_a \cdot f_a(\theta) + i_b \cdot f_b(\theta) + i_c \cdot f_c(\theta)) \tag{3.22}$$

Where:

- T_e = Electromagnetic torque
- K_t = Torque constant
- Generally, $K_t = K_e$ for BLDC motors

Input Signals:

- i_a, i_b, i_c: Phase currents
- Pal_a, Pal_b, Pal_c: Position-dependent signals

3.6.4 Mechanical System Block

The mechanical subsystem models the rotor dynamics and mechanical load.

Mechanical Equation:

$$J \cdot \frac{d\omega_m}{dt} = T_e - T_L - B \cdot \omega_m \tag{3.23}$$

Where:

- J = Moment of inertia
- T_e = Electromagnetic torque
- T_L = Load torque
- B = Viscous friction coefficient
- ω_m = Mechanical angular velocity

Angular Position:

$$\theta_m = \int \omega_m \, dt \tag{3.24}$$

Relationship between mechanical and electrical angles:

$$\theta_e = P \cdot \theta_m \tag{3.25}$$
$$\omega_e = P \cdot \omega_m \tag{3.26}$$

Where P = Number of pole pairs

Input Signals:

- T_e: Electromagnetic torque
- T_L: Load torque

Output Signals:

- W_m: Mechanical angular velocity
- θ: Rotor position

3.6.5 Integrator Block (1/s)

The integrator block converts angular velocity to angular position:

$$\theta(t) = \int_0^t \omega(\tau)\, d\tau + \theta_0 \tag{3.27}$$

This provides the rotor position feedback necessary for proper commutation timing.

3.7 Trapezoidal Control of BLDC Motors

3.7.1 Introduction

Trapezoidal control, also known as **six-step commutation** or **block commutation**, is one of the simplest and most widely implemented techniques for controlling **Brushless DC (BLDC) motors**. The name derives from the trapezoidal shape of the back-electromotive force (back-EMF) waveform produced by BLDC motors.

This method is especially popular in cost-sensitive and high-speed applications such as electric scooters, drones, fans, and robotic actuators, due to its implementation simplicity, high efficiency, and minimal sensor requirements.

3.7.2 Operating Principle

In trapezoidal control, only two of the three stator windings are energized at any given time while the third phase is left floating. This creates a rotating magnetic field that interacts with the rotor's permanent magnets. The control strategy follows a six-step commutation pattern, where each step spans 60° electrical, resulting in a full electrical revolution over six steps.

The control ensures that the phase currents are synchronized with the rotor position to produce constant torque.

3.7.3 Trapezoidal Back-EMF Characteristics

The BLDC motor used in trapezoidal control is designed with a trapezoidal back-EMF profile. Key properties include:

- Flat-topped back-EMF waveform over 120° electrical per phase.
- Phase-shifted by 120° between phases A, B, and C.
- Enables quasi-constant torque generation when combined with square-wave current injection.

3.7.4 Six-Step Commutation Sequence

The six-step sequence determines which two phases are energized and which is left floating, depending on the rotor position.

Table 3.2: Six-Step Commutation Table

Step	Phase A	Phase B	Phase C
1	+V	-V	Floating
2	+V	Floating	-V
3	Floating	+V	-V
4	-V	+V	Floating
5	-V	Floating	+V
6	Floating	-V	+V

Here, **+V** implies the high-side switch is ON and low-side is OFF; **-V** implies the opposite; and **Floating** means both switches are OFF.

3.7.5 Rotor Position Detection

Accurate rotor position detection is critical for correct commutation timing. Two main techniques are used:

Sensor-Based Control

Hall-effect sensors embedded in the motor stator detect rotor magnetic polarity. The three sensor signals generate a 3-bit code used to determine the current step in the commutation sequence.

Table 3.3: Hall Sensor States and Commutation Steps

Hall A	Hall B	Hall C	Step
1	0	1	1
1	0	0	2
1	1	0	3
0	1	0	4
0	1	1	5
0	0	1	6

Sensorless Control

Sensorless trapezoidal control estimates rotor position using the back-EMF of the unenergized phase. The zero-crossing of this signal is detected and used to infer rotor position.A delay equivalent to 30° electrical is then applied to determine the next commutation step.

3.7.6 Current Waveform and Torque Production

The phase current in trapezoidal control ideally takes the form of a rectangular waveform aligned with the flat portion of the back-EMF. This maximizes torque production, given by:

$$T_e = k_T \cdot I$$

Where:

- T_e is the electromagnetic torque,
- k_T is the torque constant,

- I is the phase current.

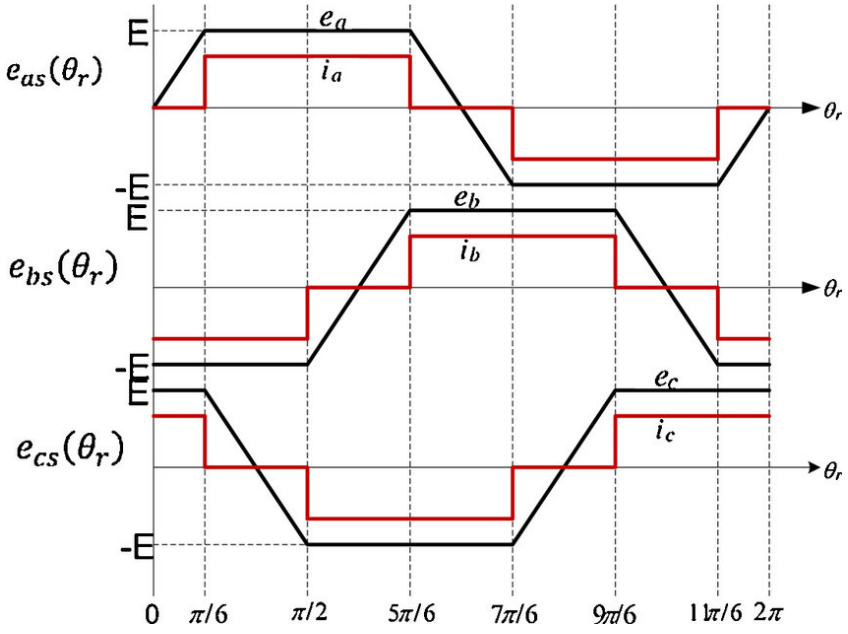


Figure 3.2: Alignment of rectangular current with trapezoidal back-EMF

3.7.7 Speed Control via PWM

Speed control is achieved by adjusting the PWM duty cycle on the low-side switches. The average voltage across the motor phases is given by:

$$V_{avg} = D \cdot V_{dc}$$

Where:

- D is the PWM duty cycle,
- V_{dc} is the supply voltage.

An increase in duty cycle leads to higher average voltage, increasing motor speed and torque.

3.7.8 Advantages

- **Simplicity**: Easy to implement in both hardware and software.
- **Low computation**: No need for complex transformations or observers.
- **Sensorless capable**: Back-EMF sensing eliminates the need for Hall sensors.
- **Fast response**: Suitable for high-speed dynamic systems.

3.7.9 10. Disadvantages

- **Torque ripple**: Due to the discontinuous current and commutation steps.
- **Low-speed limitations**: Sensorless operation is unreliable at low speeds.
- **Increased noise**: Mechanical and electrical noise due to switching.

- **Lower efficiency**: Less optimal current utilization compared to FOC.

Table 3.4: Comparison of Trapezoidal, Sinusoidal, and FOC Control Techniques

Feature	**Trapezoidal**	**Sinusoidal**	**FOC**
Back-EMF Shape	Trapezoidal back-EMF	Sinusoidal back-EMF	Sinusoidal back-EMF
Current Shape	Square wave current	Sinusoidal current	Sinusoidal current (d-q axes)
Torque Ripple	High	Moderate	Very Low
Control Complexity	Low	Medium	High
Cost	Low	Medium	High
Low-Speed Performance	Poor (sensorless)	Moderate	Excellent
Sensorless Capability	Yes (zero-crossing)	Yes (with advanced algorithms)	Yes (observers/estimators)
Efficiency	Moderate	High	Very High
Computational Demand	Very Low	Medium	High
Applications	Low-cost: e-bikes, fans	General-purpose: pumps, appliances	High-performance: EVs, robotics

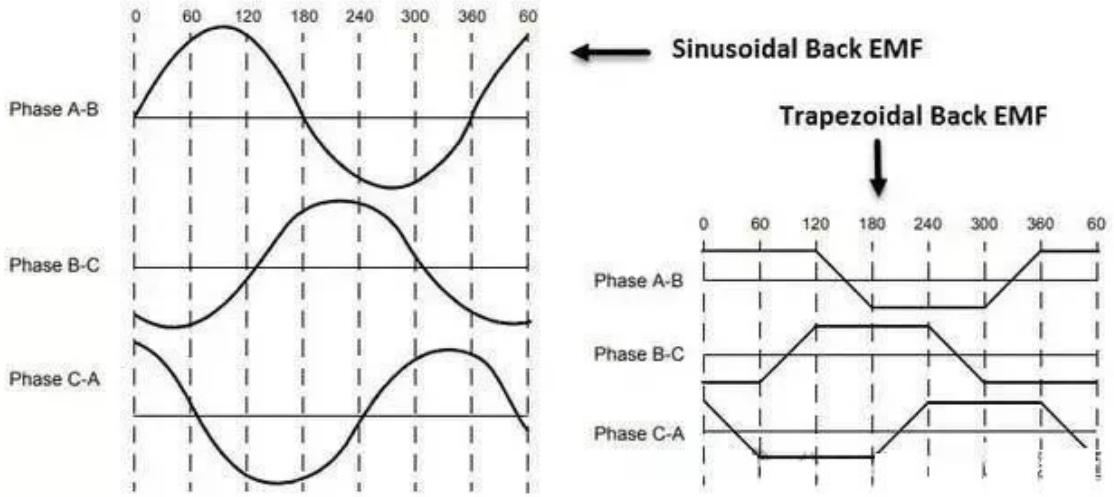


Figure 3.3: Trapezoidal vs. Sinusoidal

3.7.10 Summary

Trapezoidal control is a robust and cost-effective method for BLDC motor drive. It trades control precision and smooth torque for simplicity and fast implementation. In applications where low noise and high performance are critical, sinusoidal control or Field-Oriented Control (FOC) may be preferred. Nevertheless, trapezoidal control remains an essential technique in the electric drive engineer's toolkit.

3.8 Speed Control

3.8.1 Model

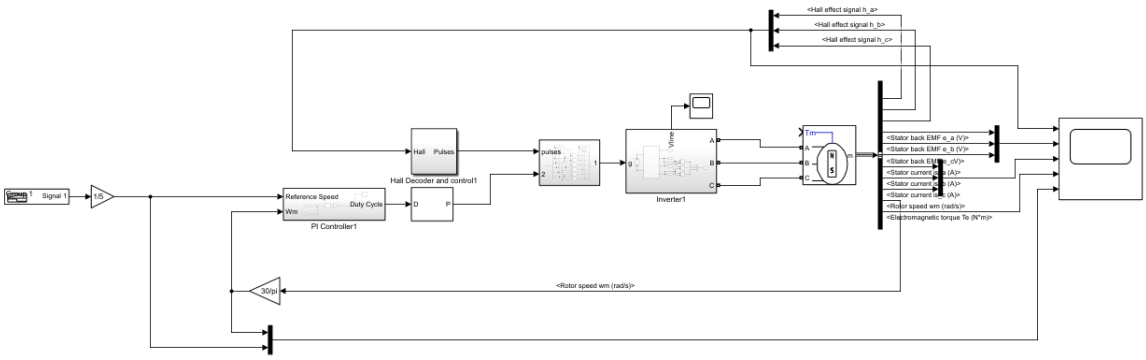


Figure 3.4: Speed Control

3.8.2 Hall effect

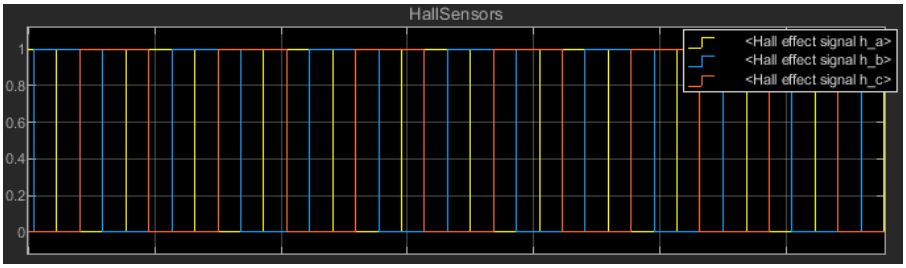


Figure 3.5: Hall Sensor

3.8.3 Back-EMF

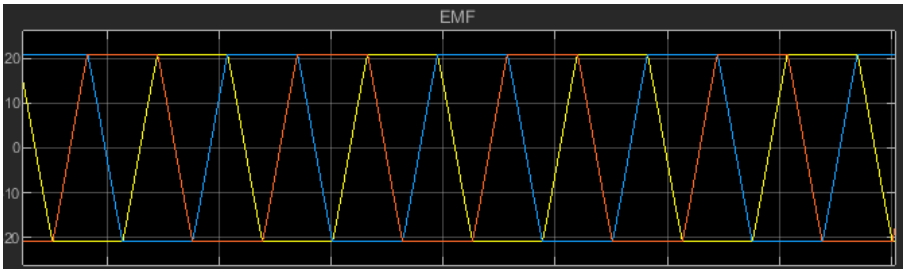


Figure 3.6: Back-EMF

3.8.4 Torque-Speed

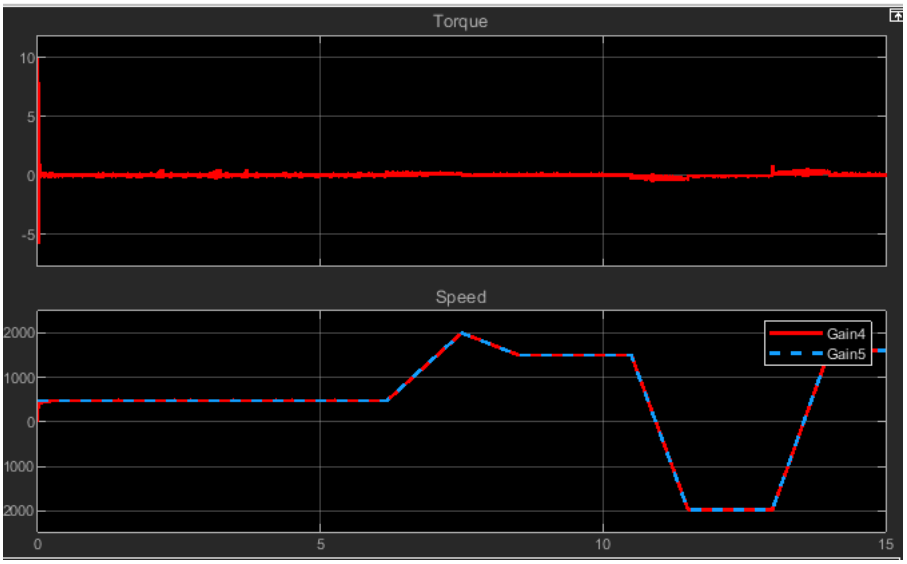


Figure 3.7: Torque-Speed

3.8.5 Output

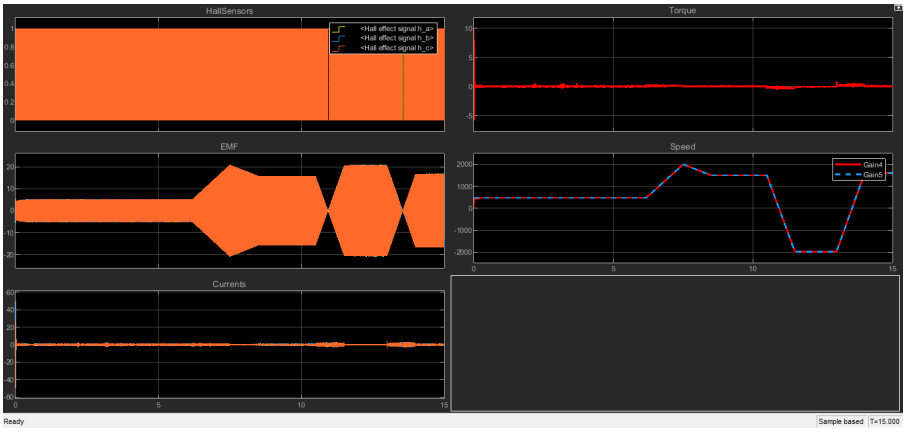


Figure 3.8: Output

Chapter 4

Practical Implementation of BLDC Motor

4.1 abstract

This chapter presents a comprehensive, step-by-step development workflow for Brushless DC (BLDC) motor control systems utilizing Model-Based Design (MBD) in MATLAB/Simulink. The workflow encompasses three key stages: Model-in-the-Loop (MIL), Software-in-the-Loop (SIL), and Hardware-in-the-Loop (HIL), enabling simulation, code generation, and real-time validation of control algorithms. Core features include detailed modeling of BLDC dynamics, Hall-effect-based speed feedback, PI controller design and tuning, and verification through Model Advisor and Code Advisor tools. The framework supports embedded deployment on microcontrollers such as the TI C2000 series, ensuring a reliable and scalable solution for real-time motor control applications.

4.2 Introduction

Brushless DC (BLDC) motors are widely used in automotive, industrial, and robotic applications due to their high efficiency, compact size, and superior torque characteristics. Developing robust and efficient control systems for BLDC motors requires a structured and verifiable design methodology. This chapter introduces a Model-Based Design (MBD) workflow that incorporates progressive testing phases using MATLAB/Simulink and embedded targets, including:

- **Model-in-the-Loop (MIL)**: Algorithm verification through simulation.
- **Software-in-the-Loop (SIL)**: Validation of generated code in a simulated environment.
- **Hardware-in-the-Loop (HIL)**: Real-time verification on embedded hardware.

By iteratively validating the system from simulation to deployment, the methodology ensures functional correctness, safety, and real-time capability before hardware implementation.

4.3 Motor Parameters Estimation

4.3.1 Stator Resistance (R_s)

Method: Apply a DC voltage across any two stator terminals of the motor (e.g., phases A and B) and measure the resulting DC current.

Explanation: Under DC excitation, the motor's inductance has no effect in steady state,so only the stator resistance limits the current.

Calculation:

$$R_{ab} = \frac{V}{I} = 2R_s \tag{4.1}$$

Thus,

$$R_s = \frac{V}{2I} \tag{4.2}$$

Note: Apply a low DC voltage to prevent excessive current that may damage the windings.

4.3.2 Stator Inductance (L_s)

Method: Replace the DC supply with a low-frequency AC supply across two motor terminals. Measure the RMS voltage (V) and current (I) to calculate the total impedance (Z).

Calculation:

$$Z = \frac{V}{I} = 2\sqrt{R_s^2 + X_L^2} \tag{4.3}$$

Solving for the inductive reactance:

$$X_L = \sqrt{\left(\frac{Z}{2}\right)^2 - R_s^2} \tag{4.4}$$

The stator inductance is then given by:

$$L_s = \frac{X_L}{2\pi f} \tag{4.5}$$

where:

- f is the frequency of the AC supply.

Note: A low frequency (e.g., 10 Hz) minimizes core losses and improves measurement accuracy.

4.3.3 Number of Poles (p)

Method: Rotate the motor using an external mechanical source at a known mechanical speed. Measure the frequency of the generated back-EMF.

Calculation:

$$N_s = \frac{120f}{p} \tag{4.6}$$

Solving for the number of poles:

$$p = \frac{120f}{N_s} \tag{4.7}$$

where:

- N_s is the mechanical speed in RPM.
- f is the electrical frequency in Hz.

Note: Distinguish between mechanical speed (RPM) and electrical frequency (Hz). The number of poles determines how many electrical cycles occur per mechanical revolution.

Summary Table

Table 4.1: Summary of Motor Parameter Estimation Methods

Parameter	Method Summary	Formula
Stator Resistance (R_s)	Apply DC, measure V and I	$R_s = \frac{V}{2I}$
Stator Inductance (L_s)	Apply low-freq AC, measure V, I, remove R_s	$L_s = \frac{\sqrt{\left(\frac{Z}{2}\right)^2 - R_s^2}}{2\pi f}$
Number of Poles (p)	Rotate motor, measure mechanical speed and f	$p = \frac{120f}{N_s}$

4.4 Model-in-the-Loop (MIL): Functional Simulation

4.4.1 BLDC Motor Modeling

The motor is modeled using either Simscape Electrical or custom Simulink blocks to reflect its electrical and mechanical behavior:

- **Electrical subsystem**: Phase resistances, inductances, and trapezoidal back-EMF generation.
- **Mechanical subsystem**: Rotor inertia, damping, and load torque.
- **Inverter model**: Three-phase PWM bridge using MOSFETs/IGBTs.
- **Position feedback**: Hall-effect sensor logic for 6-step commutation.

Governing Equations

$$V_{emf} = K_e \cdot \omega \quad \text{(Back-EMF)} \tag{4.8}$$

$$T_e = K_t \cdot I \quad \text{(Electromagnetic Torque)} \tag{4.9}$$

4.4.2 PI Controller Design

A Proportional-Integral (PI) controller is used for closed-loop speed control:

$$u(t) = K_p \cdot e(t) + K_i \int e(t)dt \tag{4.10}$$

where:

- $e(t)$: speed error,
- $u(t)$: PWM duty ratio command.

Tuning Strategy

1. Perform open-loop analysis to determine system dynamics.
2. Apply classical tuning (Ziegler–Nichols) or automated tuning.
3. Include anti-windup to handle actuator saturation.

4.4.3 Speed Estimation via Hall Sensors

The motor's speed is estimated using the time interval between transitions of the Hall-effect sensors:

$$\text{Speed (RPM)} = \frac{60}{N \cdot \Delta t} \tag{4.11}$$

where:

- N: pole pairs,
- Δt: time between Hall edge events.

4.4.4 Model Advisor Validation

The **Model Advisor** in Simulink provides comprehensive diagnostic checks to ensure control system models adhere to industry standards before code generation. This tool is critical in the *Model-in-the-Loop (MIL)* phase for identifying design flaws that could compromise embedded code quality.

Key Validation Categories

- **Compliance with Modeling Standards:**
  - **MAAB (MathWorks Automotive Advisory Board):** Ensures model consistency, readability, and maintainability
  - **ISO 26262 Part 6:** Enforces functional safety practices for automotive embedded software
- **Algebraic Loop Detection:**
  - Identifies unintended direct feedthroughs causing simulation instability
  - Recommends solutions (delay blocks, memory elements) for deterministic real-time behavior
- **Solver Configuration Checks:**
  - Verifies use of fixed-step solvers for embedded targets
  - Ensures proper handling of sample time mismatches and rate transitions
- **Data Type and Signal Verification:**
  - Enforces explicit data typing (`uint16`, `int32`, `boolean`)
  - Validates signal dimension consistency
- **Block Compatibility Analysis:**
  - Flags unsupported blocks (event-based, variable-size signals)
  - Suggests code-generation-compatible alternatives

Validation Outcome

These checks ensure models are:

- Structurally optimized for automatic code generation
- Free from common embedded implementation pitfalls
- Compliant with automotive functional safety standards

4.5 Software-in-the-Loop (SIL): Code Verification

4.5.1 Code Generation with Embedded Coder

The control algorithm is auto-coded using Embedded Coder:

- Converts Simulink models to ANSI-C.
- Supports fixed- or floating-point targets.
- Optimized for embedded deployment (e.g., TI C2000, STM32).

4.5.2 SIL Testing Process

- Replace simulation controller with compiled C code.
- Run simulation to verify functional equivalence with MIL.
- Compare outputs using scope plots, numerical metrics, or assertions.

4.5.3 Code Advisor Analysis

The **Code Advisor** evaluates auto-generated C code quality and efficiency during *Software-in-the-Loop (SIL)* and *Hardware-in-the-Loop (HIL)* verification stages.

Core Evaluation Metrics

- **MISRA C Compliance:**
  - Enforces guidelines for safety-critical systems (automotive/aerospace)
  - Prohibits risky constructs (pointer arithmetic, recursion)
- **Static Code Analysis:**
  - **Memory Usage:** Estimates peak stack/heap requirements
  - **Variable Tracking:** Identifies unused/unsafe global variables
  - **Dead Code Detection:** Flags non-executable code segments
- **Performance Profiling:**
  - Estimates CPU cycle counts per function
  - Identifies computational bottlenecks
- **Data Type Optimization:**

– Recommends fixed-point/integer replacements for floating-point
– Maps Simulink types to optimized C types (`int16_T`, `boolean_T`)

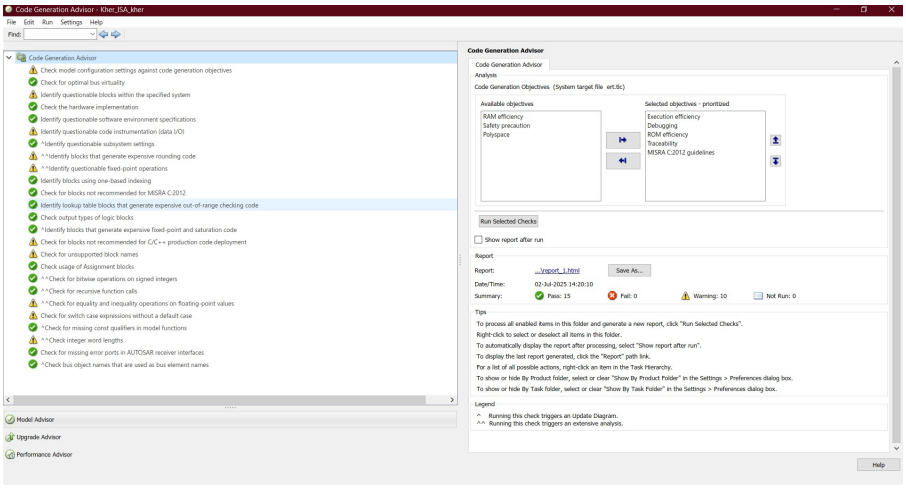


Figure 4.1: Code Advisor

Table 4.2: Model vs. Code Analysis Comparison

Aspect	**Model Advisor**	**Code Advisor**
Verification Stage	MIL	SIL/HIL
Focus	Model structure	Code implementation
Key Standards	MAAB, ISO 26262	MISRA C
Critical Checks	Algebraic loops, sample times	Memory usage, cycle counts
Optimization Targets	Solver settings, block compatibility	Data types, execution efficiency

4.6 Hardware-in-the-Loop (HIL): Real-Time Testing

4.6.1 Real-Time Target: DSP Deployment

Generated code is deployed to a real-time embedded platform:

- Target: TI F28379D or similar.
- Functions: Generate PWM, read Hall sensors via eCAP, and execute PI loop.
- Sampling rate: Typically 10 kHz control ISR.

4.6.2 Speed Measurement Firmware (Hall-Based)

Understanding the eCAP (Enhanced Capture) Module

The **Enhanced Capture (eCAP)** module is a peripheral in TI's C2000 family of microcontrollers designed to accurately measure the timing characteristics of external events. In BLDC motor control systems using Hall-effect sensors, the eCAP module plays a crucial role in speed and position estimation by capturing the timing of signal transitions.

4.6.3 Key Functionalities of eCAP

- **Edge Detection:**
  - Detects rising, falling, or both edges of an input signal
  - Essential for Hall-effect sensor applications where edges correspond to rotor sector transitions
- **Timestamp Capture:**
  - Captures the value of a free-running time base (system clock or timer) when an edge is detected
  - Stores timestamp in dedicated registers for precise event tracking
- **Multiple Capture Events:**
  - Configurable to capture up to four edges before generating an interrupt
  - Enables measurement of:
    - Time between consecutive edges
    - PWM duty cycle (high/low times)
    - Period and frequency of pulse trains
- **Time Difference Calculation:**
  - Computes Δt between captured events
  - For BLDC motors, Δt represents rotor movement between electrical positions
- **Prescaler and Time Base:**
  - Configurable prescaler improves resolution or extends measurement range
  - Time base can reset on capture or run continuously

- **Interrupt Handling:**
  - Generates interrupts after predefined edge captures
  - Enables real-time processing with deterministic timing
- **Operating Modes:**
  - **Capture Mode:** Standard edge timestamping
  - **APWM Mode:** Auxiliary PWM generation capability

4.6.4 Applications in BLDC Motor Control

The eCAP module is particularly valuable for:

- **Hall Sensor Speed Measurement:**
  - Timestamps Hall sensor transitions to calculate rotor speed
  - Speed (RPM) derived from time between sector changes
- **Sector Identification:**
  - Each unique Hall state combination identifies a rotor sector
  - Timestamping sector changes enables speed calculation
- **Diagnostic Monitoring:**
  - Detects abnormal transitions or irregular timing
  - Identifies mechanical faults (e.g., stalling) or electrical noise

4.6.5 Benefits of Using eCAP

Table 4.3: Key Features and Benefits of the eCAP Module

Feature	Benefit
High timing resolution	Enables accurate speed and period measurements
Hardware-based event detection	Reduces CPU load and improves system determinism
Flexible edge configuration	Supports various sensor types and measurement needs
Integrated interrupt generation	Facilitates real-time control loop synchronization
Multiple capture registers	Allows comprehensive signal characterization

Hardware-in-the-Loop (HIL)

4.6.6 External Mode Settings



Figure 4.2: External Mode Configuration - Hardware Resources Setup

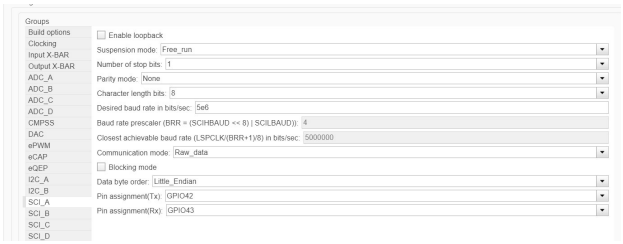


Figure 4.3: External Mode Configuration - Serial Communication Parameters

Parameter	Setting	Parameter	Value
Communication Interface	XCP on Serial	Baud Rate	5 Mbps
SCI Module	SCI_A	Stop Bits	1
Serial Port	COM4	Parity	None
Logging Buffer	1000 bytes	Data Bits	8
Verbose Mode	Enabled	Byte Order	Little Endian
Hardware Modules	ADC, CMPSS, DAC, ePWM, eCAP	TX Pin	GPIO42
		RX Pin	GPIO43
		Suspension Mode	Free Run

4.6.7 Hall Effect Sensor Signal Processing System

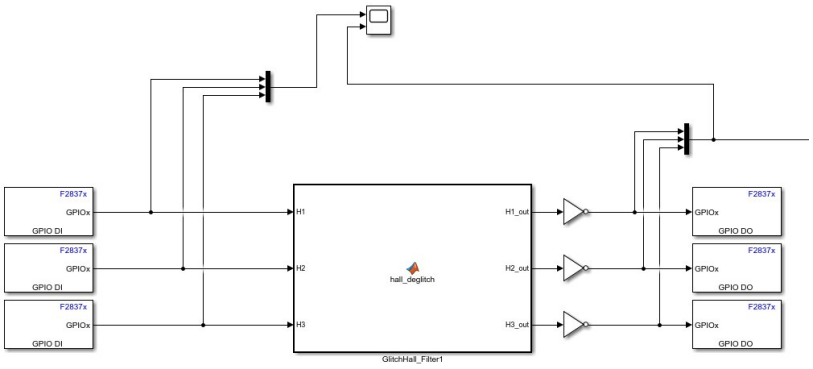


Figure 4.4: Hall Sensor Reading

This system implements a robust Hall effect sensor interface using GPIO digital input/output with signal conditioning and noise suppression capabilities.

4.6.8 Signal Acquisition

- **Input Source**: Hall effect sensor
- **Interface**: GPIO Digital Input (DI)
- **Configuration**: Pull-up resistor configuration
- **Signal Type**: Active-low digital signal

4.6.9 Signal Conditioning

- **Glitch Filtering**: Implemented to suppress electrical noise and transient spikes
- **Purpose**: Ensures signal integrity and prevents false triggering
- **Method**: Digital filtering algorithm removes short-duration noise pulses

4.6.10 Signal Inversion

- **Logic Inversion**: NOT gate operation applied to filtered signal
- **Rationale**: Compensates for pull-up resistor configuration of Hall effect sensor
- **Result**: Converts active-low input to active-high output logic

4.6.11 Output Generation

- **Output Interface**: GPIO Digital Output (DO)
- **Signal Type**: Clean, noise-free digital signal
- **Polarity**: Active-high logic level

Glitch Filtering

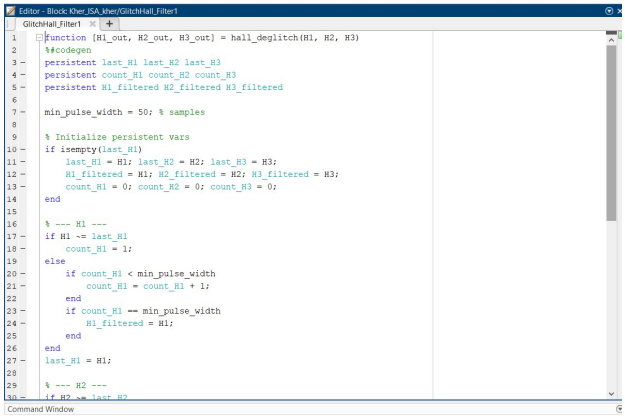


Figure 4.5: Initialization and H1 Processing Code

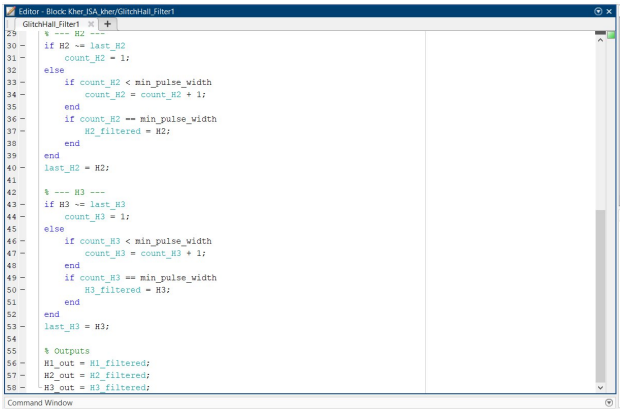


Figure 4.6: H2 and H3 Processing Code

4.6.12 Speed Measurement

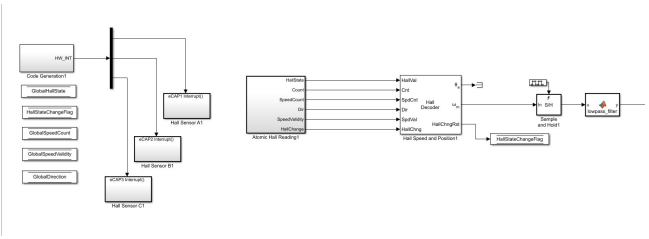


Figure 4.7: Speed Measurements

In this implementation, a combination of hardware capture and signal validation techniques is used to accurately estimate the rotor's speed and electrical position in a Hall-sensor-based BLDC motor control system. The process involves three key stages:

1. Measuring time intervals between Hall transitions using the **eCAP module**.
2. Validating the Hall state using the **Hall Validity** block.
3. Computing rotor speed and position using the **Hall Speed and Position** block.

1. Time Measurement Using eCAP

The **eCAP (Enhanced Capture)** peripheral on the TI C2000 microcontroller is configured to detect **rising edges** of Hall sensor signals. It captures timestamps of each transition using a high-resolution timer. The time difference (Δt) between two successive rising edges is computed in real-time, which forms the basis for rotor speed calculation. This hardware-based measurement ensures:

- High timing resolution.
- Low-latency, interrupt-driven operation.
- Accurate event detection even at low speeds.

2. Hall Signal Validation

Before using the captured signals for control, the raw Hall states are passed through the **Hall Validity** block. This block performs the following functions:

- Verifies that the Hall input represents a valid state (only one or two bits high at a time).
- Flags any invalid transitions (e.g., due to glitches, EMI, or sensor failure).
- Updates only when a Hall state change is detected, minimizing computational overhead.

If a valid transition occurs, the block sets the `SpeedValidity` signal to logical high (`1`). Otherwise, the signal remains low, disabling speed output from downstream processing.

3. Speed and Position Estimation

The validated Hall state, direction signal, and measured time interval are input into the **Hall Speed and Position** block. This block estimates:

- **Mechanical Speed** (ω_m): Output via the `Speed` port in revolutions per minute (RPM). The block returns zero if `SpeedValidity` is `0`, ensuring no invalid values propagate.
- **Electrical Position** (θ_e): Output via the `Position` port, based on the Hall state, rotation direction, and transition timing.

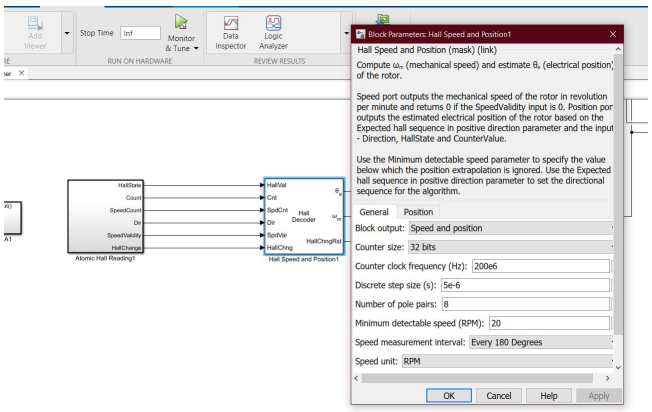


Figure 4.8: Hall Speed and Position

```
function y = lowpass_filter(u)
%#codegen

% Persistent variable to store previous output
persistent y_prev
alpha = 0.001;  % Smoothing factor (tune this: 0 < alpha < 1)

% Initialize previous output if empty
if isempty(y_prev)
    y_prev = 0;
end

% Low-pass filter formula
y = alpha * u + (1 - alpha) * y_prev;

% Update previous output
y_prev = y;
```

Figure 4.9: Speed Filter

4.6.13 Controller

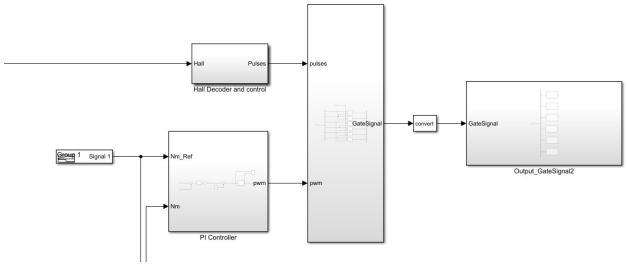


Figure 4.10: Controller

4.6.14 Outputs

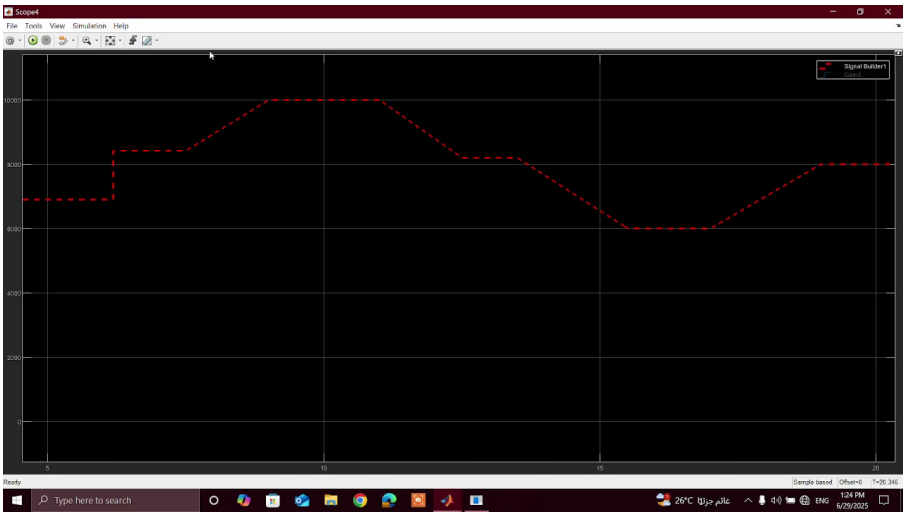


Figure 4.11: Reference Speed

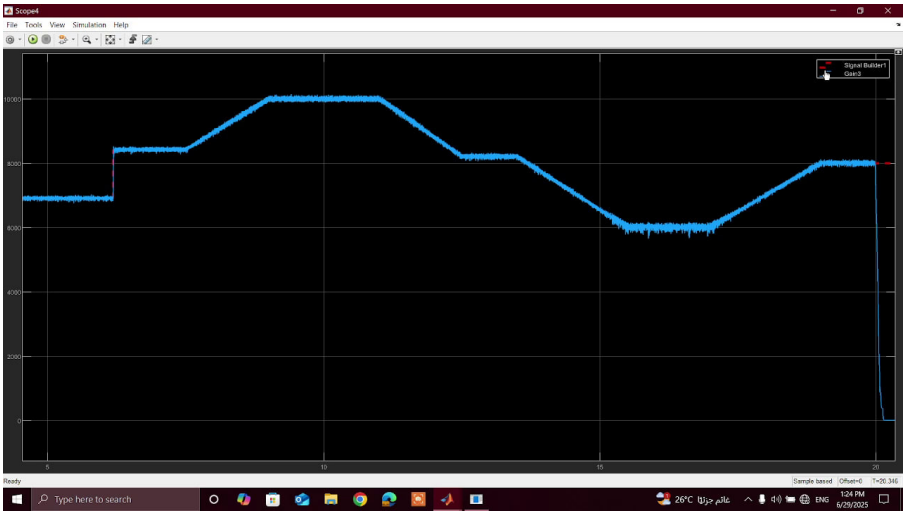


Figure 4.12: Actual Speed

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